

Trade Liberalization and Aggregate Risk in Endogenous Production Networks ^{*}

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Abstract

This paper asks how trade liberalization affects aggregate risk. I develop a multi-country model in which firms choose suppliers before shocks, trading off expected costs against diversification. Using Long-WIOD data for 24 countries and 23 industries, I invert bilateral trade wedges from expenditure shares, allowing for insurance premia. Moving border wedges from 1967 to 2024 lowers mean consumption variance by 36% and reduces risk in 20 of 24 countries. Pushing barriers below the observed range reverses the pattern: production shifts toward central suppliers, and aggregate risk rises. Market incompleteness matters through buyer-specific pricing kernels that shape an endogenous input-output network.

Keywords: Aggregate shocks, production networks, international trade, uncertainty, global financial crisis.

JEL classification: F44, F14, E32, D85, C68.

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1 Introduction

When choosing suppliers, firms trade expected costs against diversification. Liberalization expands the set of feasible suppliers, allowing buyers to diversify risk across countries. At the same time, lower trade barriers shift sourcing toward the most efficient supplier, productive hubs that serve many buyers, such as Taiwan in semiconductor manufacturing. This concentration is part of the gains from trade, efficient suppliers scale up, and buyers source at lower cost. But it can also raise aggregate risk. In general equilibrium, when a few nodes account for a large share of input–output linkages, they become central in the production network, and granular shocks generate larger aggregate fluctuations (Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi, 2012). Liberalization therefore operates through two opposing forces. The net effect on aggregate risk is therefore a quantitative question that depends on how firms value efficiency relative to risk. I answer this question with a multi-country model in which firms choose suppliers under uncertainty about productivity shocks.

I study this question in three steps. First, with firm-level data, I provide empirical evidence on the tension producers face between expected costs and diversification. Next, I extend the analysis to industry input–output data and estimate the set of moments that discipline the model’s key elasticities governing the response to changes in risk cost and risk. Finally, I embed these elasticities in a multi-country model with endogenous network formation to quantify how trade liberalization affects aggregate risk. The answer is that, over the observed liberalization in the data, the mean variance of consumption falls by 36 percent and it falls in 20 of the 24 countries. Concentration dominates only beyond observed trade barriers, on the path to free trade.

The first contribution of the paper is empirical. I measure how sales shares of public firms move in response to (i) changes in expected costs, proxied by tariffs, and (ii) changes in perceived risk. Using a firm-level panel from Compustat Global, I show that higher discriminatory tariffs on a given country–sector pair reduce the world market shares of firms in that country–sector. In addition, increases in perceived country risk reduce those shares in the affected country. Following (Fajgelbaum, Goldberg, Kennedy, Khandelwal, and Taglioni, 2024), I use discriminatory tariffs. I build them from the U.S. tariff actions 2018–2024 and partners’ retaliation at the six-digit product level, weighting each product level tariff by the country–sector’s destination sales. For perceived risk, I use the transcript-based country-risk measure of Hassan, Schreger, Schwedeler, and Tahoun (2023), built from

the frequency with which executives and analysts discuss each country in conjunction with risk language. For a firm that sells five percent of its output to a given market, a one percent tariff on that market costs 1.8 percent of its global share over four years. A one standard-difference increase in perceived risk reduces the nominal share by 10 percent at three years and the real share by 7 percent. Next, we extend the analysis to industry level data and show that a similar pattern applies at the aggregate level. Industry level regression serves as auxiliary regressions to discipline the quantitative model.

The second contribution is to extend [Kopytov, Mishra, Nimark, and Taschereau-Dumouchel \(2024\)](#) to an open economy with heterogeneous trade costs and incomplete markets. The key equilibrium object is a risk-adjusted price: the sum of the expected production cost of the supplier, a trade cost, and an insurance premium that increases with the covariance between the supplier's price and the buyer's consumption. Firms face uncertainty about global, country-specific, industry-specific, and country-industry pair (granular) productivity shocks. Before these shocks are realized, they choose suppliers to minimize risk-adjusted costs.

The key theoretical contribution is the structure of the financial market. Households can pledge a fraction of their financial income to a risk neutral world investor as in ([Biais, Hombert, and Weill, 2021](#)). Compared to [Kopytov et al. \(2024\)](#), incomplete markets eliminate the common pricing kernel. As households can pledge only a part of their financial income, the kernel that firms use to choose their suppliers is a combination of domestic and global marginal marginal utility. The stronger the friction, the less the household can share risk and the more buyers price supplier risk against their own consumption baskets. As a result, the same supplier carries different insurance premia among buyers, and network formation responds through heterogeneous, buyer-specific incentives.

The main mechanism of the model works as follows. Trade distortions fragment the global production network into domestic clusters. Reducing these distortions has two effects. First, it allows firms to diversify their supply in more countries, as in [Caselli, Koren, Lisicky, and Tenreyro \(2020\)](#). Second, it facilitates the emergence of globally central suppliers: highly productive industries that become deeply interconnected because lower trade barriers make them accessible to many trading partners.

Diversification lowers exposure to granular and country-specific shocks, thus reducing aggregate risk. However, as the network reorganizes around central suppliers, shocks to those producers are strongly propagated through the Leontief inverse to their trading partners, increasing aggregate risk.

The third contribution is quantitative. I estimate the model in 24 countries and 23 industries from 1967 to 2024, using the long-run WIOD panel extended after 2000 with the ADB MRIO tables. To measure trade costs, I extend the inversion in [Bonadio, Huo, Levchenko, and Pandalai-Nayar \(2025\)](#) to the case where expenditure shares embed insurance premia. I estimated a larger fall in trade costs for intermediate inputs compared to the one delivered by a risk-blind inversion. Quantitatively, the liberalization observed in the data is stabilizing. Moving the cross-border wedges from their 1967 to their 2024 values reduces the mean consumption variance by 36 percent and reduces it in 20 of the 24 economies. Finally, I study the effect of financial integration. The variance of consumption is 2 percent above the financial volatility in the pledgeable fraction of the baseline.

Related Literature The closest antecedents are [Kopytov et al. \(2024\)](#) and [Huo, Levchenko, and Pandalai-Nayar \(2024\)](#). [Huo et al. \(2024\)](#) measure how shocks propagate through a *given* international network. In contrast, in this paper the network is the equilibrium object responding to changes in the distribution of productivity shocks.

[Kopytov et al. \(2024\)](#) study endogenous network formation under uncertainty in a closed economy with complete markets, where the equilibrium is efficient and all firms share the same pricing kernel. This paper changes the environment in at least three key ways. First, I introduce bilateral trade costs. Therefore, my model delivers a trade cost measure that extends the inversion of the price of traded goods based on shares in [Bonadio et al. \(2025\)](#) in an environment with uncertainty, showing that expenditure shares embed both trade costs and insurance premia. Second, I allow for incomplete markets, which breaks the common kernel and makes risk premia buyer-specific. The equilibrium is then generically constrained and inefficient ([Stiglitz, 1982](#); [Geanakoplos and Polemarchakis, 1986](#); [Greenwald and Stiglitz, 1986](#)). Third, I replace quadratic adjustment costs with an entropy (KL divergence) cost. This avoids corner solutions and makes the solution of the competitive equilibrium with incomplete markets tractable.

The main quantitative results are related most closely to [Caselli et al. \(2020\)](#), who decompose trade's effect on GDP volatility into a diversification and a specialization channel and find that diversification dominates for most countries. I add a third, general-equilibrium channel, network concentration, and quantify the three. Over observed barriers, concentration offsets part of the diversification. Beyond observed barriers, it reverses the effect of trade liberalization on the average variance of GDP. My results also relate to [Grossman, Helpman, and Lhuillier \(2023\)](#), who argue that a diversification subsidy

attains the constrained optimum

Two closely related papers are [Fan and Luo \(2025\)](#) and [Kleinman, Liu, and Redding \(2025\)](#). [Fan and Luo \(2025\)](#) study how the uncertainty of the tariff *policy* reshapes sourcing; I study *productivity* shocks that propagate through the production network, so riskiness is determined by a supplier's network position rather than its bilateral tariff exposure. [Kleinman et al. \(2025\)](#) study trade of final goods under uncertainty; while I study both final and intermediate goods, where the input-output links amplify shocks through the Leontief inverse. [Castro-Vincenzi, Khanna, Morales, and Pandalai-Nayar \(2024\)](#) quantify multi-regional sourcing under risk of disruption with a fixed set of suppliers.

The paper also links with the literature on how incomplete markets modify trade gains. [Fitzgerald \(2025\)](#) shows that for volatile developing economies, balanced-trade gains are only one-third of complete market gains because agents cannot smooth consumption across states ex post. Instead, I argue that with incomplete markets, firms buy less from the most productive suppliers due to risk considerations. [Allen and Atkin \(2022\)](#) document that incomplete insurance halves the gains from trade integration for Indian farmers by distorting ex-ante crop portfolio choices. [Adamopoulos and Leibovici \(2024\)](#) find a pecuniary externality in food trade under incomplete markets that justifies large import tariffs for food-importing nations.

The empirical analysis connects to work on time-varying volatility and production networks. [Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry \(2018\)](#) document that plant-level TFP dispersion roughly doubles during recessions; an analogous pattern is found in an international panel of 24 countries and 23 industries. The time variation in network concentration is related most closely to [Carvalho and Gabaix \(2013\)](#), who show that changing Domar weights account for the Great Moderation and its partial reversal. In terms of comovement, [Kose and Yi \(2001\)](#) show that trade in intermediates raises bilateral GDP comovement beyond what final-goods trade implies; [Johnson \(2014\)](#) finds that the input trade synchronizes the gross output more than the real value added. [vom Lehn and Winberry \(2022\)](#) provide complementary evidence that the U.S. investment network has changed over time with consequences for business cycle comovement.

The paper also relates to several broader literatures: on endogenous production networks ([Arkolakis, Huneus, and Miyauchi, 2025](#); [Acemoglu and Azar, 2020](#); [Elliott, Golub, and Leduc, 2022](#); [Taschereau-Dumouchel, 2025](#); [Acemoglu and Tahbaz-Salehi, 2025](#); [Bimpikis, Candogan, and Ehsani, 2019](#); [Jiang, Rigobon, and Rigobon, 2022](#)), on aggregation and distortions in networks ([Acemoglu et al., 2012](#); [Jones, 2011](#); [Baqae and Farhi, 2019](#);

Liu, 2019; Bigio and La’O, 2020; Baqaee and Farhi, 2024; Grossman, Helpman, and Sabal, 2024), on trade and aggregate volatility (di Giovanni, Levchenko, and Méjean, 2018), on tail risk in production networks (Dew-Becker, 2023; Dew-Becker, Tahbaz-Salehi, and Vedolin, 2021), on the estimation of trade cost (Eaton and Kortum (2002); Waugh (2010)), and on the granular origins of aggregate fluctuations (Gabaix, 2011; di Giovanni, Levchenko, and Méjean, 2014). In incomplete international asset markets, exogenous (Baxter and Crucini, 1995; Kollmann, 1996; (Heathcote and Perri, 2002); Corsetti, Dedola, and Leduc, 2008) and limited in enforcement (Alvarez and Jermann, 2000; Kehoe and Perri, 2002; Biais et al., 2021). The literature on incomplete contracts (Antràs and Helpman, 2004; Antràs and Chor, 2013; Boehm and Oberfield, 2020; Macchiavello and Morjaria, 2015; Macchiavello, 2022) shows that the buyer–supplier relationships already share part of the risk that the premium prices.

Outline. Section 2 shows the trade-off in firm and industry data. Sections 3 to 5 build the model and state the mechanism. Section 6 measures it, and Section 7 reports its fit. Section 8 evaluates the effect of trade liberalization and Section 9 concludes.

2 Empirical Analysis

This section studies how firm market shares respond to changes in expected costs and country risk. Using firm-level data, I show that firms lose world market share when the tariffs they face or their country’s perceived risk rises. Then, I turn to sector-level data to motivate key model assumptions and to discipline the quantitative analysis.

2.1 Firms Respond to Risk and to Expected Cost

The firm panel merges Compustat North America and Compustat Global, yielding 40,704 listed firms over 1989–2024. The outcome variable is the firm’s world market share $s_{f,t}$, defined as firm sales divided by the total sales of all listed firms in the same sector and year. I focus on goods producers in 49 economies.

The perceived risk $h_{c,t-1}$ is measured based on the country transcript of Hassan et al. (2023). It captures how often foreign firms talk about a certain country and link it to words like “risk” or “uncertainty”. As shown by Hassan et al. (2023), this index increases around large country level shocks such as the earthquake in Japan in 2011. I averaged it over the

four quarters before the firm’s fiscal year and standardized it within the country, so one unit of $h_{c,t-1}$ is one standard deviation of the country’s own average.

The tariff $\tau_{cj,t}$ is the tariff faced by the country c and sector j , constructed as a sales-share-weighted average of the tariffs applied in the destination markets of the sector. To isolate plausibly exogenous variation, I instrument it with the change in discriminatory tariffs over the same window. These discriminatory tariffs come from the 2018–24 US trade actions and all partner retaliatory measures. They are observed at the six-digit product level and then mapped to sectors using pre-war export weights. For firm f in country c and sector j , I estimate the effect using local projections

$$\Delta_h \log s_{f,t} = \beta_h^\tau \Delta_h \log(1 + \tau_{cj,t}) + \beta_h^r h_{c,t-1} + \theta'_h \mathbf{w}_{cj,t} + \alpha_f + \delta_{j,t} + \varepsilon_{f,t+h}, \quad (1)$$

where $\Delta_h \log s_{f,t} \equiv \log s_{f,t+h} - \log s_{f,t-1}$ is the change in the logarithmic world market share of firm f from fiscal year $t - 1$ to $t + h$. $\Delta_h \log(1 + \tau_{cj,t})$ is the corresponding change in the applied tariff weighted by sales facing country c in the sector j . $\mathbf{w}_{cj,t}$ are the country’s investor “optimism” over the same four quarters, which captures expectations separately from perceived risk. In addition, the fixed effect of the firm α_f absorbs persistent firm-specific changes in market share. The sector-year fixed effect $\delta_{j,t}$ absorbs global shocks to sector j in year t . The perceived risk varies by country and year, so the specification does not include a country-year effect. See the details in Appendix A.

Figure 1 reports the two coefficients for the share of nominal and real expenditures (in sales). Panel (a) reports the effect of risk. β_h^r compares the same firm in years in which its country is perceived as more or less risky, relative to its sector in the world. Standard errors are clustered by country. An increase in the perceived risk of one-standard deviation in the country reduces the nominal share of its firms by 2.5% at impact and 15% at eight. Real shares do not move for two years and then fall 4.2% at four years and 10.9% at eight.

Panel (b) reports the effect of tariffs. A 1% increase in the sales-weighted tariff reduces the firm’s world share by 12% after one year and 53% after six. For a firm that sells 5% of its output to a market, a 1% tariff in that market costs 1.8% of the world share over four years.

Appendix A presents a set of robustness tests. The estimates are robust to the sample of countries, the control of GDP growth, country trends, the winsorization rule, and the country assignment of firms. The appendix reports lead tests, and re-does inference with a placebo that reassigns the risk paths across countries.

These estimates add the risk margin to the firm-level tariff evidence of the trade war

(Fajgelbaum et al., 2024). They also extend Hassan et al. (2023) by showing that perceived country risk reduces the world market share of the country’s firms.

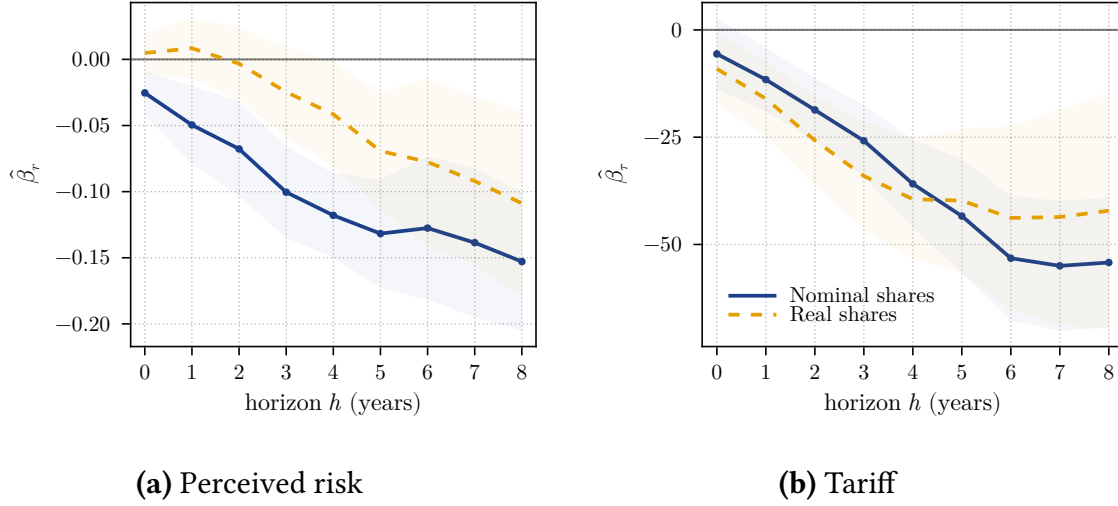


Figure 1: Country Risk and World Market Share

Notes: Equation (1); separate 2SLS regressions by horizon $h = 0, \dots, 8$. Panel (a): β_h^r , per within-country standard deviation of perceived risk. Panel (b): β_h^r , with $\Delta_h \log(1 + \tau_{c,j,t})$ instrumented by the discriminatory-tariff shock. Nominal shares solid, real shares dashed. Firm and sector-year fixed effects. Bands are 95 percent, SEs clustered by country with $t(G - 1)$ critical values.

2.2 Industry Level Analysis

Next, I study the aggregate effects of risk and tariffs using international input–output (IO) data. I combine long-WIOD (Timmer, Dietzenbacher, Los, Stehrer, and De Vries, 2015) with ADB MRIO (Asian Development Bank, 2024) to construct an annual panel of bilateral intermediate-input shares and real value added for 24 countries over 1967–2024.

Country disaster risk I begin by defining the risk of disasters at the country-level. I date disaster episodes from the Maddison historical record following Barro and Ursúa (2008). An episode is a peak-to-trough decline in logarithmic GDP per capita of at least 10% over 1870–2022. This yields 75 episodes in 23 of the 24 economies and country hazard rates between 1.3 and 3.3 percent. The Online Appendix reports additional supporting evidence of rare disasters using Q–Q plots.

Trade and risk elasticities Next, I test whether increases in perceived risk and tariff changes reallocate sourcing along global production links as follows:

$$\begin{aligned} \Delta_h \log m_{\hat{c}k,b,t} = & b^T \Delta_h \log(1 + \tau_{c,\hat{c}k,t}) + b^D e^{h_{\hat{c},t-1}} f d_{c,t-1}^{\hat{c}} D_{\hat{c},t-1} \\ & + b^N e^{h_{\hat{c},t-1}} f d_{c,t-1}^{\hat{c}} (1 - D_{\hat{c},t-1}) + \varphi_{\hat{c}k,i,t} + \varphi_{b,t} + \varphi_{\hat{c}k,c} + \mathbf{w}_{c,\hat{c},t-1} + u_{\hat{c}k,b,t} \end{aligned} \quad (2)$$

where $m_{\hat{c}k,b,t}$ is the share of the intermediate-input purchases of the buyer b sourced from the supplier country $\hat{c}$ in industry k . $\tau_{c,\hat{c}k,t}$ is the tariff rate that the country c applies to product k when it is imported from the country $\hat{c}$ at time t . It excludes the targeted tariff changes introduced in 2018–24. In addition, $h_{\hat{c},t-1}$ is perceived risk in the supplier country $\hat{c}$ (measured as standard deviations within the country). $D_{\hat{c},t-1}$ indicates that $\hat{c}$ is in a disaster episode; episodes are dated ex post in the full Maddison series. The control vector $\mathbf{w}_{c,\hat{c},t-1}$ is exposure $f d_{c,t-1}^{\hat{c}}$ interacted with both indicators. Finally, the specification includes fixed effects: $\varphi_{\hat{c}k,i,t}$ (supplier $\hat{c}k \times$ buying-industry $i \times$ year t), $\varphi_{b,t}$ (buyer $b \times$ year t), and $\varphi_{\hat{c}k,c}$ (supplier $\hat{c}k \times$ buyer country c). See Appendix B for details.

I estimate equation (2) using shock cohorts from 2005–2020 and track the results through 2021, for horizons $h = 0$ to 6. I drop 2020 as a signal year because the perceived risk increases by more than two standard deviations in all countries at the same time. Figure 2 shows the three profiles.

Panel (a) reports the elasticity of the trade. A 1% increase in the tariff on a route lowers the buyer’s input share from that route by 0.7% on impact and by 2.1% after six years. Panels (b) and (c) report the effect of perceived risk, separately for disaster years and non-disaster years. The risk term is scaled by the buyer country’s exposure to the supplier, so the coefficient can be interpreted for a given exposure level as follows. For a buyer in the 90th percentile of exposure, a one-standard-deviation increase in the supplier country’s perceived risk lowers the supplier’s share at six years by 17% when the supplier country is in an episode ($b^D = -7.61$) and by 1.7% when it is not ($b^N = -0.75$).

The tariff path is consistent with the short- and long-run trade elasticities of Boehm, Levchenko, and Pandalai-Nayar (2023). The risk coefficients extend Boehm, Flaaen, and Pandalai-Nayar (2019) and Carvalho, Nirei, Saito, and Tahbaz-Salehi (2021) by showing that buyers reallocate sourcing away from a supplier country when its perceived risk rises, before any disaster is realized.

Appendix B lists the data sources, defines the variables, and writes the regression and

its identification. It also presents a set of robustness tests. The appendix also includes descriptive evidence on changes in the global production network during 1967–2024. The aggregate risk varies over time and carries a global component that increases in years of crisis, and the production network changed in its concentration and critical locations.

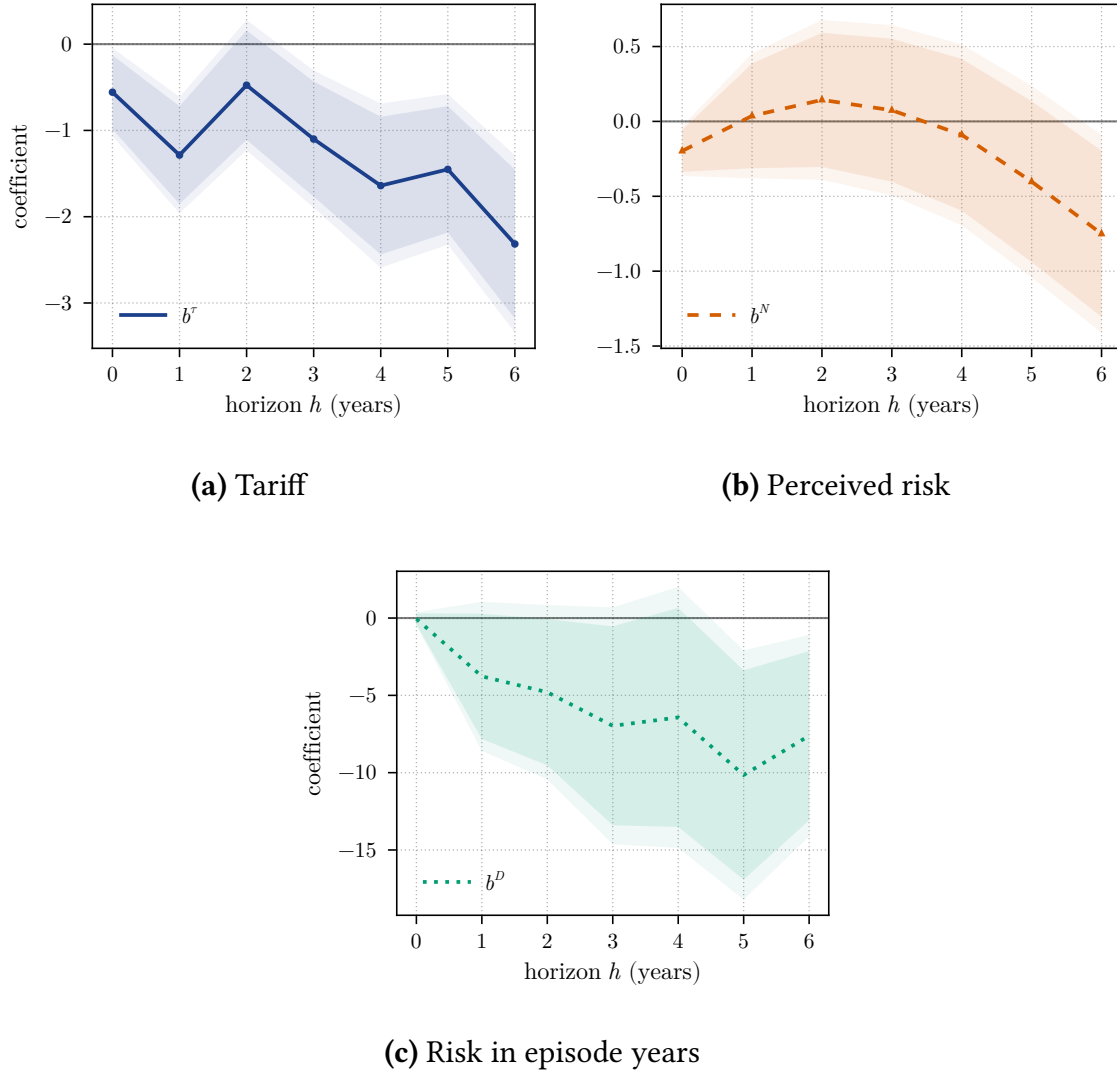


Figure 2: industry Responses by Horizon

Notes: The tariff coefficient b^τ , the non-episode risk coefficient b^N and the episode risk coefficient b^D of equation (2) at horizons $h = 0, \dots, 6$, full bilateral panel, shock cohorts 2005–2020 with outcomes through 2021 (the 2020 signal year excluded), standard errors clustered by route; each panel on its own scale. Shaded bands are 90 and 95 percent.

Risk sharing. Before closing the section, I tested the degree of risk sharing among the 24 economies in my sample. International risk sharing is limited. The test of [Asdrubali, Sørensen, and Yosha \(1996\)](#), absorption growth in income growth relative to the world, gives a slope of 1.01 (standard error 0.06) in the 24 countries over 1968–2024, against 0.42 under complete markets among households ([Appendix B.3](#)).

2.3 Disciplining the Model

When the perceived risk to the country increases, exporters from that country lose a world market share. This pattern points to a model in which a higher perceived risk makes a supplier less attractive. Therefore, I model sourcing as a choice over suppliers that trades off expected input costs against a risk premium.

Timing and target moments. In the data, perceived country risk outside disaster episodes has no contemporaneous effect on bilateral purchasing shares, and its impact builds over longer horizons. Similarly, consistent with the near-zero short-term trade elasticity in [Boehm et al. \(2023\)](#), the effect of tariffs strengthens with the horizon. Together, these patterns motivate modeling sourcing as an ex-ante decision. In addition, estimated elasticities are the moments that the model targets to identify its key parameters.

Financial markets and country risk. International risk sharing is limited, so the markets are incomplete in the model. I allow households to have limited access to an international market to ensure part of their financial income. I also include rare disaster shocks and identify them from peak-to-trough collapses in country-level GDP per capita. Finally, I introduce incomplete information about the risk of disaster shocks and introduce public signals about the probability of disaster shocks disciplined using the information provided by [Hassan et al. \(2023\)](#).

3 Environment

There are C countries and J global industries. In each country–industry pair there is one representative intermediate-goods producer, so there are $N = C \times J$ differentiated intermediate inputs. Each country also has a representative final-goods producer that

aggregates the N intermediates into a final good consumed by a representative household. Households supply labor L_c to domestic industries, consume, and own domestic firms.

Information and timing. Firms choose the production network with incomplete information. Before production, they observe C public signals about the probability of country-level disaster shocks and the implied conditional distribution of productivity (mean θ and covariance $\Sigma_{\epsilon,t}$). Then they choose production techniques to maximize the expected profits. After shocks realize, markets clear and firms and households choose quantities. There are no intertemporal choices, and periods are linked only through beliefs.

3.1 Households

Preferences. The representative household in the country c derives its utility from consumption and leisure. Preferences have the following form:

$$W_c = \sum_t \frac{\left(Z_{c,t} - \sum_i^J \frac{(L_{ci,t})^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}} \right)^{1-\gamma}}{1-\gamma} \quad (3)$$

Following [Greenwood, Hercowitz, and Huffman \(1988\)](#), preferences are quasi-linear in the labor aggregate, so labor supply does not have an income effect. The parameter γ is the curvature of utility in the consumption composite. It is the elasticity that governs how much a buyer pays to avoid a volatile supplier. The parameter ψ is the Frisch elasticity of the labor supply. Households choose consumption after the uncertainty is realized. Thus, in each period and for each state of the world, the budget constraint is:

$$P_{c,t} Z_{c,t} = \sum_i W_{ci,t} L_{ci,t} \quad (4)$$

where $P_{c,t}$ is the price index of the final good and $W_{ci,t}$ is the nominal wage in industry (c, i) . Trade is balanced, and profits are zero in equilibrium. The optimality condition for labor in each industry is given by the following expression.

$$L_{ci,t} = \left(\frac{W_{ci,t}}{P_{c,t}} \right)^\psi \quad (5)$$

It gives the standard optimal condition that relates the supply of labor to the real wage.

Financial market. Households can only insure their financial income. Before shocks are realized, the household c pledges a fraction $\vartheta \in [0, 1]$ of the income of the firms in the country to a risk-neutral financial intermediary on the exchange for an asset of the state continent, as in [Biais et al. \(2021\)](#). The intermediary must balance its budget in all states.

The extremes correspond to the benchmarks. $\vartheta = 0$ delivers financial autarky as in [Heathcote and Perri \(2002\)](#). Also note that the complete-market economy of [Kopytov et al. \(2024\)](#) requires that $\vartheta = 1$ and labor income be insurable.

3.2 Intermediate Goods Producers

First Stage. Before productivity shocks are realized, the representative firm in the country c , the industry i chooses a production technique $\eta_{ci,t}$. The firm operates a Cobb-Douglas technology:

$$F_{ci,t}(\eta_{ci,t}) = e^{\epsilon_{ci,t}} \left(\frac{L_{ci,t}}{\mu_{ci,t}} \right)^{\mu_{ci,t}} \left(\frac{e^{A_{ci,t}(\eta_{ci,t})}}{1 - \mu_{ci,t}} \prod_{\hat{c},k} \left(\frac{X_{ci,\hat{c}k,t}}{\eta_{ci,\hat{c}k,t}} \right)^{\eta_{ci,\hat{c}k,t}} \right)^{1-\mu_{ci,t}} \quad (6)$$

where $X_{ci,\hat{c}k,t}$ is the intermediate input from industry k in country $\hat{c}$, $L_{ci,t}$ is labor, and $\epsilon_{ci,t}$ is an exogenous productivity shock. The parameter $\mu_{ci,t} \in (0, 1)$ is the Cobb-Douglas labor share, which varies over time. The vector $\eta_{ci,t} = \{\eta_{ci,\hat{c}k,t}\}$ specifies the production technique, limited to the unit simplex $\sum_{\hat{c},k} \eta_{ci,\hat{c}k,t} = 1$. The normalization of each input by its Cobb-Douglas coefficient follows [Acemoglu and Azar \(2020\)](#), and ensures that the technique $\eta_{ci,t}$ affects productivity only through $A_{ci,t}(\eta_{ci,t})$, the productivity of the intermediate bundle.

Deviating from the ideal technique $\eta_{ci,t}^0$ reduces productivity. The ideal technique is a property of the industry, not of the country in which it is located. In particular, I assume $\eta_{ci,\hat{c}k,t}^0 = \eta_{i,\hat{c}k,t}^0$ for every period, so the industry blueprint i lists the same technology wherever it operates and changes over time.

$$A_{ci,t}(\eta_{ci,t}) = -\kappa^I \sum_{\hat{c},k} \eta_{ci,\hat{c}k,t} \log \frac{\eta_{ci,\hat{c}k,t}}{\eta_{ci,\hat{c}k,t}^0} \quad (7)$$

The adjustment cost $\kappa^I > 0$ governs the cost of deviating from the ideal source. The entropy function keeps all shares strictly positive for any finite κ , avoiding corner solutions. In the Online Appendix I show an example of how the optimal network at this cost compares to the quadratic cost in [Acemoglu and Azar \(2020\)](#) and describe how this cost helps to

make the solution of the competitive equilibrium more tractable.

Trade is subject to iceberg costs. The price of an intermediate good $(\hat{c}, k)$ paid by buyers in country c is $P_{c,\hat{c}k,t} = P_{\hat{c},\hat{c}k,t} \tau_{c,\hat{c}k,t}^I$.

Second Stage. In the second stage, given the production function, firms choose their demand for each input after observing the prices. The first-order conditions yield the standard unit cost function:

$$K_{ci,t}(\eta_{ci}, P) = e^{-\epsilon_{ci,t}} W_{ci,t}^{\mu_{ci,t}} \left(e^{-A_{ci,t}(\eta_{ci,t})} \prod_{\hat{c},k} P_{c,\hat{c}k,t}^{\eta_{ci,\hat{c}k,t}} \right)^{1-\mu_{ci,t}} \quad (8)$$

Because the technology exhibits constant returns to scale and markets are competitive, the technique choice is therefore a cost-minimization problem, and the first-stage problem is:

$$\eta_{ci,t}^* \in \arg \min_{\eta_{ci,t} \in \mathcal{A}} \mathbb{E}[\Lambda_{c,t} Y_{ci,t} K_{ci,t}(\eta_{ci}, P)] \quad (9)$$

where $\Lambda_{c,t}$ is the pricing kernel of the household.

3.3 Final Goods Producers

Each country has a representative final goods producer that combines intermediate goods into a consumption bundle. The production function is the following.

$$F_{c,t}(\alpha_{c,t}) = e^{A_{c,t}(\alpha_{c,t})} \prod_{\hat{c},k} \left(\frac{X_{c,\hat{c}k,t}}{\alpha_{c,\hat{c}k,t}} \right)^{\alpha_{c,\hat{c}k,t}} \quad (10)$$

where $\alpha_{c,t}$ lies in the unit simplex and deviating from the ideal technique $\alpha_{c,t}^0$ reduces productivity as in equation (7), with adjustment cost κ^F . The final goods firms choose $\alpha_{c,t}$ before the shocks are realized to maximize expected profits. The trade costs for the final goods are $\tau_{c,\hat{c}k,t}^F$.

3.4 Stochastic Structure

Uncertainty comes from TFP shocks. The productivity of each industry decomposes into global, country, industry, and granular components:

$$\epsilon_{ci,t} = g_t + \chi_{c,t} + \zeta_{i,t} + u_{ci,t} \quad (11)$$

Each component follows an independent GARCH(1) process with persistence ρ_k , a constant a mean $\theta_{k,ci}$, and variances $\sigma_{k,ci,t}$ vary over time.

Disaster risk. In addition to the Gaussian components, productivity is exposed to rare disasters such as wars, deep crises, and pandemics. Each country carries its own disaster state: in every period, with probability $\bar{\pi}_{c,t}$, country c 's productivity falls by $\xi_{c,t} \sim \mathcal{N}(d_c, \zeta_c^2)$, independently across countries and over time. Firms face incomplete information on disaster risk.

Signals and beliefs. Let $S_{c,t} \in \{0, 1\}$ denote the country c 's latent hazard regime (with $S_{c,t} = 1$ the high-hazard regime), with previous $\Pr(S_{c,t} = 1) = \bar{q}_c \in (0, 1)$. Conditional on the regime, the country faces a disaster indicator $D_{c,t} \in \{0, 1\}$ with

$$\Pr(D_{c,t} = 1 \mid S_{c,t} = s) = \pi_c^s, \quad \pi_c^0 < \pi_c^1.$$

The same latent regime also shifts the distribution of the public signal observed at the end of $t - 1$:

$$h_{c,t-1} \mid S_{c,t} = s \sim \mathcal{N}(m_s, v^2), \quad m_0 + m_1 = 0,$$

independently across countries. Firms observe $h_{c,t-1}$ and use it to form the posterior regime probability $q_{c,t} \equiv \Pr(S_{c,t} = 1 \mid h_{c,t-1})$ and, therefore, the posterior disaster probability $\pi_{c,t}$.

Lemma 1 (Beliefs). *The posterior disaster probability $\pi_{c,t} \equiv \Pr(D_{c,t} = 1 \mid \{h_{c',t-1}\}_{c'})$ depends only on $h_{c,t-1}$ and satisfies*

$$\pi_{c,t} = \pi_c^0 + (\pi_c^1 - \pi_c^0) q_{c,t}, \quad \text{logit } q_{c,t} = \text{logit } \bar{q}_c + \lambda h_{c,t-1}, \quad \lambda \equiv \frac{m_1 - m_0}{v^2},$$

where $q_{c,t}$ is the posterior probability of the high regime. Depending on the signals, the $D_{c,t}$ are independent Bernoulli draws with probabilities $\pi_{c,t}$.

Proof. See the Online Appendix. □

The normalization $m_0 + m_1 = 0$ centers the signal between its two regime-specific

means, so that $h = 0$ leaves beliefs at the prior. The posterior probabilities are log-linear in the signal, with slope λ . Intuitively, λ increases when the signal distributions under $S_{c,t} = 1$ and $S_{c,t} = 0$ are further apart relative to their noise variance. Because the signal is public, all firms (domestic and foreign, in every country) share the same posterior. The information structure therefore adds exactly one observable state variable per country, with no higher-order uncertainty, which is important to keep tractability in the choice of the production network.

3.5 Equilibrium

Now I define a competitive equilibrium. Since all firms are competitive, equilibrium prices equal costs. The equilibrium of the economy with the endogenous network is defined as follows.

Definition 1. (Competitive Equilibrium) An equilibrium is a sequence of allocations for intermediate firms $\{Y_{ci,t}, X_{ci,\hat{c}k,t}, \eta_{ci,t}\}_{i,c}$; a sequence of allocations for final good producers $\{Y_{c,t}, X_{c,\hat{c}k,t}, \alpha_{c,t}\}$; a sequence of allocations for households $\{Z_{c,t}, L_{ci,t}\}_c$; and a sequence of prices $\{P_{ci,t}, W_{ci,t}, P_{c,t}\}$ such that every agent solves its problem and goods and labor markets clear. Techniques $\{\eta_{ci,t}, \alpha_{c,t}\}$ are chosen before ϵ realizes, given $(\theta, \Sigma_{\epsilon,t}, \{\pi_{c,t}\}_c)$.

3.6 Definitions and Notation

Before studying the equilibrium, I define some important objects in the analysis.

Input-Output and Final Goods Matrices. The IO matrix η_t is the matrix $N \times N$ whose $(ci, \hat{c}k)$ th element is the Cobb-Douglas coefficient controlling the intensity at which industry ci uses inputs from industry $\hat{c}k$. The Final Goods matrix α_t stores the Cobb-Douglas coefficients in the final goods production function; rows for buyer industries within the same country share the same values, since there is one representative final good producer per country. Let μ_t be the diagonal $N \times N$ matrix of labor shares $\mu_{ci,t}$.

Domar Weights. The Domar weight of industry ci is its share of world GDP measured in sales:

$$\omega_{ci,t} = \frac{P_{ci,t} Y_{ci,t}}{GDP} \quad (12)$$

I also define the Domar weight of the country $\Omega_c = \sum_i \omega_{ci,t}$ and the Domar weight of the industry $\hat{\Omega}_i = \sum_c \omega_{ci,t}$.

4 Equilibrium Network Formation

I analyze the model backward. First, I characterize the prices given a network. Next, I use these results to characterize the optimal choice of network.

4.1 Equilibrium Given a Network

The aggregate state in the second stage is $s = \{\alpha^*, \eta^*, \Sigma, \tau\}$ and the stochastic state is ϵ . Throughout, I denote $\mathbf{x}$ as the vector that contains all values of logarithmic $X_{ci,t}$.

Domar Weights. The relative size of each industry in the global economy is summarized by its Domar weight as follows:

Lemma 2. *The Domar weight vector $\omega(s)$ is the unique solution to the linear system*

$$\omega(s) = [\alpha^{*\top} \mu + \eta^{*\top} (I - \mu)] \omega(s)$$

together with the normalization $\sum_{c,i} \mu_{ci} \omega_{ci}(s) = 1$.

Proof. See Appendix C.1. □

The term $\alpha^{*\top} \mu \omega$ accounts for revenue from sales of final goods: each country's final expenditure equals its labor income $\sum_i \mu_{ci} \omega_{ci}$. The term $\eta^{*\top} (I - \mu) \omega$ represents revenue from the sales of intermediate goods. Market shares increase either because other intermediate-good producers increase their use of the good or because final-good producers increase their expenditure shares.

The relative size of the final versus intermediate goods trade depends on the vector of labor share μ . Higher labor shares direct more production income to households, increasing

the role of the final goods network $\alpha^\star$ relative to the intermediate input network $\eta^\star$. Because profits are zero in equilibrium, the transfers of the financial intermediary are also zero, and with balanced trade no other payment moves purchasing power across countries. Given the Cobb-Douglas structure and the world-GDP numeraire, Domar weights do not change with stochastic shocks ϵ_t .

Prices. Equilibrium prices are determined by the network and the realization of productivity shocks.

Lemma 3. *Let $\mathbf{p}$ be the logarithmic price vector and ϵ be the vector of TFP shocks, with the nominal world GDP as the numeraire. Then:*

$$\mathbf{p}(s, \epsilon) = \mathcal{L}(s) (B(s) - \epsilon),$$

where the extended Leontief inverse is:

$$\mathcal{L}(s) = \left(I - \mu \frac{\psi}{1 + \psi} \alpha^\star - (I - \mu) \eta^\star \right)^{-1},$$

and the deterministic price component is:

$$\begin{aligned} B(s) = & - (I - \mu) A(\eta^\star) - \frac{\psi}{1 + \psi} \mu A(\alpha^\star) \\ & + \left[\left(\frac{\psi}{1 + \psi} \mu \alpha^\star \right) \circ \log \tau^F + ((I - \mu) \eta^\star) \circ \log \tau^I \right] \mathbf{1} + \frac{1}{1 + \psi} \mu (\log \mu + \log \omega(s)), \end{aligned}$$

where $A(\eta^\star)$ and $A(\alpha^\star)$ stack the adjustment costs of each buyer, $\circ$ is the entry product, and $\log \tau^F, \log \tau^I$ are the $N \times N$ matrices of the trade costs.

Proof. See Appendix C.2. □

Whereas the matrices $\alpha^\star$ and $\eta^\star$ record *direct* links between producers, the Leontief inverse records *direct and indirect* exposures throughout the network. This Leontief inverse generalizes the classical one by adding the term $\mu \frac{\psi}{1 + \psi} \alpha^\star$. This additional term comes from the household labor supply condition and captures how a TFP shock in any industry affects wages and, through wages, production costs in all other industries.¹ The elasticity

¹Huo et al. (2024) derive this extended Leontief inverse in a model with CES production.

parameter ψ determines the relative strength of the two transmission channels. When labor supply is inelastic, wages respond only weakly, so changes in intermediate input prices become the main transmission channel. By contrast, when labor supply is elastic, final-good prices have a larger impact on wages, making final-good trade the key transmission channel.

The vector $B(s)$ summarizes how the network tilts prices in the absence of shocks. The terms $A(\alpha_c^\star)$ and $A(\eta_{ci}^\star)$ capture how costly supplier reallocation reduces effective productivity and raises prices in industries that deviate from their ideal technology. The trade cost component makes prices higher in industries that source intensively from foreign inputs. The terms $\log \mu$ and $\log \omega(s)$ capture the fact that industries that use labor more intensively and industries with larger market shares must pay higher wages to compensate households, which supply labor with elasticity ψ .

The final good price index aggregates supplier prices as follows.

Corollary 1 (Final-good price index). *The logarithmic price index of the final product in country c is:*

$$p_c(s, \epsilon) = -A(\alpha_c^\star) + \alpha_c^\star (\mathbf{p}(s, \epsilon) + \log \tau_c^F),$$

where $\log \tau_c^F$ is the row of final-goods trade costs faced by the country c . A decrease in unit size in the productivity of the country industries c' increases it by

$$\ell^{c,c'}(s) = \sum_{\hat{c}, k} \alpha_{c, \hat{c}k}^\star \sum_i \mathcal{L}_{\hat{c}k, c'i}(s).$$

Proof. See Appendix C.9. □

This price index p_c and the elasticity $\ell^{c,c'}$ are the object that summarize each country's exposure, direct and indirect, to shocks anywhere in the global economy.

Expectations. The implied distribution of prices given the network is given by:

Lemma 4. $\mathbf{p}(s, \epsilon)$ is normally distributed conditional on the disaster indicators, and

$$E_t[\mathbf{p}(s, \epsilon)] = \mathcal{L}(s) \left(B(s) - \theta + \sum_{c'} \pi_{c',t} d_{c'} \mathbf{1}_{c'} \right),$$

$$\text{Var}_t[\mathbf{p}(s, \epsilon)] = \mathcal{L}(s) \left(\Sigma_\epsilon + \sum_{c'} (\pi_{c',t} \zeta_{c'}^2 + \pi_{c',t}(1 - \pi_{c',t}) d_{c'}^2) \mathbf{1}_{c'} \mathbf{1}_{c'}^\top \right) \mathcal{L}(s)^\top,$$

where $\mathbf{1}_{c'}$ is the vector that selects the industries of country c' .

Proof. See Appendix C.4. □

The Lemma 4 characterizes the mean and variance of the equilibrium input prices. Under Cobb–Douglas production, logarithmic prices are affine in the vector of productivity shocks ϵ through the Leontief inverse $\mathcal{L}(s)$.

The mean is the expected unit cost for each supplier. The expected productivity is the Gaussian mean θ minus the expected disaster loss in each source country c' ($\pi_{c',t} d_{c'}$), so a disaster increases the expected cost. This country-level component hits all industries in c' and then transmits to downstream industries through $\mathcal{L}(s)$.

The variance decomposes into Gaussian and disaster risk. The Gaussian risk is captured by Σ_ϵ . Disaster risk is common within a country and has two components: uncertainty about the size of the disaster $\pi_{c',t} \zeta_{c'}^2$ and uncertainty about the occurrence of the disaster $\pi_{c',t}(1 - \pi_{c',t}) d_{c'}^2$. The network amplifies and propagates both components.

Pricing Kernel. A firm's payoff has two owners. The household of its country holds the fraction $1 - \vartheta$ that is not pledged, and the financial intermediary holds the pledged fraction ϑ on behalf of the households of all countries. The kernel that prices the payoff combines the households' marginal utilities.

Lemma 5 (Pricing kernel). *The logarithmic marginal utility of the numeraire of the household of c is*

$$\log M_c(s, \epsilon) = (\gamma - 1) p_c(s, \epsilon) + C_c(s),$$

where $C_c(s) = -\gamma \log \left(\frac{1}{1+\psi} \sum_i \mu_{ci} \omega_{ci}(s) \right)$, and the kernel that prices the payoff of a firm

in country c is the mixture:

$$\Lambda_c(s, \epsilon) = (1 - \vartheta) M_c(s, \epsilon)/E_t[M_c(s, \epsilon)] + \vartheta \sum_o \xi_o M_o(s, \epsilon)/E_t[M_o(s, \epsilon)],$$

where $\xi_c \equiv \sum_i \mu_{ci} \omega_{ci}(s)$. Under the density measure $M_c(s, \epsilon)/E_t[M_c(s, \epsilon)]$, the risk-adjusted disaster probability of the country c' $\hat{\pi}_{c,c',t}$ is:

$$\frac{\hat{\pi}_{c,c',t}}{1 - \hat{\pi}_{c,c',t}} = \frac{\pi_{c',t}}{1 - \pi_{c',t}} \exp \left\{ (\gamma - 1) \ell^{c,c'} d_{c'} + \frac{1}{2} (\gamma - 1)^2 (\ell^{c,c'})^2 \varsigma_{c'}^2 \right\}.$$

Proof. See Appendix C.5. □

The Lemma 5 separates what shifts a household's marginal utility *level* from what drives its *state-by-state variation*. The term $C_c(s)$ is a deterministic shifter: it moves the marginal utility up or down with the deterministic allocation of the country, but does not load on the shocks. Because the endogenous labor supply enters directly only in the deterministic component, it does not change the insurance value of a supplier.

Incomplete markets enter through the pricing kernel Λ_c . The unpledged share of a firm is priced with the *local* marginal utility M_c , while the pledged share is priced with the *world* marginal utility, an income-weighted average. Here, ϑ controls the weight on local versus world pricing, and ξ_o are the intermediary's income weights.

Disaster risk matters because it shifts value to the tail states. For household c , the object that enters the valuation is not the physical probability $\pi_{c',t}$ of a disaster in the country c' , but the risk-adjusted probability $\hat{\pi}_{c,c',t}$ implied by reweighting the states with the marginal utility (normalized) of the household $M_c/E_t[M_c]$. This density places extra weight on states in which the cost of living of the household is high. A disaster in c' raises the price index of the household c by $\ell^{c,c'}$, so it increases the marginal logarithmic utility by $(\gamma - 1)\ell^{c,c'}$. The exponential term in the lemma is exactly this marginal-utility tilt (including uncertainty about disaster size) applied to the physical odds. When $\gamma > 1$, disasters in countries that increase the cost of living of the household c receive more weight: $\hat{\pi}_{c,c',t} > \pi_{c',t}$.

4.2 Optimal Production Networks

A buyer trades the expected cost for insurance. A supplier that is less expensive in expectation is attractive. Firms in both final and intermediate industries choose a supplier

vector to minimize expected production costs discounted by their kernel, and the problem of the first stage can be characterized as follows.

Proposition 1 (Risk-Adjusted Prices). *Let $m = \{\alpha_c, \eta_{ci}\}$ for the final and intermediate producers of goods, respectively. In equilibrium, their choice satisfies*

$$m^*(s) \in \arg \max_{m \in \Delta} -\kappa_b^m \sum_{\hat{c}, k} m_{\hat{c}k} (\log m_{\hat{c}k} - \log m_{\hat{c}k}^0) - \sum_{\hat{c}, k} m_{\hat{c}k} R_{b, \hat{c}k}(s),$$

where the risk-adjusted price of supplier $(\hat{c}, k)$ for buyer b in country c is

$$R_{b, \hat{c}k}(s) = E_t[p_{\hat{c}k}] + \log \tau_{b, \hat{c}k}^m + (1 - \vartheta) \Pi_{\hat{c}k}^c + \vartheta \sum_o \xi_o \Pi_{\hat{c}k}^o,$$

and, for every country o ,

$$\Pi_{\hat{c}k}^o = (\gamma - 1) \text{Cov}_t(p_o, p_{\hat{c}k}) + \sum_{c'} T_{o, c', t} \sum_i \mathcal{L}_{\hat{c}k, c' i},$$

where the contribution of the highest cumulants of the disaster of the country c' is

$$T_{o, c', t} = (\hat{\pi}_{o, c', t} - \pi_{c', t}) (d_{c'} + (\gamma - 1) \ell^{o, c'} \zeta_{c'}^2) - (\gamma - 1) \ell^{o, c'} \pi_{c', t} (1 - \pi_{c', t}) d_{c'}^2.$$

Proof. See Appendix C.6. □

The Proposition 1 nests the optimality condition of Kopytov et al. (2024). When all firm payoffs are priced by a single world kernel ($\alpha_c = \bar{\alpha}$ and $\vartheta = 1$), sourcing trades off expected costs against a single common covariance-based risk premium; there is no disaster risk ($\pi_{c', t} = 0$ for all c'); and trade is frictionless ($\log \tau_{b, \hat{c}k}^m = 0$).

The risk-adjusted price has four components. $E_t[p_{\hat{c}k}]$ is the expected unit cost of the supplier (including its expected disaster losses, directly and through upstream inputs). $\log \tau_{b, \hat{c}k}^m$ is the trade cost on the route. The remaining terms are risk premia: the supplier's payment is priced in part by the home household (weight $1 - \vartheta$) and in part by the pooled world investors (weight ϑ).

Each household-level premium has two pieces. The covariance term $(\gamma - 1) \text{Cov}_t(p_o, p_{\hat{c}k})$ captures that a supplier is risky for the household o if it is expensive precisely when the domestic price index p_o is high and provides insurance if it is cheap in those states.

The term $T_{o,c',t}$ captures the extra premium from the left-tail skewness of disasters, in proportion to the supplier's exposure to disaster states upstream. In addition, incomplete information changes the effect of the risk of a disaster. A higher public signal $h_{c',t-1}$ increases the posterior probability of the disaster $\pi_{c',t}$, which increases the expected losses of the disaster within $E_t[p_{\hat{c}k}]$ and, while the posterior stays below one half, the premium on its occurrence. These optimality conditions resemble the standard portfolio choice in the finance literature. Critically, as it is well studied in the finance literature, the additional cost of disaster shocks ($T_{o,c',t}$) helps the model to calculate the empirical response of firms to risk while keeping the risk aversion parameter under plausible values.

The pledged fraction ϑ indexes the structure of the financial market. At $\vartheta = 0$ (financial autarky as in [Heathcote and Perri \(2002\)](#)), each buyer prices suppliers with its own household kernel, so the same supplier can carry different premia for buyers in different countries because their consumption baskets differ. At $\vartheta = 1$, risk is discounted with a *common-world* kernel, equal to an income-weighted average of the kernels of the country's households. This eliminates cross-buyer dispersion in premia for a given supplier.

However, even at $\vartheta = 1$, markets remain incomplete because the risk of labor income is not insurable. Households still face a different risk of consumption and therefore different marginal utilities. Hence, the equilibrium generally differs from the complete allocation of the market planner in [Kopytov et al. \(2024\)](#), in which a *single* stochastic discount factor prices all benefits. Completing markets here would require additional state-contingent international transfers that fully insure labor income, doing so would make Domar weights state-dependent and would break the log-linearity of prices.

The equilibrium is instead generically inefficient similar to ([Stiglitz, 1982](#); [Greenwald and Stiglitz, 1986](#)). Each buyer prices a supplier's covariance with its own basket, and none prices the covariance its choice adds to the other buyers' baskets. Inefficiency due to incomplete markets falls on third parties, so a contract between a buyer and its supplier cannot remove it. The solution of the optimal network is:

Lemma 6 (Optimal Network). *Let $m \in \{I, F\}$, $\kappa_b^m > 0$ and $m^0 > 0$. In equilibrium, the network is the softmax:*

$$m_{\hat{c}k}^*(s) = \frac{m_{\hat{c}k}^0 \exp(-R_{b,\hat{c}k}(s)/\kappa_b^m)}{\sum_{(\hat{c}',k')} m_{\hat{c}'k'}^0 \exp(-R_{b,\hat{c}'k'}(s)/\kappa_b^m)},$$

where κ_b^m is the adjustment cost (κ_i^I for intermediates, κ^F for final goods).

Proof. See Appendix C.7. □

The equilibrium network is a softmax function of risk-adjusted prices, with κ_b^m acting as the temperature: large κ_b^m keeps the shares close to the ideal m^0 , while small κ_b^m concentrates the shares on the cheapest supplier. The equilibrium solves a fixed point: risk-adjusted prices $R_{b,\hat{c}k}$ depend on optimal networks, and optimal networks depend on risk-adjusted prices. Under the entropy cost, the expenditure shares are positive as long as $m^0 > 0$.

4.3 Existence and Uniqueness

The competitive equilibrium is a fixed point that links the aggregate network to firms' best responses: each firm's sourcing decision alters the covariance structure of prices, which then feeds back into the optimal choices of all other firms. This general-equilibrium feedback can, in principle, generate multiple equilibria. Because the competitive equilibrium does not admit a planner representation in incomplete markets, the approach of Kopytov et al. (2024), who establishes uniqueness via a planning problem, is not available here. Instead, I write the competitive equilibrium as the fixed point of a consistency map: each producer chooses its technique against the prices of an aggregate network it takes as given, and in equilibrium the aggregate of those choices is that network. Existence and a sufficient condition for uniqueness follow from that map.

Proposition 2 (Existence and Uniqueness). *Let the disaster hazards be zero, $\kappa^m > 0$ for $m \in \{I, F\}$, $\tau \leq \bar{\tau} < \infty$, and let $\omega(s) \geq \underline{\omega} > 0$ on it. Then a competitive equilibrium $(s^*, p(s^*, \epsilon))$ exists and s^* is unique if*

$$\bar{\kappa} > 2(|\gamma - 1| L_{pp}(\vartheta) + L_p),$$

where $\bar{\kappa} = \max_m \kappa_b^m$, L_p and L_{pp} are the Lipschitz constant of expected prices and risk, respectively, in the norm of the largest change of any buyer's share vector.

Proof. See the Online Appendix. □

The explicit expressions for L_p and $L_{pp}(\vartheta)$ in the Online Appendix are upper bounds

on these constants. Existence follows from Brouwer's theorem: prices are uniquely pinned down by the Leontief inverse for any given network, the entropy adjustment cost makes each producer's best response a smooth function of risk-adjusted prices, and the consistency map sends the product of simplices into a compact convex set on which prices and Domar weights are bounded. So the map is continuous on a compact convex set and must have a fixed point.

Uniqueness requires controlling general-equilibrium feedback. The right-hand side of the condition is an upper bound on how sensitive risk-adjusted prices can be to changes in the aggregate network. This sensitivity decreases when agents are more risk averse (γ is higher) and shocks are more volatile ($\tilde{\sigma}^2$ is higher).

If adjustment costs $\bar{\kappa}$ exceed this bound, the firms' sourcing responses are sufficiently muted that the best-response operator is a contraction, implying a unique equilibrium. Intuitively, high adjustment costs keep firms close to the ideal technology m^0 , dampening the response of firms to prices. The bound is the threshold at which this "anchoring" is strong enough to guaranty contraction.

When $\bar{\kappa}$ is below the bound, the best-response map need not be a contraction because expected prices can generate positive feedback. If a buyer re-sources toward cheaper inputs, its marginal cost falls, making it a cheaper supplier for its customers; this lowers their expected input prices and encourages further re-sourcing. Risk provides the opposing force: more concentrated sourcing increases exposure and the associated risk premium, creating negative feedback.

4.4 Welfare and Production

Next, I describe the effect of changes in the network $\{\alpha^\star, \eta^\star\}$ on macroeconomic outcomes.

Production. The value added in each country-industry pair as a function of the network and the realization of productivity shocks is the following.

Lemma 7. *Let a_{ci} be the log of VA, where $VA = e^{\epsilon_{ci} + A(\eta_{ci}^\star)} L_{ci}^{\mu_{ci}}$. Then:*

$$a_{ci}(s, \epsilon) = \epsilon_{ci} + A(\eta_{ci}^\star) + \mu_{ci} l_{ci}(s, \epsilon)$$

where logarithmic labor supply $l_{ci}(s, \epsilon)$ is:

$$l_{ci}(s, \epsilon) = \frac{\psi}{1 + \psi} \left[-p_c(s, \epsilon) + \log \mu_{ci} + \log \omega_{ci}(s) \right]$$

Proof. See Appendix C.3. □

The Lemma 7 decomposes a_{ci} into three components. The first, ϵ_{ci} , is the TFP shock. The second, $A(\eta_{ci}^*)$, is the endogenous productivity determined by the choice of network. The third is the endogenous response to the labor supply. A TFP shock anywhere in the network propagates into the domestic price index p_c , changing the real wage and changing the labor supply and value added in all country-industry pairs. Huo et al. (2024) emphasize this price-index channel as a source of international transmission.²

Welfare. The welfare of the household in the country c in terms of the equilibrium network, the expected price index, and the price variance is as follows.

Lemma 8. *Let $\mathcal{W}_c(s)$ be the log certainty equivalent of the household's consumption. Then*

$$\begin{aligned} \mathcal{W}_c(s) = & \log \left(\frac{1}{1 + \psi} \sum_i \mu_{ci} \omega_{ci}(s) \right) - \mathbb{E}_t[p_c(s, \epsilon)] + \frac{1}{2}(1 - \gamma) \text{Var}_t(p_c(s, \epsilon)) \\ & - \frac{1}{\gamma - 1} \sum_{c'} G_{c,c',t}, \end{aligned}$$

where $G_{c,c',t}$ is the contribution of the disaster of country c' to the higher order moments of p_c ,

$$G_{c,c',t} = \log \frac{1 - \pi_{c',t}}{1 - \hat{\pi}_{c,c',t}} - (\gamma - 1) \ell^{c,c'} \pi_{c',t} d_{c'} - \frac{1}{2}(\gamma - 1)^2 (\ell^{c,c'})^2 (\pi_{c',t} \zeta_{c'}^2 + \pi_{c',t} (1 - \pi_{c',t}) d_{c'}^2),$$

Proof. See Appendix C.8. □

Welfare aggregates four forces that operate through the network. The first term captures the relative size of the domestic economy: households in larger economies face

²The Lemma 7 is closely related to their result, but departs from it because they characterize deviations from the steady state in a CES model.

higher labor demand, work more, and consume more. The second term, $\mathbb{E}[p_c]$, measures the cost of the consumption basket: lower expected prices increase purchasing power and increase consumption. The third term captures country-specific risk. With incomplete markets, $\text{Var}_t(p_c)$ measures the volatility of consumption and reduces welfare when $\gamma > 1$. The last term is the cost of the risk of disaster beyond its mean and variance. A disaster only lowers productivity, so the consumption of the household is skewed towards large losses, and $G_{c,c',t}$ prices that skewness and every higher moment. $G_{c,c',t}$ is zero when $\pi_{c',t} = 0$ or when the household is not exposed to c' .

5 Network and Aggregate Risk

Trade liberalization can reduce or increase aggregate risk. When a buyer starts importing, it replaces one supplier with two and diversifies granular (or granular) shocks, but it also changes the transmission of shock in the global economy by changing the Leontief inverse $\mathcal{L}(s)$. This section illustrates this mechanism in the model. I first connect the concentration force to network centrality and then illustrate the trade-off in a two-country example.

5.1 Centrality

I first define centrality, the object through which every common shock reaches a price.

Definition 2 (Centrality). The centrality of the country-industry pair (c, i) is the sum of the rows of the Leontief inverse:

$$v_{ci} = \sum_{\hat{c}k} \mathcal{L}_{ci,\hat{c}k}$$

The exposure of the country c to central industries is:

$$v_c = \sum_{\hat{c}k} \alpha_{c,\hat{c}k}^* v_{\hat{c}k}$$

The logarithmic price of a node is a Leontief-weighted sum of upstream shocks, so its exposure to a given shock is the relevant row sum of the Leontief inverse over the set of nodes hit by that shock as in [Acemoglu et al. \(2012\)](#).

With independent global, country, industry, and granular components, the price vari-

ance decomposes into four corresponding terms.

$$\text{Var}(p_{ci}) = \underbrace{\sigma_g^2 v_{ci}^2}_{\text{global}} + \underbrace{\sum_{c'} \sigma_{\chi, c'}^2 (\ell_{ci}^{c'})^2}_{\text{country}} + \underbrace{\sum_j \sigma_{\zeta, j}^2 (\ell_{ci}^j)^2}_{\text{industry}} + \underbrace{\sum_{c'k} \sigma_{u, c'k}^2 \mathcal{L}_{ci, c'k}^2}_{\text{granular}} \quad (13)$$

Here, centrality is the loading on the common shocks. Global shock loads through the full row sum (v_{ci}), country and industry shocks through block row sums ($\ell_{ci}^{c'}$ and ℓ_{ci}^j), and granular shocks through individual entries $\mathcal{L}_{ci, c'k}$. With disasters, the country-shock block takes the additional term of the disaster variance proportional to $(\ell_{ci}^{c'})^2$.

Equation 13 highlights how concentration increases variance. Each contribution to $\text{Var}(p_{ci})$ is a squared load. Concentrating sourcing makes the relevant loadings more uneven, and therefore raises these squared terms. The effect is strongest for common shocks (global, country, and industry) because their loadings are row sums of $\mathcal{L}$, which are mechanically larger than individual entries. Spreading purchases across suppliers flattens the sum of quadratic terms and lowers the variance. Also, because global shock loads with the concentration index (v_{ci}^2) lower concentration reduce variance.

5.2 Trade and Aggregate Risk: An example

To see how liberalization can change aggregate risk, consider two countries, Home (H) and Foreign (F), each with an intermediate producer. Foreign is both more productive ($\theta_F > \theta_H$) and more intermediate-intensive ($\mu_H > \mu_F$), so it is an “star” supplier.

To isolate intermediate-trade forces, there is no trade in final-goods ($\alpha_{H,H} = \alpha_{F,F} = 1$ due to the high trade cost). Home chooses its intermediate sourcing shares $\eta_H = \{\eta_{H,H}, \eta_{H,F}\}$ subject to iceberg costs τ^{HF} ; Foreign’s trade cost is kept fixed. The shocks are $\epsilon_c = g + u_c$, with g, u_H, u_F independent and $\text{Var}(u_H) = \text{Var}(u_F) = \sigma_u^2$. In this economy, Home’s price variance splits into a global term and a granular term:

$$\text{Var}(p_H) = \underbrace{\sigma_g^2 v_H^2}_{\text{Global}} + \underbrace{\sigma_u^2 (\mathcal{L}_{HH}^2 + \mathcal{L}_{HF}^2)}_{\text{Granular}} \quad (14)$$

Global shock loads through Home’s centrality exposure $v_H \equiv \mathcal{L}_{HH} + \mathcal{L}_{HF}$, while granular risk loads through individual Leontief entries $\mathcal{L}_{HH}^2$ and $\mathcal{L}_{HF}^2$.

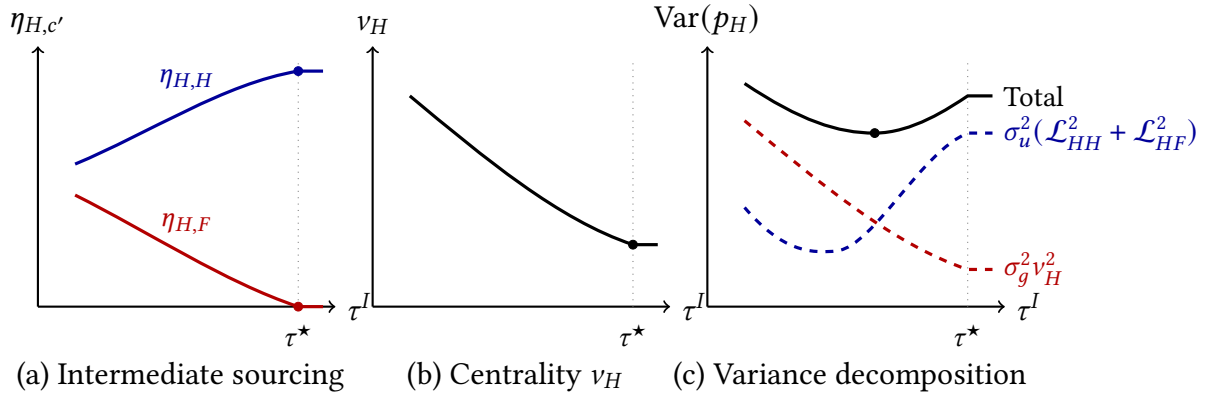


Figure 3: Liberalization in the Two-Country Example

Notes: Two-country economy with intermediate trade only ($\alpha_{H,H} = \alpha_{F,F} = 1$), $\theta_F > \theta_H$, $\mu_H > \mu_F$ and $\sigma_{u_c}^2 = \sigma_u^2$ in both countries. τ^* marks the trade cost at which Home's foreign sourcing becomes negligible; shares are interior for every finite τ^I and the curves are schematic.

For $\tau^{HF} \geq \tau^*$, trade costs are prohibitive, so Home's foreign sourcing is negligible, $\mathcal{L}_{HF} = 0$ and Home's price depends almost entirely on its domestic chain, with $\mathcal{L}_{HH} = 1/\mu_H$ and $v_H = 1/\mu_H$. In this region, Home is essentially exposed only to its own granular shock, amplified by its domestic transmission of shocks.

Once τ^{HF} drops below τ^* , Home begins to source from Foreign suppliers. Along the path toward free trade, both Leontief entries ($\mathcal{L}_{HH}$, $\mathcal{L}_{HF}$) move, and the variance decomposition features two opposing mechanisms. The granular component is a sum of squares $\mathcal{L}_{HH}^2 + \mathcal{L}_{HF}^2$, so it falls when import improves *diversification*. That is, when Home spreads exposure more evenly across domestic and foreign granular shocks. But this force is non-monotone, once sourcing becomes tilted toward Foreign, further liberalization reduces diversification, and the sum of squares starts to rise.

By contrast, the global component is the square of a sum, $\sigma_g^2 v_H^2$, and rises monotonically as Home concentrates on the most productive partner. When Foreign has the lower labor share (hence the longer chain), Home's centrality v_H increases as it reallocates spending abroad. Figure 3 summarizes these dynamics: as τ^{HF} falls from τ^* toward free trade, diversification initially lowers volatility, but increasing centrality eventually dominates and increases exposure to the global shock.

6 Quantitative Analysis

This section describes the calibration of the key elasticities γ , κ , and λ , using indirect inference. In addition, I show how I measure trade costs and technological change, as well as how I discipline the distribution of productivity shocks.

Data are input-output tables for 24 countries and 23 industries from 1967 to 2024, as in Section 2. The labor shares $\mu_{ci,t}$ are taken from the data period by period. The Appendix D.4 collects the details of each step in the following. In addition, ψ and ϑ are externally calibrated.

6.1 The Network Response: $(\kappa, \gamma, \lambda)$

I estimate the adjustment cost κ , risk aversion γ , and signal discriminability λ by indirect inference, using the equation (2) as auxiliary regression. The regression summarizes, in three coefficients, how bilateral sourcing shares respond in general equilibrium to a tariff shock and to a risk shock inside and outside disaster episodes.

I run the same regression, with the same instrument, fixed effects, winsorizing, and clustering on (i) the data and (ii) model-generated shares. The targets are the $h = 5$ coefficients in the sample with the outcomes through 2018. In the model, I start with the estimated equilibrium for each period and then simulate equilibria perturbed by the same shifters as in the data: tariff changes drawn from the MFN pool and belief shocks. The estimate chooses $(\kappa, \gamma, \lambda)$ to minimize the root of the sum of squared gaps between the data and the model coefficients, scaled by the standard errors of the data.

The instrumented tariff response identifies κ . In the simulated panel, b^τ matches the softmax slope $-1/\kappa$ to within 9%. The level of response to belief outside disaster episodes is identified γ . The ratio of response to belief within to outside disaster episodes is identified by λ . The belief state affects choices through the posterior probability of disaster in the Lemma 1 and is priced in proportion to the buyer's exposure, so the difference in the response inside and outside the episode is informative on λ .

6.2 Trade Costs and the Blueprint

Two comparisons based on the first-order condition in Lemma 6 are useful to identify (i) trade wedges and (ii) the technology blueprint.

The blueprint. Under the assumption that firms in the same industry share a common blueprint across countries, buyer's purchase shares across suppliers in different industries are informative about the underlying blueprint.

Proposition 3 (Identification of the blueprint). *Fix a buyer country c , two buying industries j, j' with a common adjustment cost κ^I , and two suppliers $\hat{c}k, \hat{c}'k'$ with $m > 0$ and $m^0 > 0$. Then*

$$\log \frac{m_{\hat{c}k,(c,j)}}{m_{\hat{c}k,(c,j')}} - \log \frac{m_{\hat{c}'k',(c,j)}}{m_{\hat{c}'k',(c,j')}} = \log \frac{m_{\hat{c}k,j}^0}{m_{\hat{c}k,j'}^0} - \log \frac{m_{\hat{c}'k',j}^0}{m_{\hat{c}'k',j'}^0}.$$

With normalization $\sum_{\hat{c}k} m_{\hat{c}k,j}^0 = 1$ for every j , this determines $\log m^0$.

Proof. See Appendix C.11. □

The double difference is free of all equilibrium objects and of κ and γ . Since there is one final-good buyer per country; α^0 is recovered by a one-way demean of log final shares across destinations, averaged per supplier, and normalized to sum to one across suppliers.

Trade costs. Similarly, if we assume that trade costs are common between the buying industries within a destination, as in Bonadio et al. (2025), the buyer's purchase shares between two suppliers are informative about the trade cost.

Proposition 4 (Identification of trade costs). *Let $\mu_{ci} > 0$ and $m^0 > 0$. Fix a buyer b in c , final or intermediate, and two suppliers $\hat{c}k, \hat{c}'k'$ with $m_{\hat{c}k,b} > 0$ and $m_{\hat{c}'k',b} > 0$. Then*

$$\begin{aligned} \log \tau_{c,\hat{c}k}^m - \log \tau_{c,\hat{c}'k'}^m &= -\kappa_b^m \log \frac{m_{\hat{c}k,b}}{m_{\hat{c}'k',b}} + \kappa_b^m \log \frac{m_{\hat{c}k,j}^0}{m_{\hat{c}'k',j}^0} - (\mathbb{E}[p_{\hat{c}k}] - \mathbb{E}[p_{\hat{c}'k'}]) \\ &\quad - (1 - \vartheta)(\Pi_{\hat{c}k}^c - \Pi_{\hat{c}'k'}^c) - \vartheta \sum_o \xi_o (\Pi_{\hat{c}k}^o - \Pi_{\hat{c}'k'}^o). \end{aligned}$$

Proof. See Appendix C.10. □

The foreign wedge measures the extra cost on the route ($\hat{c} \rightarrow c, k$) needed to match the observed tilt toward the domestic supplier (c, k). For example, if the model (expected

prices, risk premia, and blueprint) predicts that buyers in c should source substantial steel from $\hat{c}$ but the data show that they buy primarily domestic steel, the gap is rationalized by a positive wedge.

At $\gamma = 1$ the inversion is the gravity inversion of [Bonadio et al. \(2025\)](#). At $\gamma > 1$ the premium term, built from the observed network and the covariance recovered from it, filters insurance-driven sourcing out of the wedge.

I also allow domestic wedges, defined analogously within c . If, conditional on prices, risk and blueprint, buyers in c use domestic steel less than domestic machinery, the domestic steel wedge is the cost that rationalizes that shortfall. The proposition identifies differences between the wedges up to a normalization. I normalize the share-weighted mean wedge each buying industry pays is zero, so that estimated productivity shocks do not depend on τ .

The model cannot match all shares exactly because I restrict trade costs to be buyer-invariant within a destination: $\tau_{c,\hat{c}k}$ is the same for every buying industry j in c . For example, autos and construction in c importing steel from $\hat{c}$ face the same wedge, so one wedge must fit up to 23 import shares. The model therefore matches the average share towards $(\hat{c}, k)$, and the remaining within- c dispersion is a diagnostic of the restriction ([Appendix D.4](#)).

6.3 Stochastic Structure

Using the value added from WIOD and the [Lemma 7](#) on the observed network in every period, I estimate a series of panel productivity shocks at that year's observed network,

$$\hat{\epsilon}_t = \left(I + \frac{\psi}{1 + \psi} \boldsymbol{\mu}_t \boldsymbol{\alpha}_t^* \mathcal{L}_t \right)^{-1} \mathbf{a}_t, \quad (15)$$

Next, extending the country and sector decomposition of [Caselli et al. \(2020\)](#) with a global and a granular component, I decompose them as

$$\epsilon_{ci,t}^D = g_t^D + \xi_{c,t}^D + \omega_{i,t}^D + l_{ci}^D \quad (16)$$

Then I take their mean to estimate their means θ_k . The conditional variances of the four components are then GARCH(1,1) on their innovations, with the two GARCH parameters pooled across series.

Disaster risk and beliefs. I measure disasters from annual real GDP per capita in the Maddison Project (1870–2022), using the definition of peak-to-trough episodes of Barro (2006); Barro and Ursúa (2008). For each country c with at least one episode, this yields a previous annual disaster probability (hazard) between 1.3 and 3.3 percent. Ireland has no episodes, so its hazard and thus its posterior disaster probability is zero. The size of the disasters ranges from 0 to 0.55 log points (largest for Greece, Austria, China, the Netherlands and India). I assume that disasters are independent between countries. In the model, each episode becomes a one-period tail event whose cost equals the observed cumulative peak-to-trough loss; the remaining year-to-year fluctuations within the episode stay in the Gaussian component.

The beliefs come from Hassan et al. (2023). $h_{c,t}$ is the annual mean perceived risk of foreign companies from country c (standardized within country). Appendix D.4 gives construction details.

7 Quantitative Results

In this section, I report (i) the four moment targets used in estimation and (ii) the parameters used in the quantitative exercise.

7.1 Parameters

Table 1 reports the four moments targeted in estimation, the regression coefficients from Section 2. For each coefficient, it shows the estimate in the data (with its standard error) and the corresponding coefficient implied by the estimated model.

Targeted moments			
Moment	Identifying variation	Data [se]	Model
b^τ	instrumented tariff change	−1.045 [0.403]	−1.012
b^N	perceived risk $\times$ exposure, no episode	−0.738 [0.268]	−0.855
b^D	perceived risk $\times$ exposure, in episode	−6.480 [4.426]	−6.127
b^h	perceived risk $\times$ Leontief exposure, pooled	−0.089 [0.026]	−0.076

Table 1: Estimation targets: moments in the data and the model

Notes: Data standard errors are in brackets.

Across the four targets, the model matches the data to within half a standard error: 0.08 for b^r , 0.44 for b^N , 0.08 for b^D and 0.50 for b^h .

Table 2 summarizes the values of the parameters. Three inputs are taken from outside the model: labor shares μ_j , the Frisch elasticity $\psi = 0.72$ of [Huo et al. \(2024\)](#), and the pledgeable fraction $\vartheta = 0.08$, the primitive of [Biais et al. \(2021\)](#), imposed from the share of disaster losses insured internationally in [Ito and McCauley \(2022\)](#); Section 8.3 reports the results at 0, 0.08 and 0.29, the upper bound of Appendix B.3.

External		
Parameter	Value	Related to
μ_j	[0.22, 0.68]	Labor shares from WIOD
ψ	0.72	Frisch elasticity (Huo et al. (2024))
ϑ	0.08	Pledgeable fraction, imposed (Ito and McCauley, 2022)
Estimated: network response		
κ	0.95 (0.38)	Adjustment cost; MFN tariff changes
γ	10 (9.4)	Risk aversion; perceived-country-risk response
λ	5.0 (31.3)	Signal discriminability; episode split of the risk response
Estimated: stochastic structure		
ρ	0.886	Avg. AR(1) persistence (innovation extraction only)
ϱ	0.19	GARCH – innovation weight
P	0.49	GARCH – persistence
$\bar{\pi}_c$	[1.3%, 3.3%]	Country disaster hazard (Maddison 1870–2022)
d_c	[0, 0.55]	Disaster size, log TFP per episode (mean 0.21)

Table 2: Parameters

Notes: Brackets report the 10th–90th percentile range across country-industries. Standard errors (where shown) are in parentheses.

7.2 Estimated Uncertainty

The estimated shock volatilities have three broad features. First, the global component is the most persistent: it drifts slowly over time and is noticeably higher in the late sample than in the 1980s–1990s. Since its variance follows a GARCH process, large events translate into elevated volatility that can linger and may peak with a short delay.

Second, country and industry shocks are much more dispersed between units. For both blocks, a typical country/industry has a variance of the order of 10^{-4} , while the upper tail is several times larger and rises after 2010 (and, for some industries, again after 2020).

Third, granular (supplier-level) shocks are larger and have increased over the last decade: their median and upper-tail variances are roughly an order of magnitude above the country/industry blocks. Appendix Figure 10 reports the detailed time-series paths and percentile ranges for each block.

7.3 Model Fit

Throughout this section and the next, the aggregate risk in country c is the consumption variance $\text{Var}(p_c)$, the variance of the logarithmic final-good price index. Because nominal income is deterministic given the network (Lemma 2), this is also the variance of real consumption.

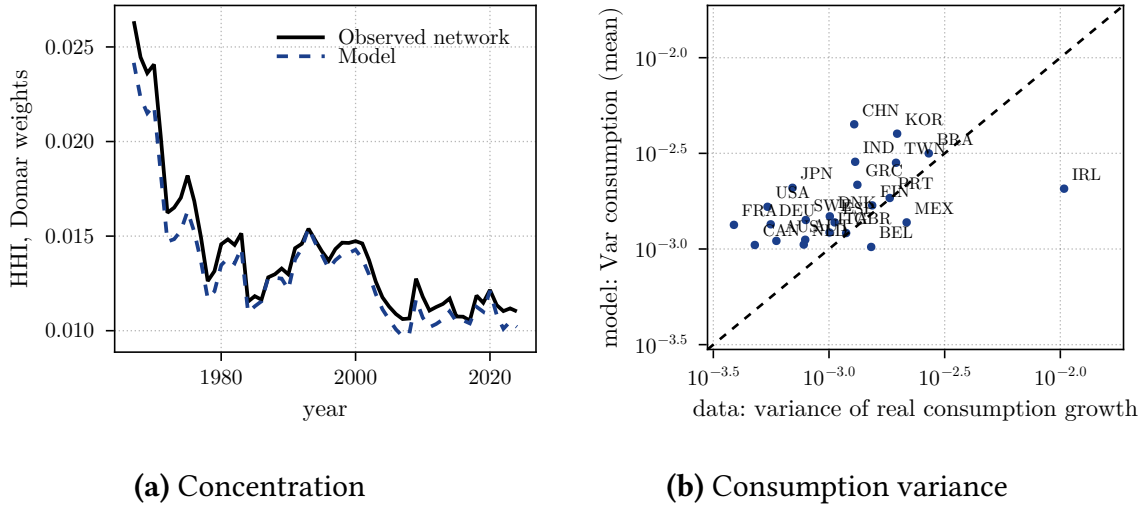


Figure 4: Concentration and Aggregate Risk: Data vs. Model

Notes: (a) Herfindahl index of Domar weights, 1967–2024 (solid: data; dashed: model). (b) Country scatter (log scales): data absorption-growth variance vs. model $\text{Var}(p_c)$, 1968–2024; 45° line shown.

I compare two moments in the data and in the model. First, concentration: the model Herfindahl index of Domar weights tracks the index implied by the observed input–output network (Figure 4). Second, aggregate risk: against the variance of real absorption growth,

total final demand, the model variance correlates 0.20 across countries, 0.46 in logs, at a level 1.2 times the data's (Figure 4b).

The inversion matches the observed network tightly: pooled across country–industry years, the model-implied Domar weights have correlation 0.994 with the data, and predicted log sourcing shares have correlation 0.942 on routes with positive trade. The remaining gap reflects the restriction that each destination assigns a single route wedge $\tau_{c,\hat{c}k}$ that is common between the buying industries.

The Online Appendix repeats the fit allowing trade deficits and shows that Domar weights track the data expenditure shares closely if we allow trade deficits. I assume no trade deficits in the baseline model in order to have cleaner counterfactual (Details in the Appendix).

8 Counterfactual Analysis

This section asks how lower trade costs change aggregate risk. Over the liberalization range observed in the data, risk falls. Financial integration raises the level of aggregate risk but barely changes the risk response to liberalization.

8.1 Removing the Borders: the Controlled Experiment

The first exercise eliminates trade distortions. Starting from the trade cost of 1967, I move the cross-border trade wedges towards zero. Figure 5 presents the results. The total variance along the free-trade path is U-shaped. It falls first and reaches its minimum after removing roughly 40 percent of barriers, then increases sharply and ends being 10 times larger in free trade.

Figure 5 also breaks the total variance into its components. Up to the minimum of the total-variance curve, the composition shifts toward global risk: the country component falls from 49 to 30 percent of the total risk and the granular component from 24 to 16 percent, while the global component increases from 22 to 45 percent. Intuitively, as barriers fall, the rows of the Leontief inverse spread across countries, and exposure tilts toward common (global) shocks.

Beyond the minimum, all components increase in levels, and the composition shifts toward granular risk. The granular component increases the most, about fifty times, and

accounts for more than half of the total risk under free trade; the industry share increases from 5 to 20 percent, while the global share declines to 15 percent. At that point, many buyers load on the same few upstream suppliers, so shocks to those suppliers effectively become global shocks.

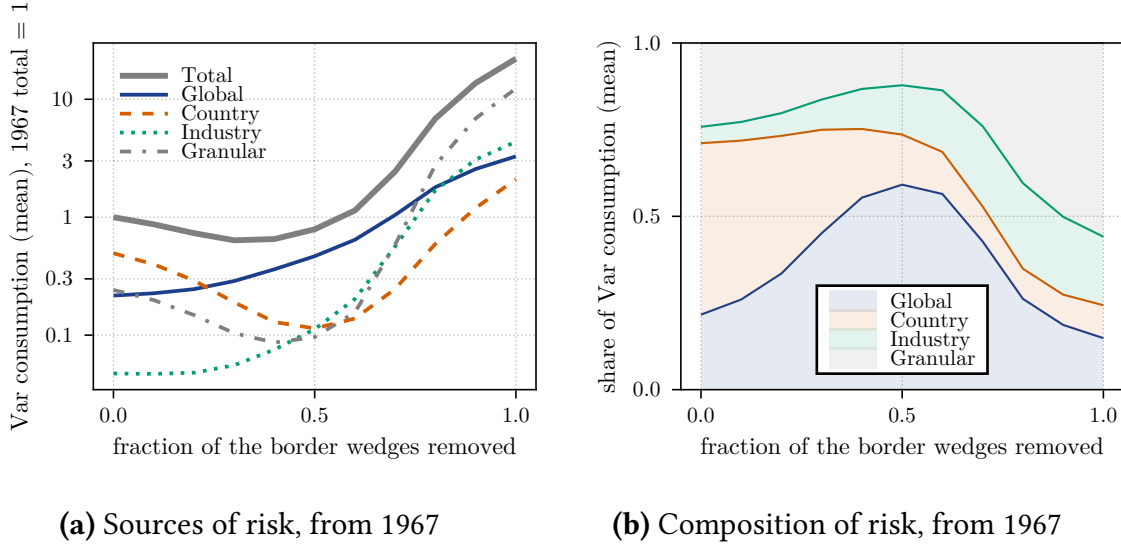


Figure 5: Removing the Borders: the Sources of Risk

Notes: Starting from 1967, we scale cross-border wedges to zero in 10 steps (free trade) and re-solve each step. Plots: cross-country mean $\text{Var}(p_c)$ decomposed into global/country/industry/granular (Section 5). (a) components vs. 1967 total (log), (b) shares. $\vartheta = 0.08$.

8.2 The Liberalization the Data Record

Next, we evaluate the effect of the observed liberalization of trade between 1967 and 2024.

Observed changes Figure 6 summarizes how measured trade wedges evolve along the observed liberalization, treating 2024 as the endpoint. The decline is concentrated in intermediate inputs. The median intermediate wedge falls by 0.81 between 1967 and 2024; it is below its 1967 level in 52 of 58 years and trends downward at 0.14 per decade. The Final-goods wedges, on the contrary, show no clear trend. They fluctuate between -0.30 and $+0.55$ and end only 0.17 below 1967.

Part of the measured intermediate decline reflects risk premia. If I re-invert the wedges imposing $\gamma = 1$ (no risk pricing), the intermediate decline shrinks to 0.22 and the final-goods wedge rises by 0.34.

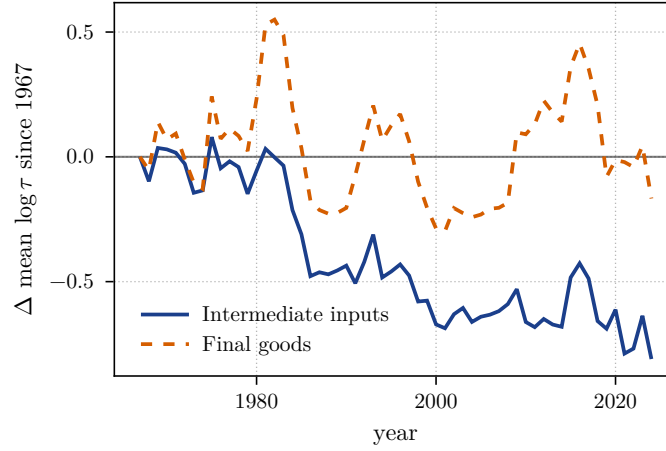


Figure 6: Measured Trade Costs, 1967–2024

Notes: Change in the mean log wedge relative to 1967 over the balanced panel of routes priced in every sample year (54,055 intermediate routes, 2,676 final-goods routes), at the calibrated (κ, γ) .

Risk and concentration To isolate the effect of the observed border changes, cross-border wedges remain fixed at their 1967 values while letting all other fundamentals change (shocks, blueprints, and domestic wedges). Figure 7 presents the results. The two series track each other through the 1980s, then separate in the 1990s and remain apart thereafter. Relative to the counterfactual with the 1967 borders, the observed liberalization reduces the mean variance of consumption by about 10% by 2024. The network also becomes less concentrated. In 2024, the Herfindahl index of Domar weights is 17% lower under the 2024 borders than under the 1967 borders. In the 2024 economy, with all else fixed, the observed border wedges lower the variance in 16 of 24 economies. The largest declines are in Greece and Ireland (about one third) and in Korea and Finland (about one fifth). Variance rises in a small set of economies, notably France, the Netherlands, Spain, and Britain, and is essentially unchanged in Sweden, Taiwan, Austria, and Italy.

The Shapley decompositions attribute the decline to intermediate-input trade: the intermediate margin lowers risk in all 24 economies (the diversification term in equation (14)). Final-goods trade is more mixed, lowering risk in 16 economies but raising it in 8 as demand shifts toward globally central producers whose prices load more on global shocks.

Compared to Caselli et al. (2020), who find lower volatility in about two thirds of their

economies, the decline is more widespread in my model. The key difference is that their network is exogenous, while in my model liberalization also changes supplier choice and hence centrality, allowing the final-goods margin to raise risk for some economies and making the sign flip once liberalization moves beyond the range observed in the data.

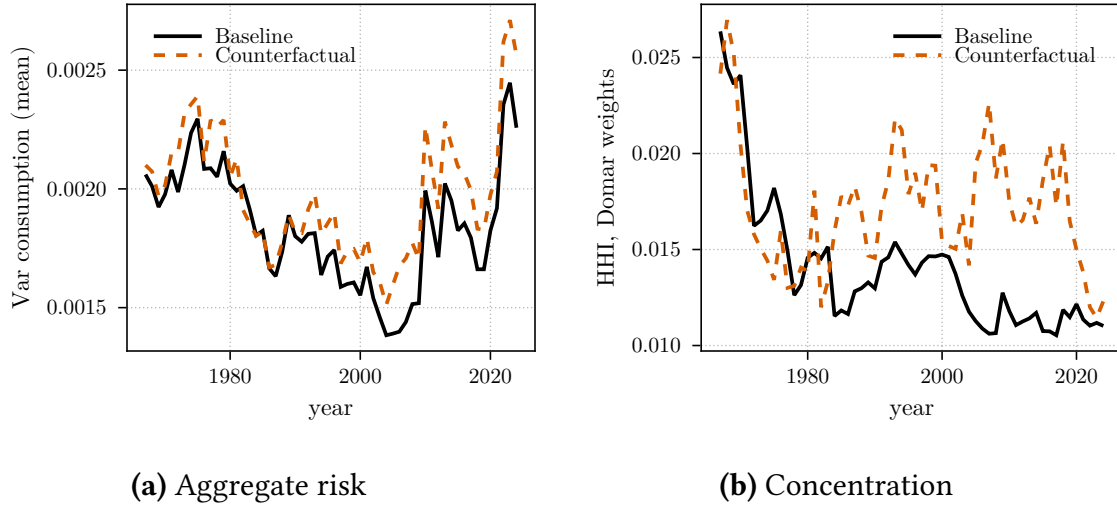


Figure 7: Aggregate Risk and Concentration Over Time

Notes: Solid: the model solved year by year at the observed wedges. Dashed: counterfactual holding cross-border wedges fixed at 1967 while letting domestic wedges, shocks, and the blueprint vary by year.

Robustness. These results are stable to alternative endpoints and alternative interpretations of the wedges. Taking 2010 or 2022 (instead of 2024) as the liberalized endpoint delivers similar effects: variance falls by about 10% and concentration declines as well (Herfindahl decreases 39% and 16%). If I re-invert at $\gamma = 1$ (so risk premia are absorbed into wedges), the 1967→2024 border changes still lower variance by 12% and concentration by 16%; along the observed path the two specifications agree because both match the 1967 and 2024 networks. However, the distinction between wedges and premia matters mainly when extrapolating beyond the data: under $\gamma = 1$, pushing from 1967 toward free trade yields larger reconcentration and risk increases more without premia (variance 29 vs. 22 times; concentration 10 vs. 7 times).

Allowing for observed trade deficits as exogenous transfers leads to comparable conclusions in the 1967 economy (variance decrease -37% , concentration goes from 0.017 to 0.010). See details in the Online Appendix.

8.3 Financial Market and the Effect of Liberalization

This subsection studies how financial integration changes the effect of trade liberalization. I proxy integration with the pledgeable fraction ϑ . I compared the 1967-to-2024 liberalization under three market structures: financial autarky ($\vartheta = 0$), baseline calibration ($\vartheta = 0.08$), and high integration ($\vartheta = 0.29$). The higher ϑ increases the level of aggregate risk.

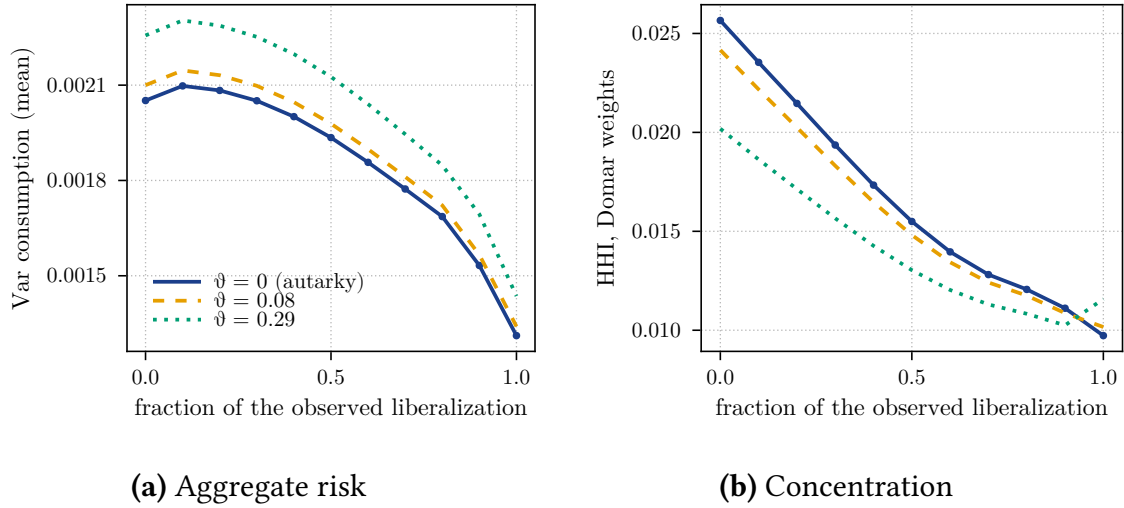


Figure 8: The Liberalization at Three Values of ϑ

Notes: The 1967 economy with its cross-border wedges moved to their 2024 values in ten equal steps, solved at $\vartheta = 0, 0.08$ and 0.29 . Panel (a): mean $\text{Var}(p_c)$. Panel (b): Herfindahl index of Domar weights.

Moving from financial autarky ($\vartheta = 0$) to high integration ($\vartheta = 0.29$) raises the mean variance in consumption by approximately 10%. In addition, liberalization reduces the Herfindahl index more under autarky than under high integration (62% vs. 43%).

As the borders open, the three ϑ paths converge because trade integration itself makes the marginal valuations of the countries more similar. The term ϑ in the risk-adjusted price captures the wedge between the premium implied by the world kernel and the premium implied by the buyer's local kernel. Under the 1967 wedges, consumption baskets are mostly domestic, so their exposures are very dispersed (mean correlation 0.10 across countries) and the world-local premium gap is large (0.46). By 2024, the baskets load on a common set of global suppliers, the mean correlation rises to 0.55, and the cross-country dispersion in the kernels shrinks to 0.18. With less heterogeneity in kernels, changing ϑ

matters less, so the concentration gap narrows from 21% to 8% and the risk gap from 10% to 8% along the liberalization path.

Welfare Figure 9 shows the decomposition of the welfare gains from liberalization. Each panel reports the gains under financial autarky ($\vartheta = 0$) and high integration ($\vartheta = 0.29$), relative to the calibrated baseline $\vartheta = 0.08$.

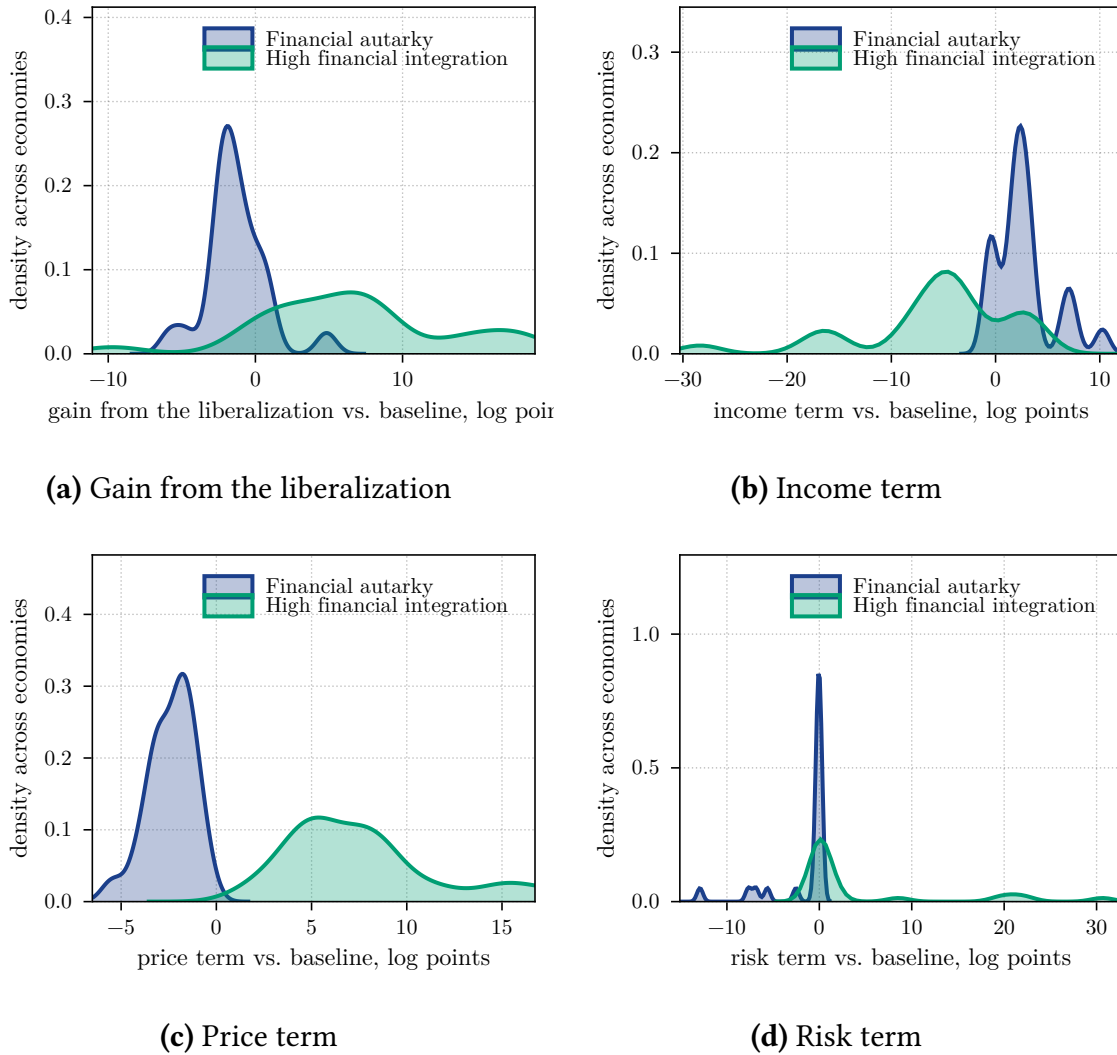


Figure 9: Market Structure and the Gains From the Liberalization

Notes: Kernel densities across the 24 economies of the welfare gain from moving the cross-border wedges of the 1967 economy to their 2024 values, under financial autarky ($\vartheta = 0$) and under high financial integration ($\vartheta = 0.29$), each minus the same object at the baseline $\vartheta = 0.08$.

Panel (a) the welfare gains from the 1967→2024 liberalization are greater when the financial markets are more integrated. The difference is driven by the expected price of the consumption basket. When a buyer can pledge part of its firms’ payoffs to world investors, so it applies a smaller risk premium to suppliers whose prices comove with the world basket rather than with its own. These are precisely the central productive suppliers that open borders make accessible. Higher integration therefore shifts sourcing towards them and decreases $\mathbb{E}[p_c]$. The price term moves in the same direction in all 24 economies and accounts for most of the shift in panel (a).

The income term captures the redistribution of production between countries, and higher integration generates higher welfare losses from redistribution. As integration concentrates on sourcing on the most productive suppliers, a few winner countries gain world income share while many smaller countries lose it. Because welfare is logarithmic in the share, a small country that loses market share can suffer a larger welfare loss than the winners’ gains. The risk term contributes little. Liberalization reduces consumption variance by a similar proportion in each ϑ , and variance is small in consumption units.

9 Conclusions

Trade liberalization changes supplier choice in two opposing ways. By expanding the set of feasible suppliers, it creates scope for diversification; by making the most productive suppliers more accessible, it also strengthens incentives to concentrate. The force that dominates depends on how firms trade expected costs against risk.

In both the data and the model, buyers respond to perceived supplier risk by shifting sourcing away from riskier countries, reducing those producers’ market shares. This disciplined risk response lets me measure the cost–insurance trade-off that governs network formation. When expected-cost motives dominate, liberalization concentrates sourcing on a few highly productive suppliers, and aggregate risk can rise; when insurance motives dominate, liberalization spreads sourcing across origins, and aggregate risk falls.

During the liberalization observed in the data, diversification dominates. Moving border wedges from 1967 to 2024 reduces the mean variance in consumption by 36% and reduces it in 20 of the 24 economies. Beyond the observed liberalization towards free trade, the variance rises once the network concentrates in few very productive suppliers. In addition, higher financial integration increases the welfare gains from trade liberalizations.

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A Additional Empirical Analysis: Firm Level

This appendix defines every variable of equation (1) and gives its source. The baseline estimates at every horizon, the robustness exercises, and the placebos are in the Online Appendix.

A.1 Data and Variables

Firm panel. Compustat North America and Compustat Global, from WRDS: 594,637 firm-years for 40,704 listed firms in 118 economies, fiscal years 1989–2024. Fiscal 2025–26 is partially filed and excluded. Sales are converted to US dollars. A firm’s sector is one of the 14 goods sectors of the input–output data, assigned from its three-digit NAICS code and from its SIC code where NAICS is missing. Its country is the one recorded in the panel: headquarters for firms outside North America, country of incorporation for North American firms; The Online Appendix reassigns the latter by headquarters. The estimation sample is goods producers in 49 economies, shock cohorts 1996–2019. The 49 are the 24 economies of the input–output data (Australia, Austria, Belgium, Brazil, Canada, China, Denmark, Finland, France, Germany, Greece, India, Ireland, Italy, Japan, Korea, Mexico, Netherlands, Portugal, Spain, Sweden, Taiwan, United Kingdom, United States) and 25 more for which the sales weights and the tariff schedules below exist (Argentina, Chile, Colombia, Czechia, Egypt, Hong Kong, Hungary, Indonesia, Israel, Malaysia, New Zealand, Nigeria, Norway, Pakistan, Peru, Philippines, Poland, Russia, Saudi Arabia, Singapore, South Africa, Switzerland, Thailand, Turkey, Vietnam).

World market share, $s_{f,t}$. Firm sales divided by the sales of all listed firms in all 118 economies, in the same sector and year. Nominal shares are sold in dollars. Real shares deflate each firm’s sales by its country-sector dollar price index and divide by the sum of deflated sales over the firms that have an index, so the real denominator runs over the economies with an index rather than the 118 of the nominal one, and real shares use only cohorts with $t + h$ inside the index’s coverage. The index is a gross-output price index in dollars, chained across two sources. Through 2014 it is the gross output deflator of the WIOD Socio-Economic Accounts (2016 release): gross output in national currency over the chained quantity index, 43 countries, 2000–2014, converted at the WIOD exchange rate. From 2015 to 2024 it grows with the gross-output deflator of the ADB multi-regional input–

output tables, current-price over constant-2010-price gross output in dollars, chained to the WIOD level at 2014 with the geometric mean of the 2010–14 ratio of the two. The index exists for 39 of the 49 economies: 32 with the WIOD level, and seven (Hong Kong, Malaysia, Pakistan, the Philippines, Singapore, Thailand and Vietnam) with the ADB deflator alone from 2007, normalized at 2014. The outcome $\Delta_h \log s_{f,t} = \log s_{f,t+h} - \log s_{f,t-1}$ is winorized in the 0.5 and 99.5 percentiles within the horizon.

Perceived risk, $h_{c,t-1}$, and sentiment. The quarterly series of country-risks of [Hassan et al. \(2023\)](#), as perceived by firms headquartered in other countries, 2002Q1–2020Q4. I take $\log(1 + \text{risk})$, standardize it within country over the full sample, and average it over the four quarters before the first quarter of the firm’s fiscal year, using the firm’s fiscal year-end month. The regressor is therefore predetermined relative to the outcome, and one unit is one standard deviation of the country’s own risk. The first cohort with a risk observation is fiscal 2003, and the last in 2019. Sentiment is built the same way from the country-sentiment series of the same source. The series exist for 45 of the 49 economies (35 of the 39 with a price index). The remaining economies enter with risk and sentiment set to zero and with their own sector-year effects, so they contribute to the tariff coefficient and to the fixed effects only.

Applied tariff, $\tau_{cj,t}$. The change in tariff that the country-sector cj faces in the outcome window is

$$\Delta_h \log(1 + \tau_{cj,t}) = \sum_{i \neq c} w_{cj,i} \left[\log(1 + \tau_{i,j,t+h}^{MFN}) - \log(1 + \tau_{i,j,t-1}^{MFN}) \right] + Z_{cj,t,h},$$

where $w_{cj,i}$ is the share of the output of cj absorbed by destination i , from the OECD Inter-Country input–output tables for 2016, and i runs across the 49 economies. The most-favored-nation rates are from UNCTAD TRAINS, through the World Bank WITS: simple-average rates at the six-digit level of the Harmonized System, averaged without weights to the 14 sectors by the HS chapter, 1995–2018, and held at the 2018 level afterwards. Members of the European Union carry their schedule. A destination enters only in years in which the pair trades under MFN: both are WTO members and no preferential agreement is in force, from the CEPII Gravity database.

Discriminatory tariffs, $Z_{cj,t,h}$ (the instrument). The tariff in force by importer, exporter and six-digit product at every change date from January 2018, covering Sections 201, 232 and 301, the IEEPA actions, and every partner's retaliation against the United States. The annual rate of a pair and a product is the time-weighted average of the rate in force over the calendar year. The sector value is the unweighted mean over the products of $\log(1 + \text{rate})$ minus its value on 1 January 2018. It was zero before 2018 and was maintained at its 2024 level afterwards. For an exporter other than the United States, $Z_{cj,t,h}$ is the change between $t - 1$ and $t + h$ in the US discriminatory tariff on cj , times $w_{cj,US}$; for the United States it is the w -weighted change in the retaliatory tariffs of its destinations. One rule applies to all 49 economies, so who is treated is read from the schedules: in 2019 China faced up to 18 log points, Germany, France, Japan and Taiwan 7 through the steel and aluminum actions, India 7, Canada and Mexico 3. The MFN changes alone identify little: within a destination they apply to every exporter in the same way, so only the destination weights differentiate exporters, and the equation needs a differential between exporters within a sector-year.

Controls, $w_{cj,t}$. Sentiment, defined above. For US firms, two variables that are zero elsewhere: the protection their sector received, the share of Chinese goods in US absorption of sector j times the change in the Section 301 tariff on j ; and the increase in their input costs, the sum over supplying sectors k of the share of Chinese sector- k goods in the inputs of US sector j times the change in the Section 301 tariff on k . Both shares are from the 2016 input-output table.

Estimation. Two-stage least squares, one regression per horizon, with firm- and sector-year effects; the sector-year effects are separate for economies with and without the risk series. Standard errors are clustered by country, with $t(G - 1)$ critical values, $G = 49$ for nominal and 39 for real shares.

B Additional Empirical Analysis: Sectoral Level

This appendix has three parts. Section B.1 lists the data sources. Section B.2 defines the variables. Section B.3 reports the risk-sharing regression and the calibration of the pledgeable fraction. The sourcing regression of Section ?? written in full, its results at every horizon, and the descriptive evidence on aggregate risk and on the production network, 1967–2024, are in the Online Appendix.

B.1 Data Sources

Input–output tables. The Long-WIOD panel: world input–output tables for 24 economies and 23 industries (ISIC Rev. 3), 1967–2024, 552 country-industry pairs in every year. It joins three sources. The tables for 1967–2000 are the Long-run WIOD of [Woltjer, Gouma, and Timmer \(2021\)](#); those for 2001–2014 are the WIOD 2016 release ([Timmer et al., 2015](#)); those for 2015–2024 are the ADB MRIO tables ([Asian Development Bank, 2024](#)). The 56 industries of the WIOD 2016 release and the 35 of the ADB tables are mapped to the 23 ISIC Rev. 3 industries of the Long-run WIOD. The three sources are joined at 2000 and 2014, in each case on the shares the model uses: every input and final-demand share of the later source is shifted by the gap between the two sources in the overlap year and renormalized to sum to one, labor shares are shifted by their gap, and gross output is scaled by its ratio in the overlap year, so the series carry the later source’s changes on the earlier source’s levels. Flows to and from economies outside the 24 are dropped: shares are normalized over the 552 sample nodes, gross output is sales within the system and value added is gross output net of purchases within it, so the accounting identities hold in the closed 24-country system. The ADB tables come from three releases (2007–2016, 2017, and 2018–2024) with level steps between them that the OECD ICIO tables, a single vintage over 2013–2020, do not show. Within a release the tables’ own changes are kept. Across a release boundary the change from 2016 to the first year of the later release is taken from ICIO at the country level (the foreign shares of intermediate and final purchases, the labor share, and the gross-output-to-value-added and deficit ratios) and at the country-industry level for the within-country shares of value added and gross output; inside each aggregate the composition of the later table is kept, and cross-border cells active only in the later releases and in no ICIO year are set to zero. Tables 3 and 4 list the industries and economies, which account for over 85 percent of global GDP throughout.

Tariffs. Most-favored-nation schedules from UNCTAD TRAINS, through the World Bank WITS, 1995–2021: simple-average rates at the six-digit level of the Harmonized System, averaged without weights to the 14 goods industries by HS chapter. Members of the European Union carry their schedule; the United Kingdom carries its own from 2021. Taiwan 2019–21 and Mexico 2021 are not in TRAINS. Whether a pair trades under MFN in a year — both are WTO members and no preferential agreement is in force — is from the CEPII Gravity database, which ends in 2021.

Perceived risk. The quarterly series of country-risks of [Hassan et al. \(2023\)](#), as perceived by firms headquartered in other countries, 2002Q1–2020Q4. It exists for 20 of the 24 economies.

Disaster episodes. The Maddison Project record of GDP per capita, 1870–2022. An episode is a peak-to-trough decline in logarithmic GDP per capita of at least 10percent, the rule of [Barro \(2006\)](#): 75 episodes in 23 of the 24 economies.

National accounts. Real income and real absorption for the risk-sharing regression are built from the same input–output tables. The channel decomposition uses the national accounts of the World Bank for 23 of the economies, 1971–2024.

Code	Description
AtB	Agriculture, Hunting, Forestry & Fishing
C	Mining & Quarrying
D15t16	Food, Beverages & Tobacco
D17t19	Textiles, Leather & Footwear
D21t22	Pulp, Paper, Printing & Publishing
D23	Coke, Refined Petroleum & Nuclear Fuel
D24	Chemicals & Chemical Products
D25	Rubber & Plastics
D26	Other Non-Metallic Mineral Products
D27t28	Basic Metals & Fabricated Metal Products
D29	Machinery, nec
D30t33	Electrical & Optical Equipment
D34t35	Transport Equipment
Dnec	Manufacturing, nec; Recycling
E	Electricity, Gas & Water Supply
F	Construction
G	Wholesale & Retail Trade
H	Hotels & Restaurants
I60t63	Transport & Storage
I64	Post & Telecommunications
J	Financial Intermediation
K	Real Estate, Renting & Business Activities
LtQ	Public Administration & Other Services

Table 3: Sectors in the Long-WIOD Panel

Notes: The twenty-three industries of the long-run WIOD panel; codes are ISIC Rev. 3 divisions.

AUS – Australia	AUT – Austria	BEL – Belgium	BRA – Brazil
CAN – Canada	CHN – China	DEU – Germany	DNK – Denmark
ESP – Spain	FIN – Finland	FRA – France	GBR – United Kingdom
GRC – Greece	IND – India	IRL – Ireland	ITA – Italy
JPN – Japan	KOR – South Korea	MEX – Mexico	NLD – Netherlands
PRT – Portugal	SWE – Sweden	TWN – Taiwan	USA – United States

Table 4: Countries in the Long-WIOD Panel

Notes: The twenty-four economies of the long-run WIOD panel, with the ISO-3 codes used in the figures.

B.2 Variables

Input shares, $m_{\hat{c}k,b,t}$. The flow from the supplier country-industry $\hat{c}k$ to the buyer country-industry b divided by the total intermediate purchases of the buyer. A cell enters at the horizon h if its share exceeds 10^{-5} at $t - 1$ and at $t + h$ and the two countries differ. The outcome is $\Delta_h \log m_{\hat{c}k,b,t} = \log m_{\hat{c}k,b,t+h} - \log m_{\hat{c}k,b,t-1}$, winsorized in the 0.5 and 99.5 percentiles within the horizon.

Exposure, $fd_{c,\hat{c},t-1}^{\hat{c}}$. The share of the final demand of the buyer's country c spent on goods from the supplier's country $\hat{c}$, from the table $t - 1$. It is an accounting ratio: no Leontief inverse, and no parameter enter it. Its median across routes is 0.16 percent, its 90th percentile 1.3 percent, and its 99th 4.9 percent.

Tariff, $\tau_{c,\hat{c}k,t}$. The MFN rate of country c in good k . A route of goods enters if the pair trades under MFN at $t - 1$ and at $t + h$; intra-EU and preferential pairs are excluded. Services do not have a tariff and enter with $\Delta_h \log(1 + \tau) \equiv 0$. The instrument is the one-year change of the schedule, $\log(1 + \tau_t) - \log(1 + \tau_{t-1})$, on routes on which the supplier's country holds less than 10 percent of the imports of the good from the buyer's country at $t - 1$, and zero elsewhere. Unlike the firm-level measure in Appendix A.1, this is the tariff on one route, not a sales-weighted average across destinations, and it excludes the discriminatory tariffs of 2018–24.

Perceived risk, $h_{\hat{c},t-1}$. The log of one plus the calendar-year mean of the quarterly series, standardized within country over 2002–2020. One unit is one standard deviation of the country's own risk. Suppliers in the four economies without the series are not in the regression.

Episode indicator, $D_{\hat{c},t-1}$. The year $t - 1$ lies between the year after the peak and the trough of a disaster episode of the supplier's country, and $N_{\hat{c},t-1} = 1 - D_{\hat{c},t-1}$. In the sample the episodes are Greece (2009–2019), Italy (2008–2019), Brazil (2014–2019), and Mexico (2019); episode cells are about 3percent of the sample.

Sample. Shock cohorts $t = 2005, \dots, 2020$ with outcomes through 2021, so horizon h uses cohorts through $\min(2020, 2021 - h)$. The risk series starts in 2002, the tariff and

MFN-status data end in 2021, and the signal year 2020 is excluded: perceived risk increases by more than two standard deviations in every country at once, and e^h in that one cohort is ten times its size elsewhere.

Other objects built from the tables. *Domar weights* $\omega_{ci,t}$ are the gross output of the country-industry (c, i) over the global GDP. The *final demand shares* $\alpha_{c,c',i',t}$ divide each final flow by the total final demand of the country. *Value added* $a_{ci,t}$ is the logarithmic real value added scaled by the labor share, the primary-input contribution of the Lemma 7; nominal value added is made real with a country-sector price index. The balanced panel, all 552 pairs in all 58 years, is what the factor decomposition of the Online Appendix requires.

B.3 Risk Sharing: the Consumption Test and the Calibration of ϑ

Market structure. The sourcing regressions identify *how firms price supplier risk* when choosing inputs. They do *not* identify *how much international risk sharing households obtain*. In the model, that second object is summarized by the pledgeable fraction ϑ , which governs the weight on the incomplete versus complete market pricing kernels. The sourcing margin cannot pin down ϑ because the two kernels look very similar in the data: a buyer’s own-basket exposure and its revenue-weighted world exposure have correlation above 0.95 (pooled and within each country), so sourcing shares barely move when ϑ changes. This is the network analog of Karaivanov and Townsend (2014): sharply different financial regimes can generate observationally similar allocations.

The absorption–income regression. The regression of Asdrubali et al. (1996),

$$\Delta \log A_{c,t} - \Delta \log A_{w,t} = \alpha_c + \beta (\Delta \log Y_{c,t} - \Delta \log Y_{w,t}) + \varepsilon_{c,t}, \quad (17)$$

with $A_{c,t}$ real absorption, total final demand, $Y_{c,t}$ real income and w the world, regresses growth in real absorption relative to the world on growth in real income relative to the world. Under the primitive “households consume their own income,” the slope is one. The regression is a *test* of the primitive, not an estimator of ϑ . I estimate it on the same input–output dataset used for the network (24 countries; 1968–2024; 1,368 country-years; standard errors clustered by country). Real income is the Laspeyres real-GDP index from the benchmarked value-added series; real absorption is total final demand deflated by

the table's value-added deflator (Long-WIOD through 2000, WIOD 2016 from 2001, ADB from 2015). With country fixed effects, $\beta = 1.009$ (standard error 0.062; $p = 0.887$ for a unit slope), so I do not reject a unit slope. Two deficit-based checks point the same way: $\Delta(D_c/Y_c)$ on relative income growth yields 0.008 ($p = 0.893$), and the change in the deficit share of world income on the change in the country's world-income share yields 0.008 ($p = 0.014$), significant but small. The estimate is stable across specifications (pooled 0.999; country and year effects 1.004; excluding 2008–2009 1.005).

What the regression can reject is *household* complete markets across countries. Under the complete market household allocation (with GHH preferences), the slope would be $1 - 1/(1 + \psi) = 0.42$, about nine standard errors away.

How I discipline ϑ . Final consumption is smoother than absorption: on final consumption (households + non-profits + government), the slope is 0.710 ($p < 0.001$). This smoothing is mostly domestic, not international: the trade balance does not covary with relative income, so it is financed by within-country saving and investment. In World Bank data for 23 countries, the [Asdrubali et al. \(1996\)](#) decomposition attributes only 0.02 (s.e. 0.05) of relative output shocks to income from cross-border factors during 1971–2024 (and 0.06 (s.e. 0.06) since 1991), versus 0.38 (s.e. 0.09) to saving. I therefore treat $1 - 0.710 = 0.29$ as an upper bound on the share of risk borne abroad and report counterfactuals at $\vartheta = 0.29$ as a high value.

For the baseline, I impose $\vartheta = 0.08$ from the literature on ex-ante international risk sharing: about 8% of disaster losses are reinsured internationally in advanced economies (1982–2017) ([Ito and McCauley, 2022](#)), and about 11% of idiosyncratic output shocks are absorbed through cross-border factor income and capital gains in the OECD ([Balli, Kalemli-Ozcan, and Sørensen, 2012](#)). I take the disaster measure because the priced tail risk in the model is disaster risk. The value is imposed, not estimated, so Section 8 reports the liberalization counterfactuals at $\vartheta = 0, 0.08$, and 0.29.

C Proofs of Theoretical Sections

C.1 Proof of Lemma 2

Proof. We begin with the revenue of the sector ci , which equals its sales to final-good producers plus its sales to intermediate-good producers. The optimality conditions of buyers give the expenditure of the final-good producer $\hat{c}$ on the supplier ci and the expenditure of the intermediate producer $\hat{c}k$ on the supplier ci :

$$\tau_{\hat{c},ci} P_{ci} Z_{\hat{c},ci} = P_{\hat{c}} Z_{\hat{c}} \alpha_{\hat{c},ci} \quad (18)$$

$$\tau_{\hat{c}k,ci} P_{ci} Z_{\hat{c}k,ci} = (1 - \mu_{\hat{c}k}) P_{\hat{c}k} Y_{\hat{c}k} \eta_{\hat{c}k,ci} \quad (19)$$

where $\alpha_{\hat{c},ci}$ is the share of the final demand of the buyer $\hat{c}$ from the supplier ci , and $\eta_{\hat{c}k,ci}$ is the share of the intermediate inputs of the buyer $\hat{c}k$ from the supplier ci . Summing over all buyers:

$$P_{ci} Y_{ci} = \sum_{\hat{c}} P_{\hat{c}} Z_{\hat{c}} \alpha_{\hat{c},ci} + \sum_{\hat{c}} \sum_k (1 - \mu_{\hat{c}k}) P_{\hat{c}k} Y_{\hat{c}k} \eta_{\hat{c}k,ci} \quad (20)$$

Next, we use the budget constraint of the households, $P_{\hat{c}} Z_{\hat{c}} = \sum_k \mu_{\hat{c}k} P_{\hat{c}k} Y_{\hat{c}k}$ (trade is balanced and profits are zero), to get:

$$P_{ci} Y_{ci} = \sum_{\hat{c}} \alpha_{\hat{c},ci} \sum_k \mu_{\hat{c}k} P_{\hat{c}k} Y_{\hat{c}k} + \sum_{\hat{c}} \sum_k (1 - \mu_{\hat{c}k}) P_{\hat{c}k} Y_{\hat{c}k} \eta_{\hat{c}k,ci} \quad (21)$$

In vector notation, with $\boldsymbol{\alpha}$ and $\boldsymbol{\eta}$ defined so that row = buyer and column = supplier:

$$PY = (\boldsymbol{\alpha}^\top \boldsymbol{\mu} + \boldsymbol{\eta}^\top (I - \boldsymbol{\mu})) PY \quad (22)$$

With world GDP as the numeraire, $\boldsymbol{\omega} = PY$ and $\sum_{ci} \mu_{ci} \omega_{ci} = 1$. Let $M \equiv \boldsymbol{\alpha}^\top \boldsymbol{\mu} + \boldsymbol{\eta}^\top (I - \boldsymbol{\mu})$. The sums of columns of M equal one by construction: for each buyer b , $\mu_b \sum_{ci} \alpha_{b,ci} + (1 - \mu_b) \sum_{ci} \eta_{b,ci} = \mu_b + (1 - \mu_b) = 1$. Hence M is column-stochastic, its spectral radius is one, and $\mathbf{1}^\top (I - M) = 0$, so the homogeneous system $(I - M) \boldsymbol{\omega} = 0$ has a non-trivial solution. By hypothesis, the shares are strictly positive, so M is strictly positive and hence irreducible; by Perron–Frobenius its Perron root 1 is algebraically

simple and the null space of $I - M$ is spanned by a strictly positive vector v . The set of solutions is therefore the ray $t v$, $t > 0$, and normalization $\sum_{ci} \mu_{ci} \omega_{ci} = 1$ pins down t uniquely, since $\mathbf{1}^\top \mu v > 0$. The selected solution is strictly positive.

Alternative representation with the standard Leontief inverse. Separate the intermediate and final-demand components:

$$\omega = \eta^\top (I - \mu) \omega + \alpha^\top \mu \omega.$$

The second term is final demand: each country's final expenditure equals its labor income. Define the country-level labor income $D_c = \sum_k \mu_{ck} \omega_{ck}$ and the standard IO Leontief inverse $\tilde{\mathcal{L}} = (I - \eta^\top (I - \mu))^{-1}$, which exists because the sums of columns of $\eta^\top (I - \mu)$ are $1 - \mu_b < 1$. Rearranging:

$$\omega = \tilde{\mathcal{L}} \alpha^\top \mu \omega.$$

The vector $\alpha^\top \mu \omega$ has element (ci) equal to $\sum_{c'} \alpha_{c',ci} D_{c'}$, i.e., the total final demand directed to the sector (ci) of all countries. In scalar notation:

$$\omega_{ci} = \sum_{\hat{c}, k} \tilde{\mathcal{L}}_{ci, \hat{c}k} \sum_{c'} \alpha_{c', \hat{c}k} D_{c'}. \quad (23)$$

This is the representation used in the computational implementation: the standard Leontief inverse $\tilde{\mathcal{L}}$ propagates the final demand through the intermediate network to obtain the Domar weights.

□

C.2 Proof of Lemma 3

Proof. C.2.1 Nominal GDP

The starting point of the proof is the Domar weights that we characterize in Lemma 2.

$$P_{ci} Y_{ci} = \omega_{ci} \quad (24)$$

Taking logs we have:

$$\log P_{ci} + \log Y_{ci} = \log(\omega_{ci}) \quad (25)$$

C.2.2 Labor Demand

The FOC of firms for labor are as follows:

$$W_{ci}L_{ci} = \mu_{ci}P_{ci}Y_{ci} \quad (26)$$

$$\log P_{ci} + \log Y_{ci} + \log \mu_{ci} = \log W_{ci} + \log L_{ci} \quad (27)$$

Then:

$$\log L_{ci} = -\log W_{ci} + \log \mu_{ci} + \log(\omega_{ci}) \quad (28)$$

C.2.3 Labor supply

Also, FOC of households implies:

$$P_c(L_{ci})^{\frac{1}{\psi}} = W_{ci} \quad (29)$$

$$\log L_{ci} = [\log W_{ci} - \log P_c]\psi \quad (30)$$

Combining both:

$$\log W_{ci} = \frac{\psi}{1+\psi} \log P_c + \frac{1}{1+\psi} [\log \mu_{ci} + \log(\omega_{ci})] \quad (31)$$

C.2.4 Demand of Inputs

From FOC of intermediate inputs of firms:

$$\log P_{ci} = -\epsilon_{ci} - (1 - \mu_{ci})A(\eta_{ci}^{\star}) + \mu_{ci} \log W_{ci} + (1 - \mu_{ci}) \sum_{\hat{c}} \sum_j \eta_{ci, \hat{c}j}^{\star} [\log P_{\hat{c}j} + \log(\tau_{c, \hat{c}j}^I)] \quad (32)$$

$$\log P_{ci} = -\epsilon_{ci} - (1 - \mu_{ci})A(\eta_{ci}^{\star}) + \mu_{ci} \left[\frac{\psi}{1+\psi} \log P_c + \frac{1}{1+\psi} (\log \mu_{ci} + \log(\omega_{ci})) \right] + (1 - \mu_{ci}) \sum_{\hat{c}} \sum_j \eta_{ci, \hat{c}j}^{\star} [\log P_{\hat{c}j} + \log(\tau_{c, \hat{c}j}^I)] \quad (33)$$

$$\begin{aligned} \log P_{ci} = & -\epsilon_{ci} - (1 - \mu_{ci})A(\eta_{ci}^*) + \mu_{ci} \left[\frac{\psi}{1 + \psi} \left[-A(\alpha_c^*) + \sum_{\hat{c}} \sum_j \alpha_{c,\hat{c}j}^* [\log P_{\hat{c}j} + \log(\tau_{c,\hat{c}}^F)] \right] + \right. \\ & \left. \frac{1}{1 + \psi} (\log \mu_{ci} + \log(\omega_{ci})) \right] + (1 - \mu_{ci}) \sum_{\hat{c}} \sum_j \eta_{ci,\hat{c}j}^* [\log P_{\hat{c}j} + \log(\tau_{c,\hat{c}j}^I)] \end{aligned} \quad (34)$$

Turning in matrix notation:

$$\begin{aligned} & \left[I - \frac{\psi}{1 + \psi} \boldsymbol{\mu} \boldsymbol{\alpha}^* - (I - \boldsymbol{\mu}) \boldsymbol{\eta}^* \right] \log \boldsymbol{\mu} \\ & -\epsilon - (I - \boldsymbol{\mu})A(\boldsymbol{\eta}^*) - \frac{\psi}{1 + \psi} \boldsymbol{\mu}A(\boldsymbol{\alpha}^*) + \left[\left(\frac{\psi}{1 + \psi} \boldsymbol{\mu} \boldsymbol{\alpha}^* \right) \circ \log \tau^F + ((I - \boldsymbol{\mu}) \boldsymbol{\eta}^*) \circ \log \tau^I \right] \mathbf{1} + \frac{1}{1 + \psi} \boldsymbol{\mu} (\log \boldsymbol{\mu} + \log(\boldsymbol{\omega})) \end{aligned} \quad (35)$$

Each row of $H \equiv \frac{\psi}{1 + \psi} \boldsymbol{\mu} \boldsymbol{\alpha}^* + (I - \boldsymbol{\mu}) \boldsymbol{\eta}^*$ sums to $\frac{\psi}{1 + \psi} \mu_{ci} + (1 - \mu_{ci}) = 1 - \frac{\mu_{ci}}{1 + \psi} < 1$, so $\|H\|_\infty < 1$, $\rho(H) < 1$, and $\mathcal{L}(s) = (I - H)^{-1} = \sum_{n \geq 0} H^n$ exists and is entrywise nonnegative. Applying it:

$$\begin{aligned} \log P = & \mathcal{L}(s) \left(-\epsilon - (I - \boldsymbol{\mu})A(\boldsymbol{\eta}^*) - \frac{\psi}{1 + \psi} \boldsymbol{\mu}A(\boldsymbol{\alpha}^*) + \left[\left(\frac{\psi}{1 + \psi} \boldsymbol{\mu} \boldsymbol{\alpha}^* \right) \circ \log \tau^F + ((I - \boldsymbol{\mu}) \boldsymbol{\eta}^*) \circ \log \tau^I \right] \mathbf{1} + \frac{1}{1 + \psi} \boldsymbol{\mu} (\log \boldsymbol{\mu} + \log(\boldsymbol{\omega})) \right) \end{aligned} \quad (36)$$

So, with $B(s)$ the bracket, which is deterministic because $\boldsymbol{\omega}(s)$ is (Lemma 2):

$$\boldsymbol{p} = \mathcal{L}(s) (B(s) - \epsilon), \quad (37)$$

which is the statement of the lemma. The step from $\log P_c$ to the final-good price index uses zero profits for the final-good producer, Corollary 1, which holds state by state.

□

C.3 Proof of Lemma 7

Proof. The proof starts from the definition of the primary-input composite:

$$e^{a_{ci}} = e^{\epsilon_{ci} + A(\eta_{ci})} L_{ci}^{\mu_{ci}} \quad (38)$$

Taking logs leads to the first part of the Lemma. For the second, combine the firm's labor demand $W_{ci}L_{ci} = \mu_{ci}\omega_{ci}$, which uses the world-GDP numeraire $P_{ci}Y_{ci} = \omega_{ci}$, with the household's condition $L_{ci} = (W_{ci}/P_c)^\psi$; because preferences are quasi-linear in the labor aggregate the marginal-utility factor cancels from the intratemporal condition, so labor supply depends on the real wage alone. Eliminating W_{ci} gives

$$l_{ci} = \frac{\psi}{1 + \psi} [-p_c + \log \mu_{ci} + \log(\omega_{ci})] \quad (39)$$

□

C.4 Proof of Lemma 4

Proof. By Lemma 3, $\mathbf{p} = \mathcal{L}(s)B(s) - \mathcal{L}(s)\epsilon$ is an affine function of ϵ with $\mathcal{L}(s)$ and $B(s)$ measurable with respect to s , which is known when techniques are chosen; $B(s)$ is deterministic because at $\vartheta = 0$ the Domar weights do not depend on ϵ (Lemma 2). Substituting $\epsilon = \epsilon^n - \sum_{c'} D_{c'} \xi_{c'} e_{(c')}$,

$$\mathbf{p} = \mathcal{L}(s)(B(s) - \epsilon^n) + \sum_{c'} D_{c'} \xi_{c'} \boldsymbol{\ell}^{c'}(s), \quad \boldsymbol{\ell}^{c'}(s) \equiv \mathcal{L}(s) e_{(c')},$$

where $e_{(c')}$ is the vector that selects the industries of country c' , $\mathbf{1}_{c'}$ in the lemma, so that $\boldsymbol{\ell}^{c'}$ has entries $\ell_{\hat{c}k}^{c'} = \sum_i \mathcal{L}_{\hat{c}k, c'i}$. Conditional on indicators $\{D_{c'}\}$ the right-hand side is an affine image of the jointly Gaussian vector $(\epsilon^n, \{\xi_{c'}\})$, so $\mathbf{p}$ is Gaussian, with mean $\mathcal{L}(B - \theta) + \sum_{c'} D_{c'} d_{c'} \boldsymbol{\ell}^{c'}$ and covariance $\mathcal{L}\Sigma_\epsilon \mathcal{L}^\top + \sum_{c'} D_{c'} \zeta_{c'}^2 \boldsymbol{\ell}^{c'} \boldsymbol{\ell}^{c'\top}$. For unconditional moments, $E_t[D_{c'} \xi_{c'}] = \pi_{c',t} d_{c'}$ by independence, which gives the mean of the lemma. The terms of the sum are independent of each other and of ϵ^n , so the variance is the sum of $\mathcal{L}\Sigma_\epsilon \mathcal{L}^\top$ and of $\text{Var}_t(D_{c'} \xi_{c'}) \boldsymbol{\ell}^{c'} \boldsymbol{\ell}^{c'\top}$, with

$$\text{Var}_t(D_{c'} \xi_{c'}) = E_t[D_{c'}] E[\xi_{c'}^2] - (\pi_{c',t} d_{c'})^2 = \pi_{c',t} (\zeta_{c'}^2 + d_{c'}^2) - \pi_{c',t}^2 d_{c'}^2 = \pi_{c',t} \zeta_{c'}^2 + \pi_{c',t} (1 - \pi_{c',t}) d_{c'}^2.$$

We call it $V_{c',t}^D$; factoring $\mathcal{L}$ on both sides gives the variance of the lemma, which

completes its proof. The rest of this appendix derives the moment generating function of prices, which the proofs of Lemma 5 and Proposition 1 use for the risk-adjusted probability and for $T_{o,c',t}$. Let $J_{c'} = D_{c'}\xi_{c'}$ be the jump and $K_{c'}(\beta) = \log E_t[e^{\beta J_{c'}}]$ its cumulant generating function. The mean of the jump under a kernel that loads β on it is $E_t[e^{\beta J_{c'}} J_{c'}]/E_t[e^{\beta J_{c'}}] = K'_{c'}(\beta)$, whose expansion in β has the mean of the jump as its constant and the variance as its slope, so $T_{c',t}(\beta) \equiv K'_{c'}(\beta) - E_t[J_{c'}] - \beta \text{Var}_t[J_{c'}]$ is the contribution of the cumulants of order three and higher. Its closed form follows from the moment generating function of prices, which Appendix C.6 also uses. Fix a vector b and write $\beta_{c'} = b^\top \ell^{c'}$, so that $b^\top p = b^\top \mathcal{L}(B - \epsilon^n) + \sum_{c'} \beta_{c'} D_{c'} \xi_{c'}$. The terms are independent, so the expectation of the exponential factorizes. The Gaussian factor is $\exp\{b^\top \mathcal{L}(B - \theta) + \frac{1}{2} b^\top \mathcal{L} \Sigma_\epsilon \mathcal{L}^\top b\}$ by the normal moment generating function. For each country, conditioned on $D_{c'}$ and using the same function for the size,

$$E_t[e^{\beta_{c'} D_{c'} \xi_{c'}}] = 1 - \pi_{c',t} + \pi_{c',t} e^{\beta_{c'} d_{c'} + \frac{1}{2} \beta_{c'}^2 \varsigma_{c'}^2}.$$

Taking logs and adding and subtracting $\pi_{c',t} d_{c'} \beta_{c'} + \frac{1}{2} V_{c',t}^D \beta_{c'}^2$ for every c' ,

$$\log E_t[e^{b^\top p}] = \underbrace{b^\top \mathcal{L}(B - \theta) + \sum_{c'} \pi_{c',t} d_{c'} \beta_{c'}}_{b^\top E_t[p]} + \underbrace{\frac{1}{2} b^\top \mathcal{L} \Sigma_\epsilon \mathcal{L}^\top b + \sum_{c'} V_{c',t}^D \beta_{c'}^2}_{\frac{1}{2} b^\top \text{Var}_t[p] b} + \sum_{c'} G_{c',t}(\beta_{c'}),$$

where

$$G_{c',t}(\beta) \equiv \log\left(1 - \pi_{c',t} + \pi_{c',t} e^{\beta d_{c'} + \frac{1}{2} \beta^2 \varsigma_{c'}^2}\right) - \pi_{c',t} d_{c'} \beta - \frac{1}{2} V_{c',t}^D \beta^2$$

is the log moment generating function of the jump net of its mean and variance. The first two derivatives of $\log(1 - \pi_{c',t} + \pi_{c',t} e^{\beta d_{c'} + \frac{1}{2} \beta^2 \varsigma_{c'}^2})$ in $\beta = 0$ are $\pi_{c',t} d_{c'}$ and $V_{c',t}^D$, the mean and variance of the jump, so $G_{c',t}(0) = G'_{c',t}(0) = G''_{c',t}(0) = 0$: $G_{c',t}$ collects the cumulants of the jump of order three and higher. Differentiating,

$$G'_{c',t}(\beta) = \hat{\pi}_{c',t}(\beta) (d_{c'} + \beta \varsigma_{c'}^2) - \pi_{c',t} d_{c'} - V_{c',t}^D \beta = T_{c',t}(\beta),$$

with $x_{c'}(\beta) = \beta d_{c'} + \frac{1}{2} \beta^2 \varsigma_{c'}^2$ and $\hat{\pi}_{c',t}(\beta) = \pi_{c',t} e^{x_{c'}(\beta)} / (1 - \pi_{c',t} + \pi_{c',t} e^{x_{c'}(\beta)}) = \left[1 + \frac{1 - \pi_{c',t}}{\pi_{c',t}} e^{-x_{c'}(\beta)}\right]^{-1}$. The household of the country o loads $\beta_{o,c'} \equiv (\gamma - 1) \ell^{o,c'}$ on the jump, because $\log M_o = (\gamma - 1) p_o + C_o$ and p_o rise by $\ell^{o,c'}$ per unit of jump (Corollary 1); $\beta_{o,c'}$ is used only in the proofs. The objects of the main text are these functions in that loading. The risk-adjusted probability of the Lemma 5 is $\hat{\pi}_{o,c',t} = \hat{\pi}_{c',t}(\beta_{o,c'})$, whose

odds are $\frac{\pi_{c',t}}{1-\pi_{c',t}} e^{x_{c'}(\beta_{o,c'})}$. The term of Proposition 1 is $T_{o,c',t} = T_{c',t}(\beta_{o,c'})$, and substituting $V_{c',t}^D = \pi_{c',t} \zeta_{c'}^2 + \pi_{c',t}(1 - \pi_{c',t}) d_{c'}^2$ and collecting terms,

$$T_{c',t}(\beta) = (\hat{\pi}_{c',t}(\beta) - \pi_{c',t})(d_{c'} + \beta \zeta_{c'}^2) - \beta \pi_{c',t}(1 - \pi_{c',t}) d_{c'}^2,$$

the form displayed there. Setting every $\pi_{c',t}$ to zero gives $G_{c',t} \equiv 0$, $T_{c',t} \equiv 0$ and the Gaussian case, $\mathbf{p} \sim \mathcal{N}(\mathcal{L}(B - \theta), \mathcal{L}\Sigma_\epsilon\mathcal{L}^\top)$. $\square$

C.5 Proof of Lemma 5

Proof. The proof has three steps: the household's marginal utility, the intermediary's valuation, and the kernel that prices a firm's payoff. Throughout, s' indexes the state realized after techniques are chosen and $\Pr(s')$ its probability under the common posterior of Lemma 1; world nominal GDP is the numeraire in every state.

Step 1: The household's marginal utility. No household trades a claim, so the budget constraint is (4) at every ϑ .

From the budget constraint $P_c z_c = \sum_i W_{ci} L_{ci}$ and the per-sector labor FOC $W_{ci}/P_c = L_{ci}^{1/\psi}$, we obtain

$$z_c = \sum_i \frac{W_{ci}}{P_c} L_{ci} = \sum_i L_{ci}^{1+\frac{1}{\psi}}.$$

Since the disutility of labor is $\sum_i L_{ci}^{1+1/\psi}/(1+1/\psi)$, the GHH net consumption simplifies to

$$z_c - \sum_i \frac{L_{ci}^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}} = \left(1 - \frac{1}{1+1/\psi}\right) \sum_i L_{ci}^{1+\frac{1}{\psi}} = \frac{1}{\psi+1} \sum_i L_{ci}^{1+\frac{1}{\psi}}.$$

The stochastic discount factor is

$$M_c = \frac{u'\left(z_c - \sum_i \frac{L_{ci}^{1+1/\psi}}{1+1/\psi}\right)}{P_c} = \left(\frac{1}{\psi+1} \sum_i L_{ci}^{1+\frac{1}{\psi}}\right)^{-\gamma} P_c^{-1}.$$

Combining labor demand $W_{ci} L_{ci} = \mu_{ci} \omega_{ci}$, which uses the world-GDP numeraire

$P_{ci} Y_{ci} = \omega_{ci}$, with labor supply $L_{ci} = (W_{ci}/P_c)^\psi$:

$$L_{ci} = \left(\frac{\mu_{ci} \omega_{ci}}{P_c} \right)^{\frac{\psi}{1+\psi}}, \quad L_{ci}^{1+\frac{1}{\psi}} = \frac{\mu_{ci} \omega_{ci}}{P_c},$$

where the second equality uses $\frac{\psi}{1+\psi} \cdot (1 + \frac{1}{\psi}) = 1$. Summing across sectors,

$$\sum_i L_{ci}^{1+\frac{1}{\psi}} = \frac{1}{P_c} \sum_i \mu_{ci} \omega_{ci}.$$

Substituting, the GHH net consumption becomes

$$\frac{1}{\psi + 1} \sum_i L_{ci}^{1+\frac{1}{\psi}} = \frac{1}{P_c} \frac{1}{\psi + 1} \sum_i \mu_{ci} \omega_{ci},$$

where the labor income is a nominal magnitude in units of world GDP; the Domar weights are deterministic and strictly positive (Lemma 2), so the price index factors out and the income term is known before ϵ and positive. Taking logs,

$$\begin{aligned} \log M_c &= -\gamma \left[-p_c + \log \left(\frac{1}{\psi + 1} \sum_i \mu_{ci} \omega_{ci} \right) \right] - p_c \\ &= (\gamma - 1) p_c - \gamma \log \left(\frac{1}{\psi + 1} \sum_i \mu_{ci} \omega_{ci} \right). \end{aligned}$$

Thus

$$\log M_c(s, \epsilon) = (\gamma - 1) p_c(s, \epsilon) + C_c(s),$$

where

$$C_c(s) = -\gamma \log \left(\frac{1}{\psi + 1} \sum_i \mu_{ci} \omega_{ci}(s) \right) = -\gamma \log \sum_i \mu_{ci} \omega_{ci}(s) + \gamma \log(\psi + 1)$$

depends only on the network s (through the labor-share-weighted Domar weights $\{\mu_{ci} \omega_{ci}\}$) and parameters, but not on shocks.

Step 2: The intermediary. Let $y_b(s')$ be the nominal payoff of firm b in state s' . The household that owns the firm pledges $\vartheta y_b(s')$ to the intermediary, state by state, and keeps $(1 - \vartheta) y_b(s')$. The intermediary has no resources of its own and its budget must

balance in every state, so what it receives in s' it pays out in s' : the household of c' , which holds the share $\xi_{c'}$ of the intermediary, receives $\xi_{c'} \vartheta \sum_b y_b(s')$, and $\sum_{c'} \xi_{c'} = 1$ because world nominal GDP is the numeraire. The intermediary is risk neutral: it attaches no value of its own to a unit of the numeraire in any state, and it takes a position only if its holders value it. The value it attaches to one more unit of the numeraire received in state s' is therefore the multiplier on its budget constraint in that state, which is the value of that unit to the households it is passed to,

$$\Pr(s') \sum_{c'} \xi_{c'} \hat{M}_{c'}(s'), \quad \hat{M}_{c'} \equiv M_{c'}/E_t[M_{c'}].$$

Each household's valuation is divided by $E_t[M_{c'}]$, its marginal valuation of a riskless unit of the numeraire, so that the valuations of different households are in the same unit and can be added. By Step 1 every $M_{c'}$ is positive with a finite mean, because the moment generating function of ϵ is finite under the Gaussian-plus-jump law, so $\sum_{c'} \xi_{c'} \hat{M}_{c'}$ is positive with mean one. The state prices of the intermediary are not physical probabilities: that would require it to absorb payoffs with resources of its own, which it does not have.

Step 3: The kernel that prices a firm's payoff. A firm in country c is owned by the household of c for the fraction $1 - \vartheta$ and by the intermediary for the fraction ϑ , and it maximizes its value to its owners. By Step 2 the part held by the intermediary is worth $\vartheta \sum_{c'} \xi_{c'} E_t[\hat{M}_{c'} y]$. The part kept by the household cannot be sold, so it is worth what the household would give up for it in riskless units of the numeraire: its marginal valuation of the payoff, $E_t[M_c y]$, over its marginal valuation of a riskless unit, $E_t[M_c]$, that is $(1 - \vartheta) E_t[\hat{M}_c y]$. The two parts have separate owners, so their values add, and the firm values y at

$$(1 - \vartheta) E_t[\hat{M}_c y] + \vartheta \sum_{c'} \xi_{c'} E_t[\hat{M}_{c'} y] = E_t[\Lambda_c y], \quad \Lambda_c = (1 - \vartheta) \hat{M}_c + \vartheta \sum_{c'} \xi_{c'} \hat{M}_{c'}.$$

The combination is in levels because values add across owners; nothing makes log values add. Λ_c is positive and $E_t[\Lambda_c] = 1$, so it is a density with respect to $\Pr$, a convex combination of the densities $\hat{M}_{c'}$, and for every random variable x with a finite mean

under each of them,

$$E_t[\Lambda_c x] = (1 - \vartheta) E_t[\hat{M}_c x] + \vartheta \sum_{c'} \xi_{c'} E_t[\hat{M}_{c'} x]$$

with no remainder.

No purchasing power moves. Every firm has constant returns and earns zero profits in every state, so $y_b(s') = 0$ at the optimum: the intermediary receives nothing and pays nothing, its budget balances in every state identically, and the household's budget constraint is (4) at every ϑ , which is what Step 1 uses. The household's own marginal utility is M_c at every ϑ ; Λ_c is the kernel at which the firm values a marginal change in its state-contingent costs, and that is the only place ϑ enters. Because the intermediary's shares pay nothing, they are worth nothing and the households' holdings of them are not pinned down by portfolio choice; $\xi_{c'}$ is a convention.

Step 4: The risk-adjusted disaster probability. $\hat{M}_c$ is positive with mean one, so $Q_c(A) \equiv E_t[\hat{M}_c \mathbf{1}_A]$ is a probability measure, and the probability of the disaster of country c' under it is $\hat{\pi}_{c,c',t} = E_t[\hat{M}_c D_{c'}]$. By Step 1 and Corollary 1, $\hat{M}_c \propto e^{(\gamma-1)p_c}$ and p_c rises by $\ell^{c,c'}$ per unit of the jump $D_{c'} \xi_{c'}$, so $\hat{M}_c$ contains the factor $e^{\beta D_{c'} \xi_{c'}}$ with $\beta = (\gamma - 1)\ell^{c,c'}$, and its other factors are independent of $D_{c'}$ and $\xi_{c'}$. Hence

$$\hat{\pi}_{c,c',t} = \frac{E_t[e^{\beta D_{c'} \xi_{c'}} D_{c'}]}{E_t[e^{\beta D_{c'} \xi_{c'}}]} = \frac{\pi_{c',t} E[e^{\beta \xi_{c'}}]}{1 - \pi_{c',t} + \pi_{c',t} E[e^{\beta \xi_{c'}}]}, \quad E[e^{\beta \xi_{c'}}] = e^{\beta d_{c'} + \frac{1}{2} \beta^2 \zeta_{c'}^2},$$

by the normal moment generating function, and dividing by $1 - \hat{\pi}_{c,c',t}$ gives the odds of the lemma. $\square$

C.6 Proof of Proposition 1

Proof. The proof has four parts. Part A derives the firm's first-order condition under the mixture kernel of Lemma 5 and shows that the premium is the same mixture of the households' premia. Part B computes the premium of one household within the Gaussian block, Part C adds the disaster term, and Part D collects the components into the displayed form. Throughout, s' indexes the state realized after techniques are chosen and $\Pr(s')$ its probability under the common posterior of Lemma 1; world nominal GDP is the numeraire in every state.

Part A: The firm's problem under the mixture kernel. Consider an atomistic firm b in country c with unit cost $K_b(m, P) = e^{-A_b(m)} \prod_{\hat{c}, k} (\tau_{b, \hat{c}k}^m P_{\hat{c}k})^{m_{\hat{c}k}}$, $A_b(m) = -\kappa_b^m \sum_{\hat{c}, k} m_{\hat{c}k} \log(m_{\hat{c}k}/m_{\hat{c}k}^0)$, as in Section 3. By Lemma 5 it values a state-contingent nominal payoff with $\Lambda_c = (1 - \vartheta)\hat{M}_c + \vartheta \sum_o \xi_o \hat{M}_o$, $\hat{M}_o \equiv M_o/E[M_o]$ the normalized kernel of the household of country o . Taking demand $Y_b(s')$, prices and the aggregate network as given, the firm chooses m before s' to minimize expected discounted cost,

$$\min_{m \in \Delta} E[\Lambda_c(s') Y_b(s') K_b(m, P(s'))] = e^{-A_b(m)} E\left[\Lambda_c Y_b \exp\left(\sum_{\hat{c}, k} m_{\hat{c}k} (p_{\hat{c}k} + \log \tau_{b, \hat{c}k}^m)\right)\right].$$

The moment generating function of ϵ is finite under the Gaussian-plus-jump law, so expectation and differentiation may be interchanged. Differentiating with respect to $m_{\hat{c}k}$ on the simplex (multiplier ω), dividing by the objective, and evaluating at the optimum m^* , where $K_b(m^*, P) = P_b$ by zero profits and $Y_b P_b = \text{Rev}_b$, which is deterministic,

$$E^{Q_b}[p_{\hat{c}k}] + \log \tau_{b, \hat{c}k}^m + \kappa_b^m \left(\log \frac{m_{\hat{c}k}^*}{m_{\hat{c}k}^0} + 1 \right) = \omega, \quad \frac{dQ_b}{d\text{Pr}}(s') = \frac{\Lambda_c(s') \text{Rev}_b}{E[\Lambda_c] \text{Rev}_b} = \Lambda_c(s').$$

The objective is strictly convex on the simplex, being the sum of the strictly convex entropy term and a log-moment-generating function, so the stationary point is the unique minimizer, and solving the first-order condition gives the softmax of Lemma 6 with $R_{b, \hat{c}k} = E^{Q_b}[p_{\hat{c}k}] + \log \tau_{b, \hat{c}k}^m$. Because Λ_c is a convex combination of densities, $E^{Q_b}[p_{\hat{c}k}] = (1 - \vartheta)E^{\hat{M}_c}[p_{\hat{c}k}] + \vartheta \sum_o \xi_o E^{\hat{M}_o}[p_{\hat{c}k}]$, so the premium is $(1 - \vartheta)\Pi_{\hat{c}k}^c + \vartheta \sum_o \xi_o \Pi_{\hat{c}k}^o$ with $\Pi_{\hat{c}k}^o \equiv E^{\hat{M}_o}[p_{\hat{c}k}] - E_t[p_{\hat{c}k}]$. Because $E_t[\hat{M}_o] = 1$, $E^{\hat{M}_o}[p_{\hat{c}k}] - E_t[p_{\hat{c}k}] = E_t[\hat{M}_o p_{\hat{c}k}] - E_t[\hat{M}_o] E_t[p_{\hat{c}k}] = \text{Cov}_t(\hat{M}_o, p_{\hat{c}k})$, and by the same argument $E^{Q_b}[p_{\hat{c}k}] - E_t[p_{\hat{c}k}] = \text{Cov}_t(\Lambda_c, p_{\hat{c}k})$, the form in the proposition.

Part B: The Gaussian block of a household's component. This part computes $\Pi_{\hat{c}k}^o$ for one household, written as the household of the firm's own country c at $\vartheta = 0$, within the Gaussian block, for a kernel of the form $\exp\{(\gamma - 1)p_c + \text{const}\}$ with deterministic nominal revenue; the argument uses only these two properties, so it holds for the household of any country o .

Step B.1: The firm's objective within the Gaussian block Starting from the firm's problem of Section 3, with demand Y taken as given by the atomistic firm. For any

input-bundle shares, the problem is the following.

$$\max E[M_c Y (P_{\text{rev}} - K_{\text{cost}})],$$

with $(P_{\text{rev}}, K_{\text{cost}}) = (P_c, K_c(\alpha, P))$ for the finals and $(P_{\text{rev}}, K_{\text{cost}}) = (P_{ci}, K_{ci}(\eta, P))$ for the intermediates. Since P_{rev} does not depend on the shares, choosing the shares is equivalent to *minimizing* the discounted expected cost:

$$\min E[M_c Y K_{\text{cost}}].$$

Log-moment step. Write $k_{\text{cost}} = \ln K_{\text{cost}}$ and $\lambda_c = \ln M_c$, $y = \ln Y$. Within the Gaussian block, λ_c , y and k_{cost} are affine in ϵ (Lemmas 3 and 5), so their sum is normal and the log-normal identity gives

$$\ln E[e^{\lambda_c + y + k_{\text{cost}}}] = E[\lambda_c + y + k_{\text{cost}}] + \frac{1}{2} \text{Var}(\lambda_c + y + k_{\text{cost}}).$$

Only the part that varies with the shares matters. The log unit cost varies with the shares through the index of the bundle, $k_{\text{cost}} = -A(m) + \sum_{c',k} m_{c'k} (p_{c'k} + \ln \tau_{\bullet, c'k})$, with $-A(m) = \kappa \sum_{c',k} m_{c'k} \ln(m_{c'k}/m_{c'k}^0)$ the entropy penalty of equation (7); for an intermediate producer the index enters the unit cost with the factor $1 - \mu_{ci} > 0$, which does not move the minimizer. Its full derivative with respect to the share on (c', k) is

$$\frac{\partial k_{\text{cost}}}{\partial m_{c'k}} = p_{c'k} + \ln \tau_{\bullet, c'k} + \kappa \left(\ln \frac{m_{c'k}}{m_{c'k}^0} + 1 \right),$$

of which only $p_{c'k}$ is random. Using $\frac{1}{2} \partial \text{Var}(X) = \text{Cov}(X, \partial X)$, the derivative of the log objective with respect to the share on (c', k) is

$$\underbrace{E[p_{c'k}] + \ln \tau_{\bullet, c'k}}_{\text{mean delivered price}} + \underbrace{\text{Cov}(\lambda_c + y + k_{\text{cost}}, p_{c'k})}_{\text{risk term}} + \underbrace{\kappa \left(\ln \frac{m_{c'k}}{m_{c'k}^0} + 1 \right)}_{\text{entropy gradient}},$$

where $\tau_{\bullet, c'k}$ denotes the relevant iceberg factor (final or intermediate). The first two terms are the share-wise index that multiplies each supplier; the entropy gradient is the term Lemma 6 solves for, and both reappear in Step 3.

Final-good problem. Here $y = y_c$ and $k_{\text{cost}} = k_c$. Nominal revenue is deterministic

in the numeraire $-P_c Z_c = \sum_i \mu_{ci} \omega_{ci}$ by Lemma 2 – so $y_c = -p_c$ up to a constant; and at the optimum zero profits give $k_c(\alpha_c^*) = p_c$. Hence, at the fixed point,

$$y_c = -p_c, \quad k_c = p_c.$$

Away from the optimum the objective carries the additional term $\frac{1}{2}\text{Var}(k_c(\alpha_c)) - \text{Cov}(p_c, k_c(\alpha_c))$, whose gradient vanishes at $\alpha_c = \alpha_c^*$, so the first-order condition displayed below is the exact one at every equilibrium; the objective is convex in the shares, being the sum of a strictly convex entropy term and a convex quadratic, so the stationary point is the global minimizer. Therefore

$$\text{Cov}(\lambda_c + y_c + k_c, p_{c'k}) = \text{Cov}(\lambda_c, p_{c'k}) + \underbrace{\text{Cov}(-p_c, p_{c'k}) + \text{Cov}(p_c, p_{c'k})}_{=0}.$$

Hence the per-supplier index for finals is

$$E[p_{c'k}] + \ln \tau_{c,c'k} + \text{Cov}(\lambda_c, p_{c'k}).$$

Intermediate-good problem. For intermediates we must use sectoral (own-good) demand Y_{ci} ; then $y = \ln Y_{ci} = y_{ci}$ and $k_{\text{cost}} = k_{ci}$. By the same argument, $P_{ci} Y_{ci} = \omega_{ci}$ is deterministic and $k_{ci}(\eta_{ci}^*) = p_{ci}$ at the optimum, so

$$y_{ci} = -p_{ci}, \quad k_{ci} = p_{ci}.$$

Thus

$$\text{Cov}(\lambda_c + y_{ci} + k_{ci}, p_{c'k}) = \text{Cov}(\lambda_c, p_{c'k}) + \underbrace{\text{Cov}(-p_{ci}, p_{c'k}) + \text{Cov}(p_{ci}, p_{c'k})}_{=0}.$$

Hence the per-supplier index for intermediates is

$$E[p_{c'k}] + \ln \tau_{ci,c'k} + \text{Cov}(\lambda_c, p_{c'k}).$$

Substituting the pricing kernel of Lemma 5 for λ_c yields the covariance term of the proposition.

Stacking. Summing these per-supplier indexes with weights $\alpha_{c,c'k}$ (finals) and $\eta_{ci,c'k}$

(intermediates) gives the objective indexes used in (23) and (24), respectively.

Step B.2: Pricing kernel. By Lemma 5, $\lambda_c = (\gamma - 1) p_c + C_c(s)$ with $C_c(s)$ known before ϵ .

Step B.3: Combining both For final-good producers in country c , the (log) cost index is

$$k_c(\alpha_c; \alpha^\star) = -A_c(\alpha_c) + \sum_{\hat{c}, k} \alpha_{c, \hat{c}k} (p_{\hat{c}k} + \log \tau_{c, \hat{c}k}),$$

while the equilibrium final-good price index satisfies

$$p_c(\alpha^\star) = -A_c(\alpha_c^\star) + \sum_{\hat{c}, k} \alpha_{c, \hat{c}k}^\star (p_{\hat{c}k} + \log \tau_{c, \hat{c}k}).$$

Since $C_c(s)$, $A_c(\cdot)$ and $\log \tau_{c, \hat{c}k}$ are deterministic, they drop out of covariances, and we obtain

$$\begin{aligned} \text{Cov}(\lambda_c, k_c(\alpha_c; \alpha^\star)) &= (\gamma - 1) \text{Cov}(p_c(\alpha^\star), k_c(\alpha_c; \alpha^\star)) \\ &= (\gamma - 1) \text{Cov}\left(\sum_{\hat{c}', j} \alpha_{c, \hat{c}'j}^\star p_{\hat{c}'j}, \sum_{\hat{c}, k} \alpha_{c, \hat{c}k} p_{\hat{c}k}\right) \\ &= (\gamma - 1) \sum_{\hat{c}, k} \sum_{\hat{c}', j} \alpha_{c, \hat{c}k} \alpha_{c, \hat{c}'j}^\star \text{Cov}(p_{\hat{c}k}, p_{\hat{c}'j}). \end{aligned}$$

The final-good producer in country c chooses α_c to minimize expected (log) cost adjusted for risk:

$$\min_{\alpha_c \in \mathcal{A}} \mathbb{E}[k_c(\alpha_c; \alpha^\star)] + \text{Cov}(\lambda_c, k_c(\alpha_c; \alpha^\star)).$$

Substituting the expressions above and rearranging,

$$\begin{aligned} &\mathbb{E}[k_c(\alpha_c; s)] + \text{Cov}(\lambda_c, k_c(\alpha_c; s)) \\ &= -A_c(\alpha_c) + \sum_{\hat{c}, k} \alpha_{c, \hat{c}k} \left(\mathbb{E}[p_{\hat{c}k}] + \log \tau_{c, \hat{c}k} + (\gamma - 1) \sum_{\hat{c}', j} \alpha_{c, \hat{c}'j}^\star \text{Cov}(p_{\hat{c}k}, p_{\hat{c}'j}) \right). \end{aligned}$$

Defining the *risk-adjusted delivered price* as

$$\mathcal{R}_{c, \hat{c}k}(s) := \mathbb{E}[p_{\hat{c}k}] + \log \tau_{c, \hat{c}k} + (\gamma - 1) \sum_{\hat{c}', j} \alpha_{c, \hat{c}'j}^\star \text{Cov}(p_{\hat{c}k}, p_{\hat{c}'j}),$$

and using the entropy adjustment cost of equation (7), $A_c(\alpha_c) = -\kappa^F \sum_{\hat{c},k} \alpha_{c,\hat{c}k} \log(\alpha_{c,\hat{c}k}/\alpha_{\hat{c}k}^0)$, so that the penalty in k_c is $-A_c(\alpha_c) = \kappa^F \sum_{\hat{c},k} \alpha_{c,\hat{c}k} \log(\alpha_{c,\hat{c}k}/\alpha_{\hat{c}k}^0) \geq 0$, the problem of the final-good producer is

$$\alpha_c^* \in \arg \min_{\alpha_c \in \mathcal{A}} \kappa^F \sum_{\hat{c},k} \alpha_{c,\hat{c}k} \log \frac{\alpha_{c,\hat{c}k}}{\alpha_{\hat{c}k}^0} + \sum_{\hat{c},k} \alpha_{c,\hat{c}k} \mathcal{R}_{c,\hat{c}k}^F(s).$$

For an intermediate producer in sector (c, i) , the (log) cost index is

$$k_{ci}(\eta_{ci}; s) = -A_{ci}(\eta_{ci}) + \sum_{\hat{c},k} \eta_{ci,\hat{c}k} (p_{\hat{c}k} + \log \tau_{ci,\hat{c}k}),$$

while the household SDF for country c is the same M_c as above. Using again that constants drop from covariances, we obtain

$$\begin{aligned} \text{Cov}(\lambda_c, k_{ci}(\eta_{ci}; s)) &= (\gamma - 1) \text{Cov}\left(p_c(\alpha^*), \sum_{\hat{c},k} \eta_{ci,\hat{c}k} p_{\hat{c}k}\right) \\ &= (\gamma - 1) \sum_{\hat{c},k} \sum_{\hat{c}',j} \eta_{ci,\hat{c}k} \alpha_{c,\hat{c}'j}^* \text{Cov}(p_{\hat{c}k}, p_{\hat{c}'j}). \end{aligned}$$

Thus the intermediate producer solves

$$\min_{\eta_{ci} \in \mathcal{E}} \mathbb{E}[k_{ci}(\eta_{ci}; s)] + \text{Cov}(\lambda_c, k_{ci}(\eta_{ci}; \eta^*)),$$

that is,

$$-A_{ci}(\eta_{ci}) + \sum_{\hat{c},k} \eta_{ci,\hat{c}k} \left(\mathbb{E}[p_{\hat{c}k}] + \log \tau_{ci,\hat{c}k} + (\gamma - 1) \sum_{\hat{c}',j} \alpha_{c,\hat{c}'j}^* \text{Cov}(p_{\hat{c}k}, p_{\hat{c}'j}) \right).$$

Replacing the risk-adjusted prices and writing the entropy adjustment cost as in equation (7), $A_{ci}(\eta_{ci}) = -\kappa^I \sum_{\hat{c},k} \eta_{ci,\hat{c}k} \log(\eta_{ci,\hat{c}k}/\eta_{ci,\hat{c}k}^0)$, so that the penalty in k_{ci} is $-A_{ci}(\eta_{ci})$, the problem of the intermediate producer is

$$\eta_{ci}^* \in \arg \min_{\eta_{ci} \in \mathcal{E}} \kappa^I \sum_{\hat{c},k} \eta_{ci,\hat{c}k} \log \frac{\eta_{ci,\hat{c}k}}{\eta_{ci,\hat{c}k}^0} + \sum_{\hat{c},k} \eta_{ci,\hat{c}k} \mathcal{R}_{ci,\hat{c}k}(s).$$

which is the household's component of the proposition within the Gaussian block.

Part C: The disaster term of a household's component. With disasters the log-normal identity of Part B is unavailable because ϵ is no longer Gaussian; the first-order condition is instead evaluated directly, using the cancellation $y + k = 0$ established in Step B.3, so that the derivative of $\log E_t[e^{\lambda_c + y + k}]$ with respect to the share on $(\hat{c}, k)$ is $E^Q[p_{\hat{c}k}] + \log \tau$ under the tilt $dQ/d\text{Pr} \propto e^{(\gamma-1)p_c}$. Write $\epsilon = \epsilon^n - \sum_{c'} D_{c'} \xi_{c'} e_{(c')}$, where ϵ^n collects the Gaussian components, $D_{c'}$ is the disaster indicator, $\xi_{c'} \sim \mathcal{N}(d_{c'}, \varsigma_{c'}^2)$ the size of the decline, and $e_{(c')}$ the incidence vector of country c' 's productivities; the jumps are independent of ϵ^n , of each other and over time (Section 3). Conditional on the signals the $D_{c'}$ are independent Bernoulli($\pi_{c',t}$), so for any loading a

$$E_t[e^{a^\top \epsilon}] = E[e^{a^\top \epsilon^n}] \prod_{c'} \left[(1 - \pi_{c',t}) + \pi_{c',t} e^{-s_{c'}(a) d_{c'} + \frac{1}{2} s_{c'}(a)^2 \varsigma_{c'}^2} \right], \quad s_{c'}(a) \equiv a^\top e_{(c')},$$

using the normal moment generating function for each size. The Gaussian factor delivers the Gaussian part of $E_t[p]$ and the whole of Cov_n , exactly as in the proposition without disasters. With $a_c \equiv \alpha_c^{\star\top} \mathcal{L}$ the loading of p_c on ϵ , so that $p_c = \text{const} - a_c^\top \epsilon$, the tilt $e^{(\gamma-1)p_c}$ loads $+(\gamma-1)a_c^\top e_{(c')} = +\beta_{c,c'}$ on the jump size $\xi_{c'}$, by the definitions of $\ell^{c'}$ and $\beta_{c,c'}$. Under Q the jumps remain independent across countries; the indicator $D_{c'}$ is Bernoulli with

$$\text{logit } \hat{\pi}_{c,c',t} = \text{logit } \pi_{c',t} + x_{c,c'}, \quad x_{c,c'} = \beta_{c,c'} d_{c'} + \frac{1}{2} \beta_{c,c'}^2 \varsigma_{c'}^2,$$

because the Bernoulli mixture tilted by $E[e^{\beta \xi}] = e^x$ is again a two-point distribution, and by the Gaussian exponential-tilt identity the size mean shifts to $d_{c'} + \beta_{c,c'} \varsigma_{c'}^2$. Since $p_{\hat{c}k}$ loads $+\ell_{\hat{c}k}^{c'}$ on $\xi_{c'}$,

$$E^Q[p_{\hat{c}k}] - E_t[p_{\hat{c}k}] = \sum_{c'} \ell_{\hat{c}k}^{c'} \left[\hat{\pi}_{c,c',t} (d_{c'} + \beta_{c,c'} \varsigma_{c'}^2) - \pi_{c',t} d_{c'} \right]$$

for the jump block, which is the bracket of the proposition; the posterior-mean component $\sum_{c'} \ell_{\hat{c}k}^{c'} \pi_{c',t} d_{c'}$ is already contained in $E_t[p_{\hat{c}k}]$. Setting every $\bar{\pi}_{c'}$ to zero removes each factor and recovers the Gaussian formula.

Part D: The components and the form of the proposition. Every kernel in the mixture is a household's: $\hat{M}_o \propto \exp\{(\gamma-1)p_o\}$ with p_o affine in ϵ , $p_o = \text{const} - \alpha_o^{\star\top} \mathcal{L}\epsilon$, and ϵ follows the Gaussian-plus-independent-jumps law of Section 3. Part B computes

$E^{\hat{M}_o}[p_{\hat{c}k}] - E_t[p_{\hat{c}k}]$ within the Gaussian block using only these two properties, and Part C adds the jump term, in which the household's exposure to country c' enters through $\beta_{o,c'} = (\gamma - 1) \sum_{\hat{h},k'} \alpha_{o,\hat{h}k'}^* \ell_{\hat{h}k'}^{c'}$. Both arguments hold for the household of any country o , which gives

$$\Pi_{\hat{c}k}^o = (\gamma - 1) \text{Cov}_n(p_o, p_{\hat{c}k}) + \sum_{c'} \ell_{\hat{c}k}^{c'} \left[\hat{\pi}_{o,c',t} (d_{c'} + \beta_{o,c'} \varsigma_{c'}^2) - \pi_{c',t} d_{c'} \right]$$

for every o .

The same expression follows in one line from the moment generating function of prices derived in Appendix C.4, $\log E_t[e^{b^\top p}] = b^\top E_t[p] + \frac{1}{2} b^\top \text{Var}_t[p] b + \sum_{c'} G_{c',t}(b^\top \ell^{c'})$, which also regroups it. By Corollary 1, $p_o = \text{const} + \alpha_o^{*\top} p$, so $\hat{M}_o = e^{b_o^\top p} / E_t[e^{b_o^\top p}]$ with $b_o \equiv (\gamma - 1) \alpha_o^*$, and

$$E^{\hat{M}_o}[p_{\hat{c}k}] = \frac{\partial}{\partial b_{\hat{c}k}} \log E_t[e^{b^\top p}] \Big|_{b=b_o} = E_t[p_{\hat{c}k}] + (\text{Var}_t[p] b_o)_{\hat{c}k} + \sum_{c'} \ell_{\hat{c}k}^{c'} G_{c',t}(b_o^\top \ell^{c'}),$$

the moment generating function being finite for every b . Here $(\text{Var}_t[p] b_o)_{\hat{c}k} = (\gamma - 1) \text{Cov}_t(p_o, p_{\hat{c}k})$ and $b_o^\top \ell^{c'} = \beta_{o,c'}$, so

$$\Pi_{\hat{c}k}^o = (\gamma - 1) \text{Cov}_t(p_o, p_{\hat{c}k}) + \sum_{c'} \ell_{\hat{c}k}^{c'} T_{o,c',t},$$

because $G'_{c',t}(\beta_{o,c'}) = T_{c',t}(\beta_{o,c'}) = T_{o,c',t}$, the tail premium of Lemma 4 (Appendix C.4). The two displays for $\Pi_{\hat{c}k}^o$ agree term by term: by Lemma 4, $(\gamma - 1) \text{Cov}_t(p_o, p_{\hat{c}k}) = (\gamma - 1) \text{Cov}_n(p_o, p_{\hat{c}k}) + \sum_{c'} \ell_{\hat{c}k}^{c'} V_{c',t}^D \beta_{o,c'}$, which moves the variance of each jump out of the bracket and into the covariance. This is the displayed form of Proposition 1 for every $\vartheta \in [0, 1]$. $\square$

Remarks Ownership is split mechanically. For each firm, a share ϑ of its state-contingent payoff is pledged to a global intermediary, which distributes it to households in all countries in proportion to their income shares; the remaining share $1 - \vartheta$ is held by the firm's domestic household. In this model households always consume their own (deterministic) labor income in the world numeraire given the network (Lemma 2). As a result, changing ϑ does *not* move purchasing power across states or countries: final expenditure by country, Domar weights, wages, and the price system in Lemma 3 are the same for all ϑ . What changes with ϑ is only the *pricing kernel* used to evaluate supplier risk.

Formally, the kernel is a convex combination as in equation (21) of [Biais et al. \(2021\)](#): the pledgeable share of any payoff is priced with the intermediary's state price, while the non-pledgeable share is priced with the domestic household's MRS. Here the intermediary's state price is the income-weighted average of households' normalized marginal utilities. Unlike an endowment economy, the pledged claims themselves pay zero in every state, so state prices are not recovered from trade; they are pinned by who holds the intermediary.

For the Gaussian (covariance) block, the cross-household average premium against supplier $(\hat{c}, k)$ collapses to a premium against the income-weighted average price index because covariance is linear: $\sum_o \xi_o \text{Cov}_t(p_o, p_{\hat{c}k}) = \text{Cov}_t(\sum_o \xi_o p_o, p_{\hat{c}k})$ (including jumps). The tail block does *not* collapse: it averages $T_{c',t}$ evaluated with each household's exposure $\beta_{o,c'}$, because $T_{c',t}$ is nonlinear in β .

The complete-markets pole. At $\vartheta = 1$, every buyer prices every supplier with the same world kernel: the premium on $(\hat{c}, k)$ is $\sum_o \xi_o \Pi_{\hat{c}k}^o$ for all buyers, so the dispersion of the premia between buyers (the incomplete-markets signature in Section 4) disappears. Increasing ϑ therefore compresses premium dispersion, but it does not determine the sign of any given premium change. In the Gaussian block, buyer c 's premium change on $(\hat{c}, k)$ is $\vartheta(\gamma - 1) \text{Cov}_n(\sum_o \xi_o p_o - p_c, p_{\hat{c}k})$, whose sign depends on whether the supplier co-moves more with the world basket than with buyer c 's basket. Hence pledgeability has no general implication for concentration; Section ?? works out when it raises or lowers it.

C.7 Proof of Lemma 6

We consider the problem of a generic buyer with shares $s \in \mathcal{A}$ (the unit simplex), ideal technology s^0 , adjustment cost $\kappa > 0$ and risk-adjusted prices $\mathcal{R}$. The entropy adjustment cost problem is the following:

$$s^* \in \arg \min_{s \in \mathcal{A}} \left\{ \kappa \sum_k s_k \log \frac{s_k}{s_k^0} + \sum_k s_k \mathcal{R}_k \right\},$$

subject to $\sum_k s_k = 1$ and $s_k \geq 0$ for all k . We assume $s_k^0 > 0$ and $\sum_k s_k^0 = 1$.

Step 1: Forming the Lagrangian. We introduce a Lagrange multiplier λ for the equality constraint. Since $\partial[s_k \log(s_k/s_k^0)]/\partial s_k = \log(s_k/s_k^0) + 1 \rightarrow -\infty$ as $s_k \rightarrow 0^+$ whenever $s_k^0 > 0$, the marginal cost of reducing any share diverges at the boundary, so the minimizer lies

in the interior of the simplex and the non-negativity constraints are never active. The Lagrangian is:

$$\mathcal{L}(s, \lambda) = \kappa \sum_k s_k \log \frac{s_k}{s_k^0} + \sum_k s_k \mathcal{R}_k + \lambda \left(1 - \sum_k s_k \right).$$

Step 2: First-Order Conditions. Differentiate $\mathcal{L}$ with respect to each s_k :

$$\frac{\partial \mathcal{L}}{\partial s_k} = \kappa \left(\log \frac{s_k}{s_k^0} + 1 \right) + \mathcal{R}_k - \lambda = 0.$$

Solving for s_k :

$$\log \frac{s_k}{s_k^0} = \frac{\lambda - \kappa - \mathcal{R}_k}{\kappa},$$

$$s_k = s_k^0 \exp \left(\frac{\lambda - \kappa - \mathcal{R}_k}{\kappa} \right) = s_k^0 \exp \left(-\frac{\mathcal{R}_k}{\kappa} \right) \cdot \exp \left(\frac{\lambda - \kappa}{\kappa} \right).$$

Step 3: Enforcing the Constraint. Summing over k and imposing $\sum_k s_k = 1$:

$$\exp \left(\frac{\lambda - \kappa}{\kappa} \right) \sum_k s_k^0 \exp \left(-\frac{\mathcal{R}_k}{\kappa} \right) = 1.$$

Defining the partition function $Z = \sum_k s_k^0 \exp(-\mathcal{R}_k/\kappa)$, we have $\exp((\lambda - \kappa)/\kappa) = 1/Z$. Substituting back:

$$s_k^\star = \frac{s_k^0 \exp(-\mathcal{R}_k/\kappa)}{\sum_{k'} s_{k'}^0 \exp(-\mathcal{R}_{k'}/\kappa)}.$$

This is the softmax of $-\mathcal{R}_k/\kappa$ with base measure s^0 . All shares are strictly positive whenever $s_k^0 > 0$. The result holds for both intermediate producers (with $s = \eta_{ci}$, $\kappa = \kappa^I$) and final-good producers (with $s = \alpha_c$, $\kappa = \kappa^F$), using the corresponding risk-adjusted prices $\mathcal{R}^I$ and $\mathcal{R}^F$.

C.8 Proof of Lemma 8

Proof. From the budget constraint $P_c Z_c = \sum_i W_{ci} L_{ci}$ and the per-sector labor FOC $W_{ci}/P_c = L_{ci}^{1/\psi}$, we have

$$Z_c = \sum_i \frac{W_{ci}}{P_c} L_{ci} = \sum_i L_{ci}^{1+\frac{1}{\psi}}.$$

Since the per-sector disutility of labor in the household objective is $\sum_i L_{ci}^{1+1/\psi}/(1+1/\psi)$, net consumption simplifies to

$$Z_c - \sum_i \frac{L_{ci}^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}} = \frac{1}{1+\frac{1}{\psi}} \sum_i L_{ci}^{1+\frac{1}{\psi}}.$$

Using the per-sector equilibrium relation from Lemma 3, $L_{ci}^{1+1/\psi} = \mu_{ci} \omega_{ci}/P_c$, this is a nominal magnitude deflated by the price index:

$$Z_c - \sum_i \frac{L_{ci}^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}} = \frac{1}{P_c} \frac{1}{1+\frac{1}{\psi}} \sum_i \mu_{ci} \omega_{ci},$$

and therefore

$$\log \left(Z_c - \sum_i \frac{L_{ci}^{1+1/\psi}}{1+1/\psi} \right) = -p_c(s, \epsilon) + \log \left(\frac{1}{1+\frac{1}{\psi}} \sum_i \mu_{ci} \omega_{ci} \right).$$

Write $X_c \equiv Z_c - \sum_i L_{ci}^{1+1/\psi}/(1+1/\psi)$ and $I_c(s) \equiv \frac{1}{1+\frac{1}{\psi}} \sum_i \mu_{ci} \omega_{ci}(s)$, the deterministic nominal income term; $I_c(s)$ is measurable with respect to the first-stage state alone, because Domar weights are invariant to ϵ by Lemma 2, and it is positive. Then $\log X_c = \log I_c(s) - p_c(s, \epsilon)$ and

$$\log \mathbb{E}_t [X_c^{1-\gamma}] = (1-\gamma) \log I_c(s) + \log \mathbb{E}_t [e^{(\gamma-1)p_c(s, \epsilon)}].$$

By Corollary 1, $p_c = \text{const} + \boldsymbol{\alpha}_c^{\star \top} \mathbf{p}$, so the last expectation is the moment generating function of prices at $\mathbf{b} = (\gamma-1)\boldsymbol{\alpha}_c^{\star}$, derived in Appendix C.4: the Gaussian factor gives $(\gamma-1)\mathbb{E}_n[p_c] + \frac{1}{2}(\gamma-1)^2 \text{Var}_n(p_c)$, and each country's jump gives one two-point factor,

$\log(1 - \pi_{c',t} + \pi_{c',t} e^{\beta d_{c'} + \frac{1}{2} \beta^2 \varsigma_{c'}^2})$ at the loading $\beta = b^\top \ell^{c'} = (\gamma - 1) \ell^{c,c'}$. Hence

$$\log \mathbb{E}_t[X_c^{1-\gamma}] = (1 - \gamma) \log I_c(s) + (\gamma - 1) \mathbb{E}_n[p_c(s, \epsilon)] + \frac{1}{2}(\gamma - 1)^2 \text{Var}_n(p_c(s, \epsilon)) + \sum_{c'} \log(1 - \pi_{c',t} + \pi_{c',t} e^{\beta d_{c'} + \frac{1}{2} \beta^2 \varsigma_{c'}^2})$$

Welfare is the log certainty equivalent $\mathcal{W}_c(s) = \log \text{CE}_c$, with $\text{CE}_c^{1-\gamma} = \mathbb{E}_t[X_c^{1-\gamma}]$, so $\log \text{CE}_c = \frac{1}{1-\gamma} \log \mathbb{E}_t[X_c^{1-\gamma}]$. Dividing by $(1 - \gamma)$ gives the lemma: the Gaussian terms become $-\mathbb{E}_n[p_c] + \frac{1}{2}(1 - \gamma) \text{Var}_n(p_c)$, because $\frac{1}{2}(\gamma - 1)^2/(1 - \gamma) = \frac{1}{2}(1 - \gamma)$, and the jump factors carry the coefficient $-1/(\gamma - 1)$. By Step 4 of Appendix C.5, $1 - \hat{\pi}_{c,c',t} = (1 - \pi_{c',t})/(1 - \pi_{c',t} + \pi_{c',t} e^{\beta d_{c'} + \frac{1}{2} \beta^2 \varsigma_{c'}^2})$ at $\beta = (\gamma - 1) \ell^{c,c'}$, so each factor is $\log[(1 - \pi_{c',t})/(1 - \hat{\pi}_{c,c',t})]$. To write the lemma with the full moments, add and subtract the mean and the variance of each jump: by Lemma 4, $\mathbb{E}_t[p_c] = \mathbb{E}_n[p_c] + \sum_{c'} \ell^{c,c'} \pi_{c',t} d_{c'}$ and $\text{Var}_t(p_c) = \text{Var}_n(p_c) + \sum_{c'} (\ell^{c,c'})^2 (\pi_{c',t} \varsigma_{c'}^2 + \pi_{c',t} (1 - \pi_{c',t}) d_{c'}^2)$, so $-\mathbb{E}_n[p_c] + \frac{1}{2}(1 - \gamma) \text{Var}_n(p_c) = -\mathbb{E}_t[p_c] + \frac{1}{2}(1 - \gamma) \text{Var}_t(p_c) + \sum_{c'} [\ell^{c,c'} \pi_{c',t} d_{c'} + \frac{1}{2}(\gamma - 1)(\ell^{c,c'})^2 (\pi_{c',t} \varsigma_{c'}^2 + \pi_{c',t} (1 - \pi_{c',t}) d_{c'}^2)]$; collecting these with the log ratio gives $-\frac{1}{\gamma-1} \sum_{c'} G_{c,c',t}$ with

$$G_{c,c',t} = G_{c',t}((\gamma-1)\ell^{c,c'}) = \log \frac{1 - \pi_{c',t}}{1 - \hat{\pi}_{c,c',t}} - (\gamma-1)\ell^{c,c'} \pi_{c',t} d_{c'} - \frac{1}{2}(\gamma-1)^2 (\ell^{c,c'})^2 (\pi_{c',t} \varsigma_{c'}^2 + \pi_{c',t} (1 - \pi_{c',t}) d_{c'}^2),$$

the higher-cumulant part of the jump's log moment generating function of Appendix C.4 at the household's loading, as displayed in the lemma; since $G_{c',t}(0) = 0$ and $G'_{c',t} = T_{c',t}$, it is the integral of the premium of Proposition 1 over the loading from zero to the household's. When every $\pi_{c',t}$ is zero the last sum vanishes and $\log X_c$ is exactly normal, which is the Gaussian log-normal identity $\log \mathbb{E}[X_c^{1-\gamma}] = (1 - \gamma) \mathbb{E}[\log X_c] + \frac{1}{2}(1 - \gamma)^2 \text{Var}(\log X_c)$. $\square$

C.9 Proof of Corollary 1

Zero profits for the final-good producer with the technology of Section 3: the log unit cost is $-A(\alpha_c^\star)$ plus the share-weighted log delivered prices. For the exposure, $\mathbf{p} = \mathcal{L}(s)(B(s) - \epsilon)$ by Lemma 3, so lowering the productivity of every industry of country c' by one unit, $\epsilon \rightarrow \epsilon - e_{(c')}$ with $e_{(c')}$ the vector that selects those industries, raises $\mathbf{p}$ by $\mathcal{L}(s)e_{(c')} = \ell^{c'}(s)$, whose entries are $\sum_i \mathcal{L}_{ik,c'i}$, and raises p_c by $\alpha_c^\star \ell^{c'}(s) = \ell^{c,c'}(s)$.

C.10 Proof of Proposition 4

The display is the difference of the log first-order condition of Lemma 6 across the two suppliers at the same buyer, whose partition function cancels. Read at every buying industry in which both legs are active, once between each domestic supplier and a reference industry of the destination and once between each foreign supplier and the domestic producer of its own industry, it determines the wedges of a destination up to one constant per buying industry, which the shares cannot see. That constant is set by the normalization $\sum_{\hat{c}k} m_{\hat{c}k,b} \log \tau_{c,\hat{c}k}^m = 0$ over the priced cells of buyer b , for each block $m \in \{I, F\}$. The wedges enter the expected-price recursion of Lemma 4, $\mathbb{E}[p_{ci}] = B_{ci}^0 - \theta_{ci} + \sum_{\hat{c}k} w_{ci,\hat{c}k}^I (\mathbb{E}[p_{\hat{c}k}] + \log \tau_{c,\hat{c}k}^I) + \sum_{\hat{c}k} w_{ci,\hat{c}k}^F (\mathbb{E}[p_{\hat{c}k}] + \log \tau_{c,\hat{c}k}^F)$ with $w^I = (1 - \mu_{ci})\eta^\star$ and $w^F = \frac{\psi}{1+\psi}\mu_{ci}\alpha^\star$, only through the row sums $\sum_{\hat{c}k} w_{ci,\hat{c}k}^m \log \tau_{c,\hat{c}k}^m$, which the normalization sets to zero at the observed network over the priced cells. Expected prices are therefore $\mathbb{E}[\mathbf{p}] = \mathcal{L}(s)(B^0 - \theta)$ plus the contribution of the active cells that have no comparison, which carry a wedge 50 log points above the buyer's most expensive priced supplier; that contribution is an affine map of $\mathbb{E}[\mathbf{p}]$ into itself that runs only through the shares of those cells, its spectral radius is far below one, and the fixed point is unique. Every wedge then follows from the display. No wedge is set to zero and no per-destination system arises.

C.11 Proof of Proposition 3

Differencing the log first-order condition of Lemma 6 over j, j' at a fixed supplier removes $\mathbb{E}[\mathbf{p}]$ and $\log \tau$, which carry no buying-industry index, and leaves $-\left[(1 - \vartheta)\Pi_{\hat{c}k}^c + \vartheta \sum_o \xi_o \Pi_{\hat{c}k}^o\right](1/\kappa_j - 1/\kappa_{j'})$, which vanishes when $\kappa_j = \kappa_{j'}$ because the premium is priced with the country household's kernel and also carries no buying-industry index; differencing again over suppliers removes the two partition functions. In $\log m_{\hat{c}k,j}^0 = a_{\hat{c}k} + b_j + g_{\hat{c}k,j}$, the double differences identify every interaction contrast, hence $\log m^0$ up to the additive two-way part, on each connected component. Given a , normalization $\sum_{\hat{c}k} \exp(a_{\hat{c}k} + b_j + g_{\hat{c}k,j}) = 1$ has a unique solution in b_j , its left side being continuous and strictly increasing from 0 to ∞ in b_j , so only $a_{\hat{c}k}$ remains free.

The proposition identifies only η^0 . The final-good producer has one buyer per country, so there is no second buying industry to the difference over and the double difference is not available for α^0 . The final-good blueprint is instead recovered by a one-way demean across destinations: the log first-order condition, $\log \alpha_{c,\hat{c}k} = \log \alpha_{\hat{c}k}^0 - (\mathbb{E}[p_{\hat{c}k}] + \log \tau_{c,\hat{c}k}^F +$

$\mathcal{R}_{c,\hat{c}k})/\kappa^F - \log Z_c$, is averaged over the destinations c in which the cell is active, and the destination effect $\log Z_c$ is removed by demeaning, with normalization $\sum_{\hat{c}k} \alpha_{\hat{c}k}^0 = 1$ fixing the level. What this leaves in α^0 is the destination average of each supplier's expected price, wedge, and premium, which a one-way demean does not separate from the blueprint.

D Estimation Algorithm

This appendix describes the estimation procedure and counterfactual computation. The algorithm takes as given the balanced panel of $C = 24$ countries and $J = 23$ sectors over $T = 58$ years (1967–2024), constructed by chain-linking the Long-run WIOD (1967–2000) and the WIOD 2016 release (2001–2014) at the overlap year 2000 and extending with the ADB MRIO tables (2015–2024), chain-linked at 2014 (Step 0).

D.1 Inputs

The following objects are computed from data and do not depend on structural parameters:

- Observed networks: $\eta_{ci,\hat{c}k,t}^{\text{data}}$ (intermediate shares), $\alpha_{c,\hat{c}k,t}^{\text{data}}$ (final-demand shares).
- Intermediate cost shares $\mu_{ci,t}$ and Domar weights $\omega_{ci,t}$.
- Log value added $a_{ci,t}$.
- Ideal technology $\eta_{\hat{c}k,j}^0$: the exact double-difference inversion of Proposition 3, computed year by year as described in Section 6.

The structural parameters are (ψ, γ, κ) , where ψ is the Frisch elasticity, γ is risk aversion, and κ is the softmax temperature, common across buyers, which is the hypothesis under which Proposition 3 holds.

D.2 Step 1: Recovery of $\hat{\epsilon}_t$ and GARCH Estimation of Σ_t^ϵ

The time-varying covariance matrix of TFP shocks is estimated from value-added data, independently of (γ, κ, τ) .

1. **Inversion at the observed network.** Lemma 7 implies $\mathbf{a}_t = M_t \epsilon_t + \mathbf{D}_t$, with $\mathbf{D}_t$ deterministic given the network. Each cell of $a_{ci,t}$, the log real value added scaled by the labor share, is projected on a constant, a linear trend and its labor share, which absorbs $\mathbf{D}_t$; the residual $\tilde{\mathbf{a}}_t$ is inverted year by year at that year's observed network,

$$\hat{\epsilon}_t = M_t^{-1} \tilde{\mathbf{a}}_t, \quad M_t \equiv I + \frac{\psi}{1+\psi} \text{diag}(\boldsymbol{\mu}_t) \boldsymbol{\alpha}_t^\star L_t, \quad (40)$$

so no change in the network between $t - 1$ and t enters $\hat{\epsilon}_t$.

2. **Factor decomposition.** Project $\hat{\varepsilon}_{ci,t}$ onto the factor structure:

$$\hat{\varepsilon}_{ci,t} = f_t^1 + f_{c,t}^2 + f_{j,t}^3 + f_{ci,t}^4, \quad (41)$$

by demeaning: f_t^1 is the year mean over active cells; $f_{c,t}^2$ is the country-year mean minus f_t^1 ; $f_{j,t}^3$ is the industry-year mean minus f_t^1 ; $f_{ci,t}^4$ is the remainder. Each block except the global one sums to zero within a year.

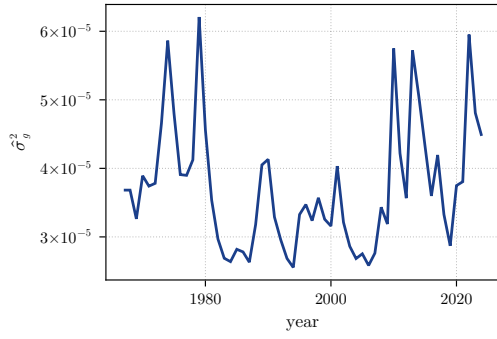
3. **AR(1) persistence.** Estimate ρ_k for each factor $k \in \{g, \chi, \zeta, u\}$ by pooled OLS, clamping $|\rho_k| \leq 0.999$. The persistence whitens the factors for the GARCH only; the mean firms price around is the node's time mean $\hat{\theta}$ of the outer loop, fixed over the sample, and $\rho_k \varepsilon_{t-1}$ enters no equilibrium.
4. **Innovations.** Compute $v_{k,t} = k_t - \rho_k k_{t-1}$ for each factor. Winsorize at $3 \times \text{MAD}$.
5. **Panel GARCH(1,1).** Estimate common (α, β) by maximizing the panel log-likelihood across all factor series simultaneously:

$$\hat{\sigma}_{k,t}^2 = (1 - \alpha - \beta) \bar{\sigma}_k^2 + \alpha v_{k,t-1}^2 + \beta \hat{\sigma}_{k,t-1}^2, \quad (42)$$

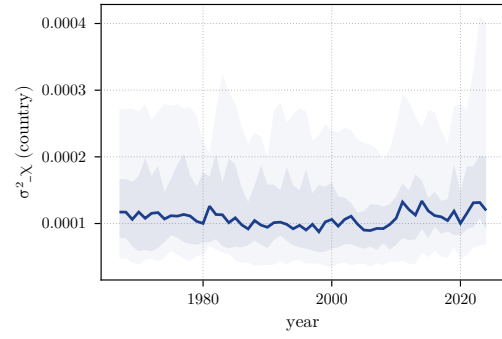
with $\hat{\sigma}_k^2(t_0) = \text{Var}(v_k)$ and the constraint $\alpha + \beta \leq 0.80$.

6. **Assemble $\hat{\Sigma}_t^\varepsilon$.** The covariance of primitive shocks is the sum of the four blocks: $\hat{\sigma}_{g,t}^2$ in every entry, $\hat{\sigma}_{\chi c,t}^2$ within each country block, $\hat{\sigma}_{\zeta j,t}^2$ within each industry block and $\hat{\sigma}_{uci,t}^2$ on the diagonal.

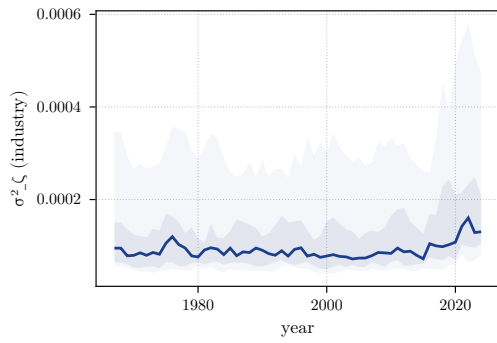
Figure 10 plots the estimated components; Section 7.2 describes them.



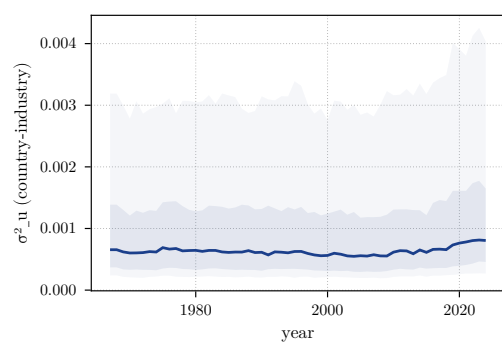
(a) Global variance $\hat{\sigma}_{g,t}^2$



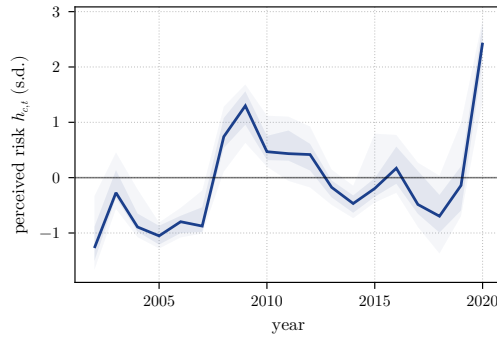
(b) Country variance $\hat{\sigma}_{\chi,c,t}^2$



(c) Industry variance $\hat{\sigma}_{\zeta,j,t}^2$



(d) Granular variance $\hat{\sigma}_{u,ci,t}^2$



(e) Perceived country risk $h_{c,t}$

Figure 10: Changes in Uncertainty

Notes: Panels (a)–(d) plot conditional variances of the global, country, industry, and country–industry components, 1967–2024 (median with 25th–75th and 10th–90th percentile bands). Panel (e) plots perceived country risk, 2002–2020 (median with the same bands).

D.3 Step 2: Trade Cost Identification

Bilateral trade costs are recovered exactly from the softmax FOCs at the observed economy each year, given the structural parameters (κ, γ, ψ) and the average TFP shock $\hat{\theta}$. No linearization around a steady state is used: every quantity below $(\eta^{\text{data}}, \alpha^{\text{data}}, \mu, \mathcal{L}, \text{Cov}(\mathbf{p}))$ is computed at the data shares for the year being inverted.

The softmax FOC for buyer (c, k) purchasing from supplier $(\hat{c}, k')$ is:

$$\log \eta_{ck, \hat{c}k'}^* = \log \eta_{ck, \hat{c}k'}^0 - \kappa^{-1} [\mathbb{E}[p_{\hat{c}k'}] + \log \tau_{c, \hat{c}k'}^\eta + R_{ck, \hat{c}k'}^\eta] - \log Z_{ck}, \quad (43)$$

where $R_{ck, \hat{c}k'}^\eta = (1 - \vartheta) \Pi_{\hat{c}k'}^c + \vartheta \sum_o \xi_o \Pi_{\hat{c}k'}^o$ is the premium of Proposition 1: each Π^o is the Gaussian covariance $(\gamma - 1) \text{Cov}(p_o, p_{\hat{c}k'})$, evaluated with the full covariance $\hat{\Sigma}^\varepsilon$ through $\alpha_o^{*\top} \mathcal{L} \hat{\Sigma}^\varepsilon \mathcal{L}^\top e_{\hat{c}k'}$, plus the disaster bracket at the prior hazards $\bar{\pi}_{c'}$ (the inversion sets $h = 0$ in every year, so the posterior equals the prior and a year's wedges carry that year's belief-driven sourcing; the Online Appendix re-extracts at the posterior of the observed signal from 2002); Z_{ck} is the partition function. The code implements exactly this mixture (21_extract_wedges.jl). The analogous FOC for α replaces $\eta \rightarrow \alpha$ and the buyer index by the country.

Asymmetric trade cost. Take the FOC at buyer (c, k) for the foreign supplier $(\hat{c}, k)$ and the domestic same-sector producer (c, k) . This is the two-level comparison of Proposition 4: at the first level every domestic supplier (c, i) is read against one reference industry of the destination, $r(c)$, and at the second level every foreign supplier $(\hat{c}, i)$ against the domestic producer of its own industry, (c, i) , in every buying industry of c in which both legs are active; the readings are averaged. The comparisons determine the wedges of a destination up to one constant per buying industry, which the shares cannot identify but which enters the buyer's cost B and through it every expected price. The constant is set so that the share-weighted mean wedge each buying industry pays is zero over its priced cells, $\sum_{\hat{c}k} \eta_{ci, \hat{c}k} \log \tau_{c, \hat{c}k}^\eta = 0$, and likewise for the final-goods block; the wedge a buying industry faces is the destination contrast plus its own level. (Fixing the reference route at a zero wedge instead makes $\mathbb{E}[p_{r(c)}]$ enter every wedge of the destination and the wedges enter B ; the resulting linear map from $\mathbb{E}[p]$ to itself has spectral radius 0.97 in 1967 and 1.17 in 2024, so its solution amplifies the data: Spain's import wedges came out at -12 and its price level 23 log points below the United States.) No wedge is set to zero. Subtracting the

two FOCs of the second level (the partition function Z_{ck} cancels because the buyer is the same), with the domestic wedge carried along:

$$\log \tau_{c,\hat{c}k}^\eta - \log \tau_{c,ck}^\eta = d_{(\hat{c},c,k)}^\eta + (\mathbb{E}[p_{ck}] - \mathbb{E}[p_{\hat{c}k}]), \quad (44)$$

where the data-only constant is

$$d_{(\hat{c},c,k)}^\eta = \kappa \log \frac{\eta_{ck,ck}^{\text{data}}}{\eta_{ck,\hat{c}k}^{\text{data}}} - \kappa \log \frac{\eta_{ck,ck}^0}{\eta_{ck,\hat{c}k}^0} + (R_{ck,ck}^\eta - R_{ck,\hat{c}k}^\eta). \quad (45)$$

The first level reads the same display with (c, i) and $r(c)$ in place of $(\hat{c}, k)$ and (c, k) . The analogous formula for τ^α replaces $\eta \rightarrow \alpha$ at the buyer side and the destination by the country c . Unlike the symmetric (Head–Ries) closed form, this derivation does not average the FOC for $\hat{c} \rightarrow c$ with the FOC for $c \rightarrow \hat{c}$; the $\mathbb{E}[p]$ terms therefore survive and $\tau_{c,\hat{c}k}^\eta \neq \tau_{\hat{c},ck}^\eta$ in general.

Solving the system for $\mathbb{E}[p]$. The contrasts (44) express the wedges in terms of data and the price expectations $\mathbb{E}[p]$, and $\mathbb{E}[p]$ depends on the wedges through the price system of Lemma 3,

$$B = B_0 + \left[((I - \mu) \boldsymbol{\eta}^\star) \circ \log \tau^I + \left(\frac{\psi}{1+\psi} \mu \boldsymbol{\alpha}^\star \right) \circ \log \tau^F \right] \mathbf{1}, \quad \mathbf{p} = \mathcal{L} (B - \hat{\theta}), \quad (46)$$

where $\circ$ is the entrywise product, $\log \tau^I$ and $\log \tau^F$ are the $N \times N$ matrices whose row (c, k) carries $\log \tau_{c,\hat{c}k'}^\eta$ and $\log \tau_{c,\hat{c}k'}^\alpha$ across suppliers $(\hat{c}, k')$, and B_0 collects the τ -free terms $(A(\boldsymbol{\eta}^\star), A(\boldsymbol{\alpha}^\star), \log \mu, \log \omega)$. The bracket is the share-weighted sum of the wedges in each row, which the normalization sets to zero over the priced cells. The wedges therefore enter B only through the active cells with no comparison, whose wedge is 50 log points above the buyer's most expensive priced supplier and moves with $\mathbb{E}[p]$ through that supplier's contrast. Writing that contribution as $S(\mathbb{E}[\mathbf{p}])$, an affine map,

$$(I - \mathbf{A}) \mathbb{E}[\mathbf{p}] = \mathcal{L} (B_0 - \hat{\theta}) + S(\mathbf{0}), \quad \mathbf{A} \equiv \mathcal{L} [S(\cdot) - S(\mathbf{0})], \quad (47)$$

is one linear solve; $\mathbf{A}$ runs only through the shares of those cells, its spectral radius is far below one, and the residual of the solve is at machine precision in every year. The wedges then follow from the contrasts and the normalization. All quantities are computed at the observed economy each year; $\hat{\theta}$ enters as an exogenous input from the outer loop. There is

no iteration in τ .

Outer fixed-point loop on $\hat{\theta}$. The inversion above takes $\hat{\theta}$ as given. After recovering τ , productivity shocks are updated via $\hat{\varepsilon}_t = M_t^+(\text{va}_t - D_t)$ where M_t, D_t are built from the data shares and the new τ , and the aggregate is $\hat{\theta} = \text{mean}_t(\hat{\varepsilon}_t)$. The outer loop iterates the pair $(\tau, \hat{\theta})$ until $\|\Delta\hat{\theta}\|_\infty < 10^{-5}$, which takes five to six iterations in every year; convergence is observed, not proven.

Identification and active set. A direction $(\hat{c}, c, k)$ is identified when the two shares of the asymmetric comparison, $\eta_{ck,ck}^{\text{data}}$ and $\eta_{ck,\hat{c}k}^{\text{data}}$, and the corresponding prior cells exceed the active threshold 10^{-5} in some buying industry; the symmetric formula below needs all four. An inactive route takes the prohibitive value of Section 6, $\bar{\tau} = 50$ in logs. An active route with no comparison, a supplier whose own industry has no active domestic producer in the destination, takes a wedge 50 log points above the buyer's most expensive priced supplier; its observed share is then not reproduced, and the code checks that no active cell is left at a default value.³

Comparison with the symmetric closed form. Imposing Head-Ries symmetry $\log \tau_{c,\hat{c}k}^\eta = \log \tau_{\hat{c},ck}^\eta$ and averaging the two reciprocal FOCs cancels the $\mathbb{E}[p]$ differences, yielding the per-cell scalar formula

$$\log \tau_{(\hat{c},c,k)}^{\eta,\text{sym}} = \frac{\kappa}{2} \log \frac{\eta_{\hat{c}\hat{c}}^{\text{data}} \eta_{cc}^{\text{data}}}{\eta_{\hat{c}c}^{\text{data}} \eta_{c\hat{c}}^{\text{data}}} - \frac{\kappa}{2} \log \frac{\eta_{\hat{c}\hat{c}}^0 \eta_{cc}^0}{\eta_{\hat{c}c}^0 \eta_{c\hat{c}}^0} + \frac{\gamma-1}{2} \Delta \text{Cov}(p), \quad (48)$$

which forces $\tau_{AB} = \tau_{BA}$ and requires no linear-system solve. The asymmetric inversion of (47)–(44) retains the directionality that the averaging collapses, at the cost of solving a single linear system per year.

³The entropy formulation guarantees all equilibrium shares are strictly positive (Lemma 6). In practice, the softmax assigns exponentially small shares to inactive links, and the floor and prohibitive default ensure numerical stability without contradicting the no-corner-solutions property.

D.4 Estimation Details

Identification of the Wedges

The inner difference of Proposition 3 removes $\mathbb{E}[p]$, $\log \tau$ and the premium $(1 - \vartheta)\Pi^c + \vartheta \sum_o \xi_o \Pi^o$; the outer difference across suppliers removes the partition functions. The double difference is free of every equilibrium object and of κ and γ , and identifies the blueprint up to the level of each supplier's column. Proposition 4 identifies the wedges from the remaining margin, differences across suppliers within a destination. The level of a supplier's wedge and the level of its blueprint are identified only as a sum; cross-destination variation in τ , the domestic-versus-foreign gap, and the variation of τ over time at a given blueprint are unaffected. At $\gamma = 1$ the inversion is the gravity inversion of Bonadio et al. (2025); at $\gamma > 1$ the premium, built from the observed network and the covariance recovered from it, removes insurance-driven sourcing from the wedge without solving for the equilibrium network.

Degrees of freedom. τ varies by origin country-sector and destination country, η^0 by supplier and buying industry. Per block, τ contributes at most $C(CJ - 1)$ contrasts, common to the buying industries of a destination, plus one level per buying industry that is normalized (the share-weighted mean wedge the buying industry pays is zero), and η^0 one blueprint per buying industry, against $(CJ)^2$ observed shares. The two load on orthogonal contrasts: η^0 on what is common across destinations within a buying industry, τ on what differs across suppliers within a destination.

Which cells identify τ . Each domestic supplier is compared with a reference industry of the destination and each foreign supplier with the domestic producer of its own industry; every buying industry of c that sources from both legs of a comparison contributes one reading of the wedge, eight at the median, and the estimator averages them. A supplier that sells at home but not to a destination has a route-level zero, rationalized by a prohibitive wedge of 50 log points up to a share of order $m^0 e^{-50/\kappa}$, and does not enter the price system. An active route whose supplier's industry has no active domestic producer in the destination has no comparison; it receives a wedge 50 log points above the buyer's most expensive priced supplier, so its observed share is not reproduced.

Small shares. A cell is active when its share exceeds 10^{-5} . The inversion runs on the unfloored data and the solver applies no floor, because a floored zero is read as a prohibitive trade cost, increasingly so in later years, when trade spreads across more and smaller links. Observed shares at or below the cutoff are treated as zeros and carry the same prohibitive wedge, so the inversion is exact on the active cells only.

The Common-Trade-Cost Restriction

Roughly ten thousand wedges per year explain over a hundred thousand share observations. Trade costs are common across buying industries within a destination, so the readings of a wedge across those industries must agree under the model; their dispersion is not fitted and is reported as a diagnostic, since it has no null benchmark and so does not test the restriction. The median standard deviation across readings of the same wedge is 0.03 log points in the 1970s and 0.69 in the 2020s; on readings whose two cells both exceed 10^{-3} of the buyer's inputs it is 0.60 and 0.71, so the trend comes from the thin cells, which are 96 to 97 percent of the readings in every decade, as networks densify and the marginal cell becomes small enough for its log to be noisy. The blueprint carries the mirror restriction, common across destinations, recovered on the orthogonal margin.

The Estimation Sequence

The blueprint comes first: the double difference depends on no parameter, it is computed per year, each year's wedges are inverted against that year's blueprint, and each counterfactual holds it at the year of the economy it starts from. The double difference leaves one level per supplier free, the term a cross-supplier comparison would need and on which the wedge is identified instead. The two-way within transform sets that level to zero, so the blueprint carries no supplier level and the wedge carries the sum of the two, which is what the shares identify; for α^0 the one-way demean across destinations does the same. The rule is the same in every year, so yearly blueprints are comparable and a change in a supplier's level moves its wedge, not its blueprint.

The wedges come second and are conditional on the kernel. In the inversion $\log \tau_{c,\hat{c}k} = G_{c,\hat{c}k} - \mathbb{E}[p_{\hat{c}k}] + \text{const}_c$, the contrast G contains the risk matrix, which depends on (κ, γ) and ϑ , and the constant sets the share-weighted mean wedge each buyer pays to zero. A τ extracted at one parameter point does not rationalize the network at another.

The parameters come third, by indirect inference on equation (??) with the wedges held

at a reference point; the wedges are then re-extracted at the estimate and the moments recomputed. The estimate is the conditional fixed point: the parameter point that minimizes the objective given wedges extracted at that point. The selected point $(\kappa, \gamma, \lambda) = (0.90, 11, 4)$ is the extraction point, so the re-extraction returns the same wedges. Every moment is within half a standard error of the data: 0.08 for the tariff, 0.44 and 0.08 for the episode split, 0.50 for the pooled belief response. The objective, the root of the sum of squared moment gaps in data standard errors, is 0.69 at the estimate; 0.73 at $\kappa = 0.85$ and 0.80 at 0.95; 0.85 at $\gamma = 10$ and 1.20 at $\gamma = 9$; 0.71 at $\lambda = 3.5$ and 0.76 at $\lambda = 6$; and 0.70 at $\lambda = 5$, which the four moments do not distinguish from the estimate. The bound $\gamma \leq 11$ maintained in the search does not bind: the objective is 0.77 at $\gamma = 12$ and 0.93 at $\gamma = 13$. The pledgeable fraction ϑ is calibrated outside the network (Section ??) and enters only as the weight on the two kernels in Proposition 1.

The Auxiliary Regression, the Instrument and the Horizon

The auxiliary regression. Equation (??) is the auxiliary model of an indirect-inference estimator: the same regression, instrument, fixed effects, winsorizing and clustering on the data and on the shares the model generates under the same shifters. No coefficient is a structural parameter. The instrumented tariff change is informative about κ : on the simulated panel b^r reproduces $-1/\kappa$ to within about six percent at the estimate (-0.99 against -1.05). The belief regressor, $e^{h_{\hat{c},t-1}} - 1$ interacted with the buyer country's exposure to the supplier's country and split by whether that country is inside a disaster episode at $t - 1$, the observable proxy for the persistent hazard regime of Lemma 1, is informative about γ through the level of the response and, given the disaster calibration, about λ through the episode ratio, because the belief state acts through the posterior disaster probability (Lemma 1) and is priced in proportion to exposure (Proposition 1). The fixed effects are supplier-node by buying-industry by year, buyer-node by year, and route. The response of a log share to the measured global variance, $-\frac{\gamma-1}{\kappa} (v_{\hat{c}k} - \bar{v}_b) v_c \Delta_h \sigma_{g,t}^2$ from the fixed-network first-order condition, with $\bar{v}_b$ the share-weighted centrality of the buyer's suppliers, is untargeted and tested in the Online Appendix.

The instrument. The tariff term is instrumented with most-favored-nation schedule changes, as in Boehm et al. (2023): the change in the importer's MFN rate, for pairs trading on MFN terms in both periods (no preferential agreement, both GATT/WTO members,

CEPII status) and for suppliers below ten percent of the importer’s sectoral imports. Schedules for the eleven tariff regimes covering the 24 countries come from TRAINS at the HS6 level, 1995–2021, aggregated to the 23 sectors (the targets use outcomes through 2018, the profiles of Section ?? through 2021); services suppliers enter with $\Delta \log(1+T) \equiv 0$. Each horizon- h coefficient is the ratio of the share projection to the projection of the regressor’s own path on the instrument; the first stage has $t \approx 20$. Standard errors are clustered by route. The maintained exclusion is that, conditional on these effects, MFN schedule changes on minor routes are orthogonal to importer-by-supplying-industry shocks; the facts below are consistent with it and do not establish it.

The horizon. All moments are read at one pre-committed horizon, $h = 5$, chosen from the persistence of the treatments: the short- against long-run elasticities of [Boehm et al. \(2023\)](#), the firm-level tariff profile that builds over six years (Figure 1), and the sectoral profile that reaches its trough at five to six years (Figure 2). At $h = 5$ the perceived-risk response is -0.59 ($p = 0.028$), the tariff response -0.90 ($p = 0.026$), the episode response -4.83 ($p = 0.29$); each would be larger at its own peak, $h = 6, 4$ and 4 .

The moments. b^τ and the pooled belief response b^h , which weights perceived risk by the Leontief consumption exposure $\ell_c^{\hat{c}}$, come from one regression; the episode split b^N , b^D of equation (??), with the model-free final-demand exposure, from a second. Both are estimated on the model-support cells, those the wedge store prices at both ends of the window and the blueprint supports, 700,241 cells at $h = 5$, the full panel of cells active at both ends under the calibrated wedge store. Two facts support the exclusion restrictions: the belief measure does not move applied tariffs ($p > 0.15$ at every horizon), and the tariff instrument does not move the belief state.

From Moments to Parameters

The projections are estimated on equilibrium data, so their coefficients are not partial derivatives of the first-order condition. At each candidate $(\kappa, \gamma, \lambda)$ I perturb the equilibrium with the measured shifters, tariff changes from the empirical MFN pool and belief draws through the candidate λ map, run the same projection on the simulated responses, and minimize the root of the sum of squared distances between simulated and data coefficients in units of the data standard errors. b^D , whose standard error is nearly as large as its point

estimate, disciplines λ weakly, which is why Section ?? reports it as the weakest moment. λ does not enter the inversion, since beliefs are at their prior when shares are inverted.

The sufficient condition of Proposition 2, in its Gaussian $\vartheta = 0$ block, bounds the search: the adjustment cost must be large relative to the sensitivity of risk-adjusted prices to the network, a bound that tightens with γ . Two restrictions are over-identifying and untargeted: the flight-to-safety response above, and the proportionality of the belief response to the predetermined exposure.

Disaster Risk and Measured Beliefs

Disaster risk. The Gaussian components are estimated on winsorized innovations with the world episodes neutralized; country episodes remain in them, winsorized. Episodes come from 152 years of real GDP per capita (Maddison Project, 1870–2022) by the peak-to-trough method of Barro (2006) and Barro and Ursúa (2008): a cumulative decline of at least 10 percent (5 percent for the population-weighted world aggregate). Country disasters are independent across countries and over time; the five world episodes (WWI, the Great Depression, WWII, COVID among them) are the years neutralized when the Gaussian variances are estimated. The prior hazard $\bar{\pi}_c$ is the country’s episode count over the 152 years, 1.3 to 3.3 percent a year, 2.1 on average, and zero for Ireland; the size d_c is the productivity decline per episode year times the mean duration, zero to 0.55 log points, 0.21 on average, largest for Greece, Austria, China, the Netherlands and India, so the jump is the episode’s cumulative loss placed in one period and the one-period tail is the episode’s tail, while the year-to-year variation within episodes is in the Gaussian block; the dispersion ζ_c is the per-year dispersion scaled by the square root of the duration. Sizes are mapped from GDP into TFP units with the elasticity of Lemma 7.

Measured beliefs. $h_{c,t}$ is the annual mean of $\log(1 + \text{risk})$ as perceived by foreign firms in Hassan et al. (2023), standardized within country; foreign perception identifies a signal about the country factor. Its first principal component accounts for 70 percent of the variance with all loadings positive. It enters the model through the posterior of Lemma 1, $\text{logit } q_{c,t} = \text{logit } \bar{q}_c + \lambda h_{c,t-1}$ with $\pi_c^0 = 0$, $\pi_c^1 = 1$, so $q_{c,t} = \pi_{c,t}$ and $\bar{q}_c = \bar{\pi}_c$; the regressions use $e^{h_{c,t-1}} - 1$, the proportional change in the posterior odds at unit discriminability, and the estimated λ sets how far the odds move. Ireland, with no Maddison disaster, carries the second-highest perceived exposure and the United States the lowest. The geopolitical

threat index of [Caldara and Iacoviello \(2022\)](#), correlated -0.50 with the common factor of $h_{c,t}$, is a robustness measure, not a component of the state. Neither measure moves measured second moments or professional GDP forecasts at any horizon.

Estimating uncertainty. The conditional variances $\sigma_{g,t}^2$, $\sigma_{\chi,c,t}^2$, $\sigma_{\zeta,j,t}^2$ and $\sigma_{u,ci,t}^2$ are GARCH(1,1) on the episode-cleaned factor innovations of the productivity shocks, each year's shocks being the inversion of Lemma 7 at that year's observed network, equation (15), with the two GARCH parameters pooled across series. The covariance of primitive shocks is the sum of the four blocks, year by year. Appendix D gives the algorithm.